

Folding (nice) rational
subsets of automatic groups.

Gazteak RSME '25

FOLDING (NICE) RATIONAL SUBSETS OF AUTOMATIC GROUPS

*L-quasiconvexly
undistorted*

Lucía Asencio Martín

+ J. Britnell, A. Duncan, D. Francœur, S. Rees

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$$\langle h_1, \dots, h_k \rangle = H \leq F(S)$$

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$$\langle h_1, \dots, h_k \rangle = H \leq F(S)$$

Given $w \in F(S) \dots w \in H ??$

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SUBGROUP MEMBERSHIP PROBLEM

$$\langle h_1, \dots, h_k \rangle = H \leq F(S)$$

Given $w \in F(S) \dots w \in H ??$



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SUBGROUP MEMBERSHIP PROBLEM

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Given $w \in F(S) \dots w \in H ??$

Stallings '83: very beautiful answer! (Foldings)

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SUBGROUP MEMBERSHIP PROBLEM

other
subsets? $\langle h_1, \dots, h_k \rangle = H \leq \cancel{F(S)}$ other
groups?

Given $w \in F(S) \dots w \in H ??$

Stallings '83: very beautiful answer! (Foldings)

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SUBGROUP MEMBERSHIP PROBLEM

other subsets? $\langle h_1, \dots, h_k \rangle = H \leq \cancel{F(S)}$ other groups?
rational subsets automatic groups

Given $w \in F(S) \dots w \in H ??$

Stallings '83: very beautiful answer! (Foldings)

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1. Context

2. $K \subseteq G$
rational automatic

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0. Defs / notation

- Languages
- FSA
- Automatic gps.

1. Context

2. $K_{\text{rational}} \subseteq G_{\text{automatic}}$

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subsets of automatic groups.

0. Defs / notation

0.1. Languages

Folding (nice) rational
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0. Defs / notation

0.1. Languages

- S is a symmetric, finite alphabet

$$S = \{a, b, c, \bar{a}, \bar{b}, \bar{c}\}$$

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0. Defs / notation

0.1. Languages

- S is a symmetric, finite alphabet
- S^* denotes finite words over S

$$S = \{a, b, c, a^{-1}, b^{-1}, c^{-1}\}$$

$$S^* = \{\varepsilon, ab, bb, ca^{-1}b, \dots\}$$

↳ empty word

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0. Defs / notation

0.1. Languages

- S is a symmetric, finite alphabet
- S^* denotes finite words over S
- A language L is a subset of S^*

$$S = \{a, b, c, \bar{a}, \bar{b}, \bar{c}\}$$

$$S^* = \{\varepsilon, ab, bb, c\bar{a}b, \dots\}$$

↳ empty word

$$L = \{\text{words starting with } a\}$$

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0. Defs / notation

0.2. Finite state automata (FSA)

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0. Defs / notation

0.2. Finite state automata (FSA) (over an alphabet S)

Directed graph:

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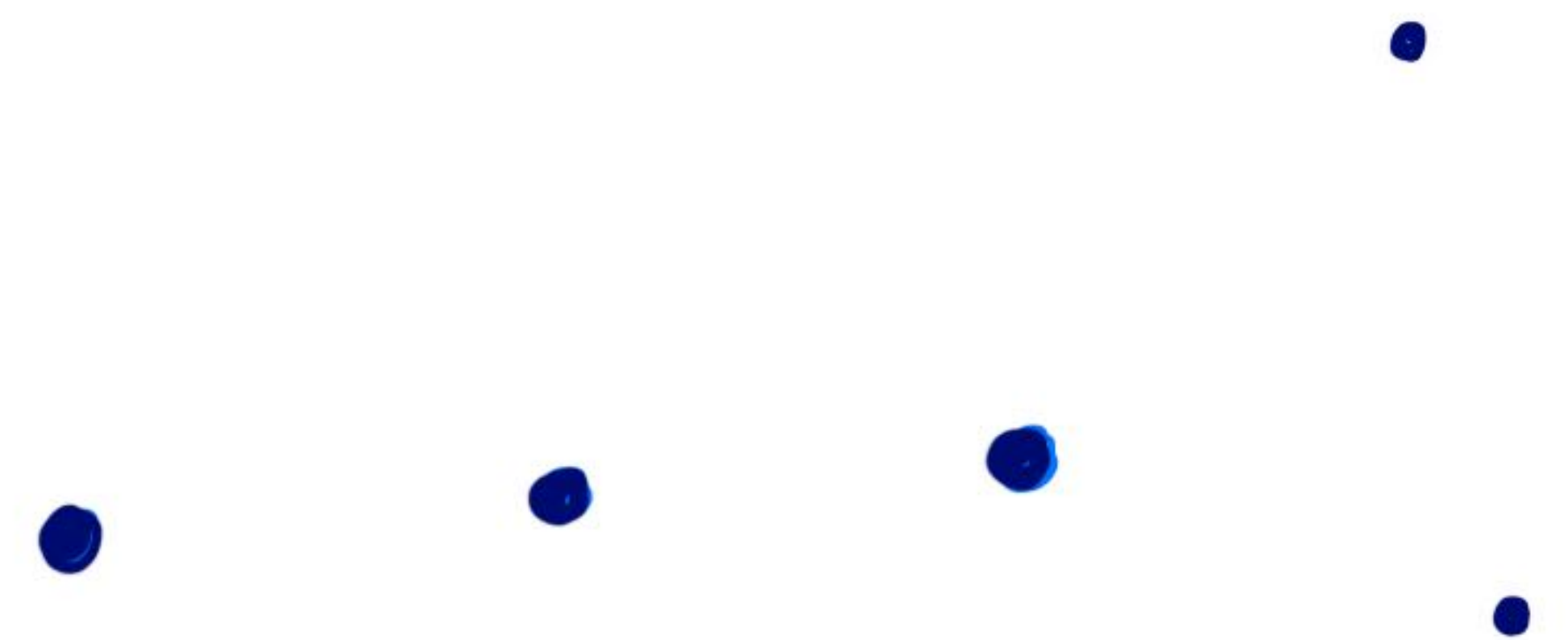
0. Defs / notation

0.2. Finite state automata (FSA) (over an alphabet S)

Directed graph:

- vertices (states)

A



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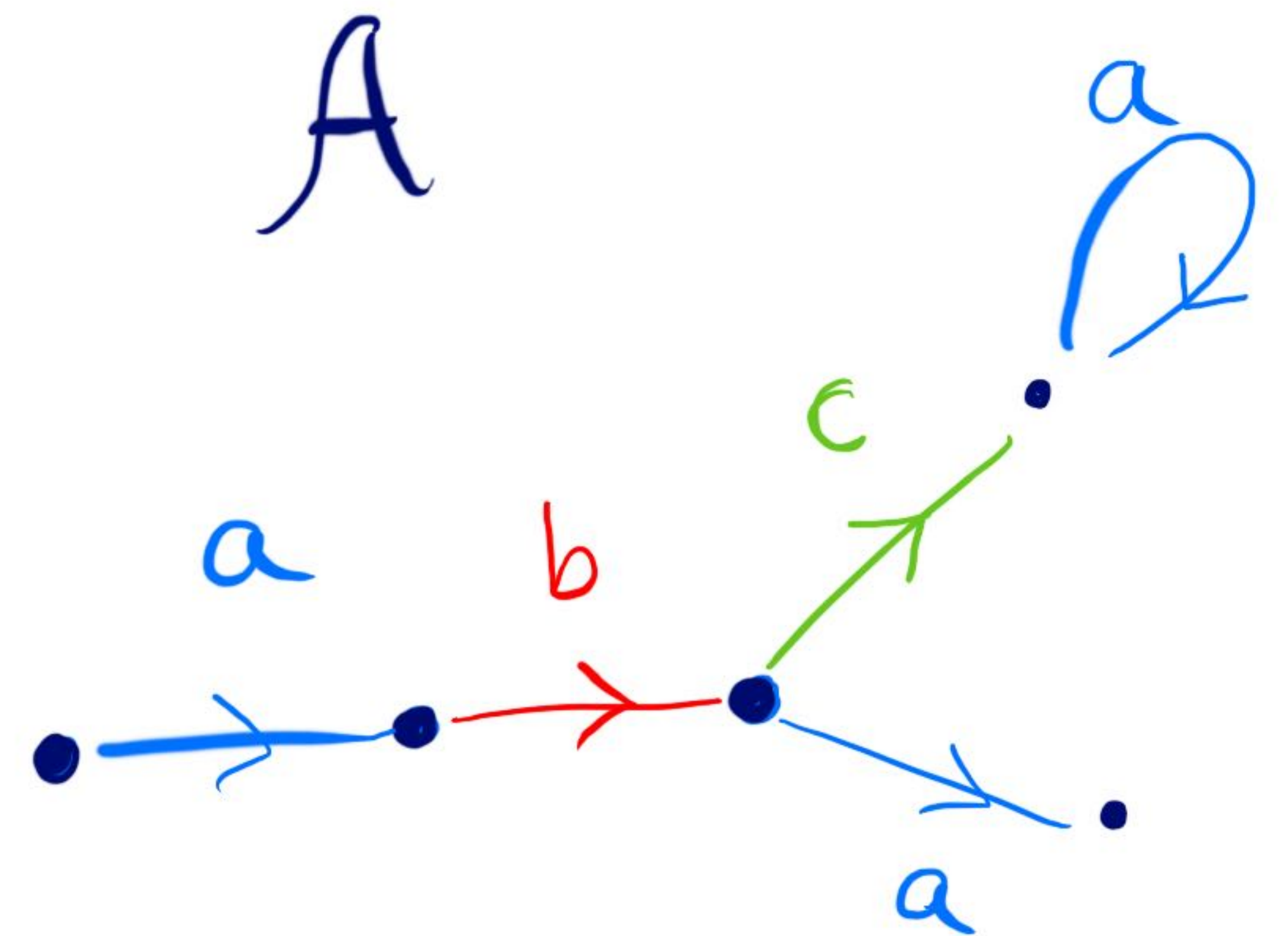
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0. Defs / notation

0.2. Finite state automata (FSA) (over an alphabet S)

Directed graph:

- vertices (states)
- S -labelled edges (transitions)



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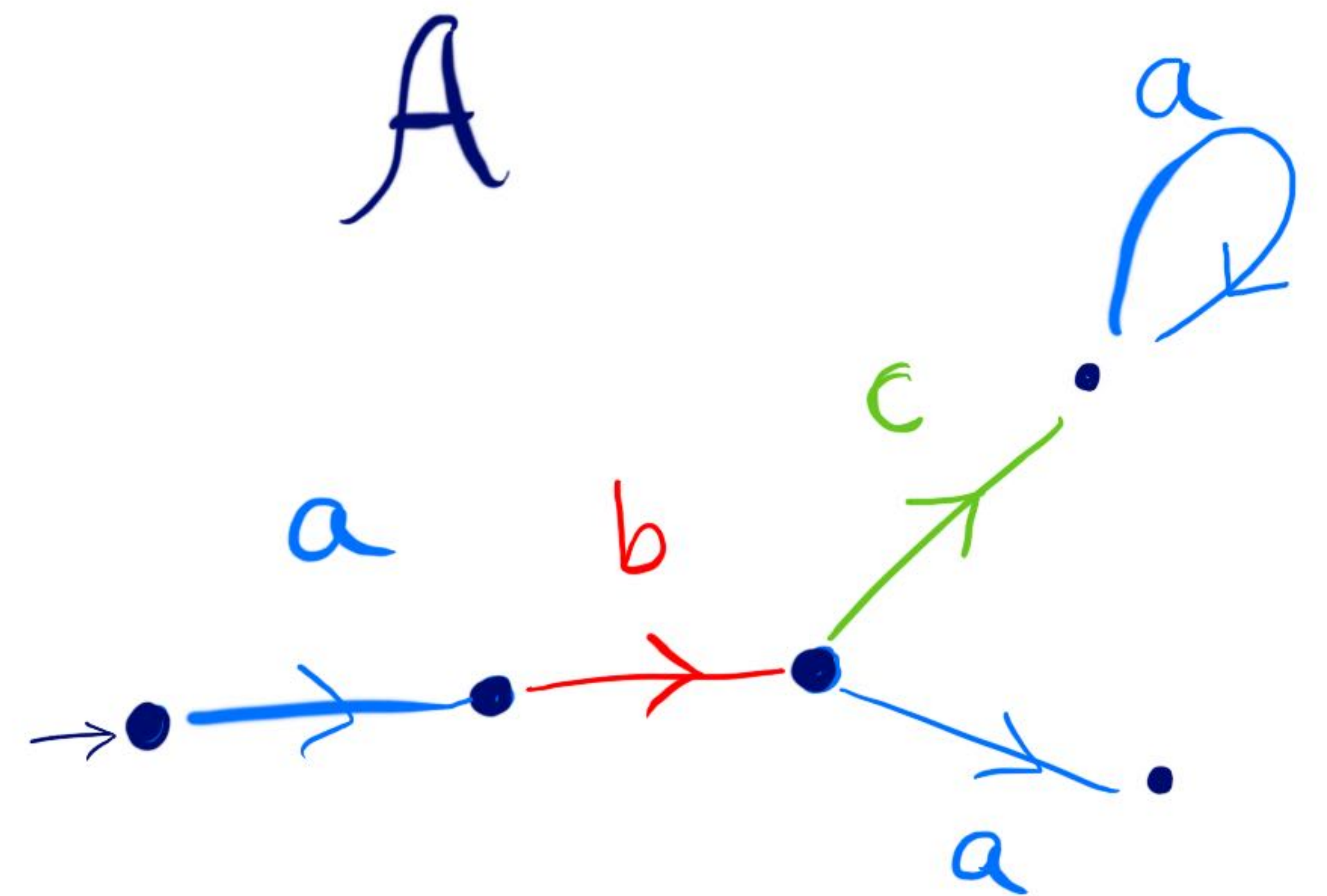
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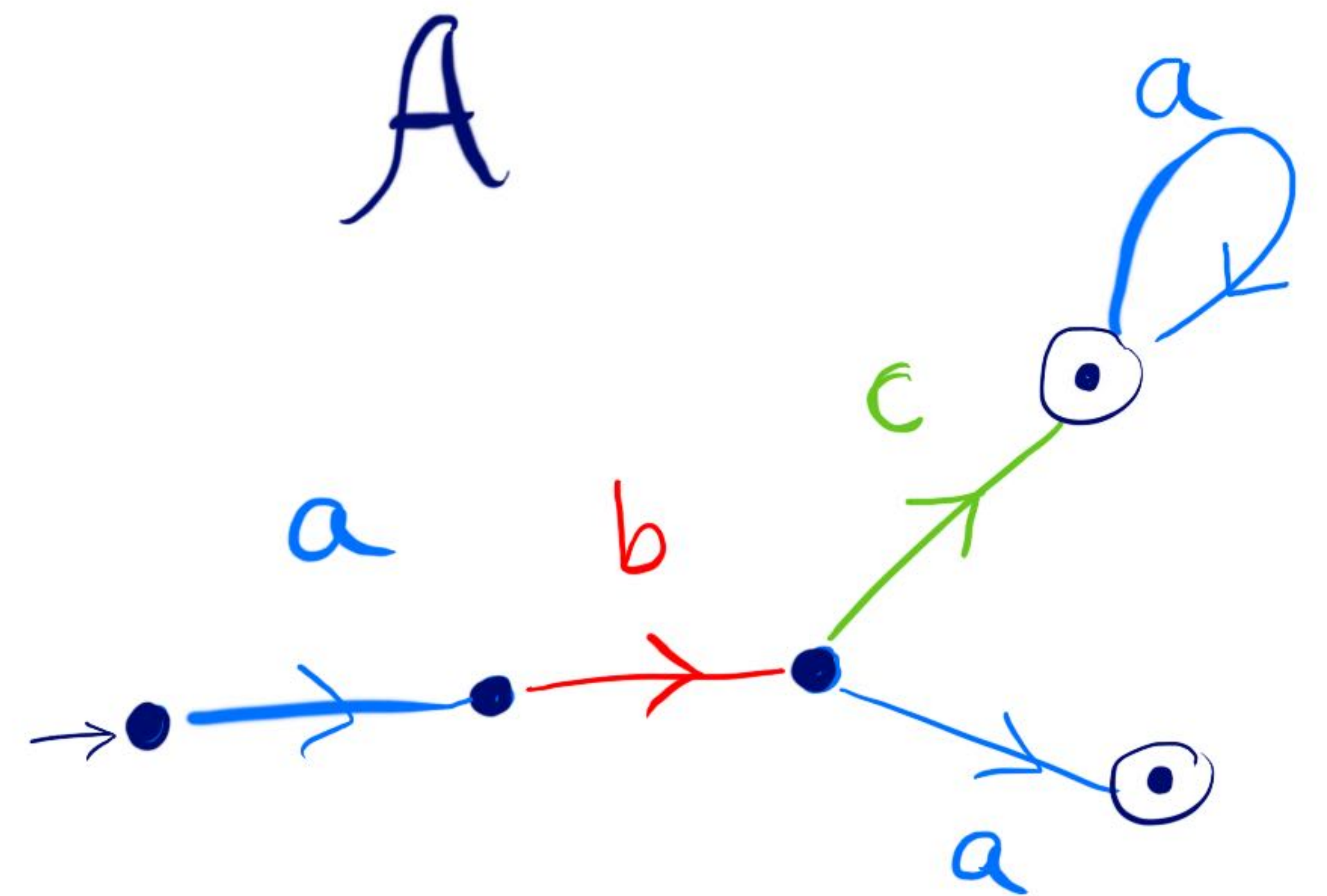


0. Defs / notation

0.2. Finite state automata (FSA) (over an alphabet S)

Directed graph:

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- Starting vertex ($\rightarrow \bullet$)
- Accepting vertices ($\bullet \odot$)

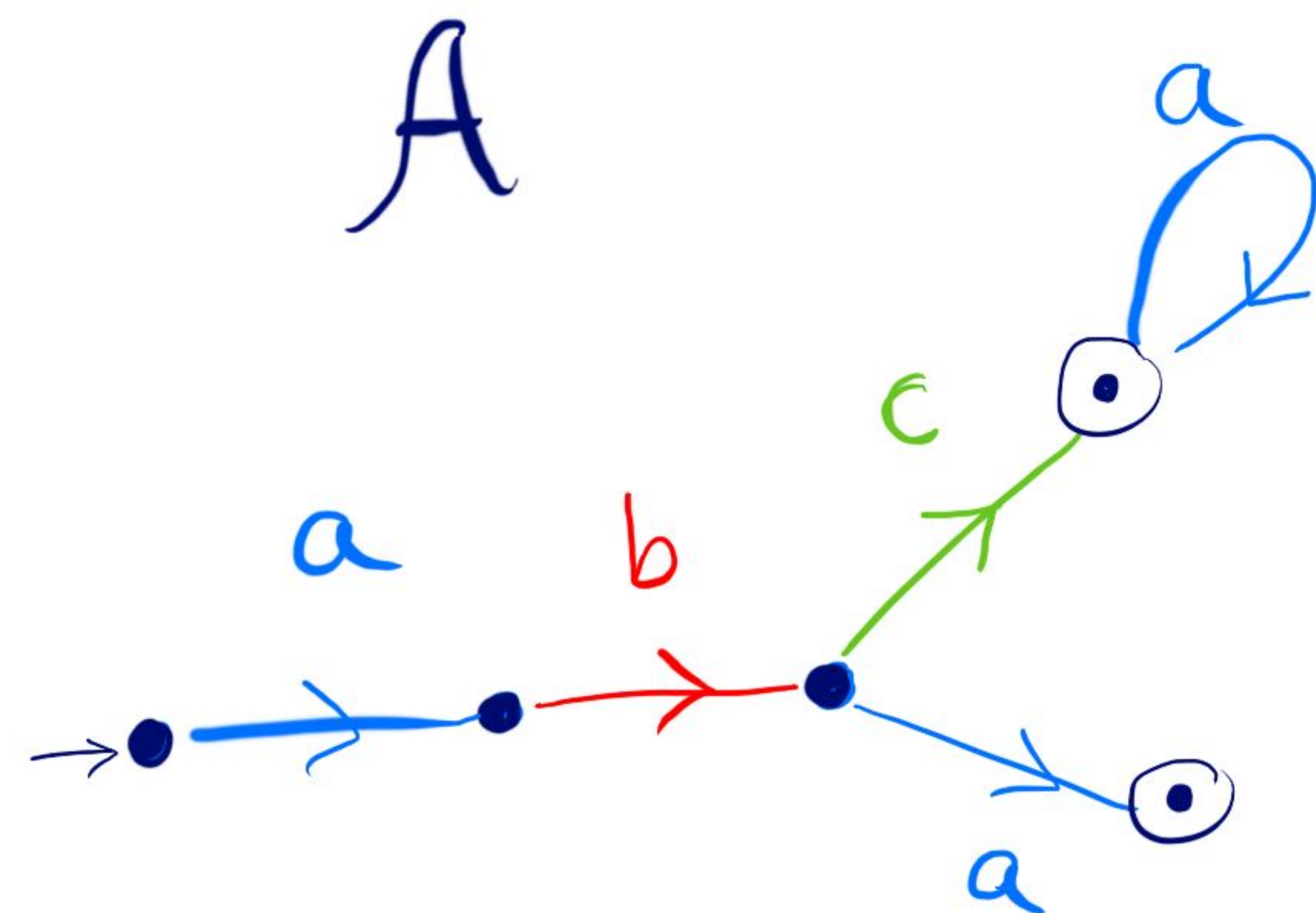


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$$L(A) = \{ aba, abc, abca, abcaa \dots \}$$

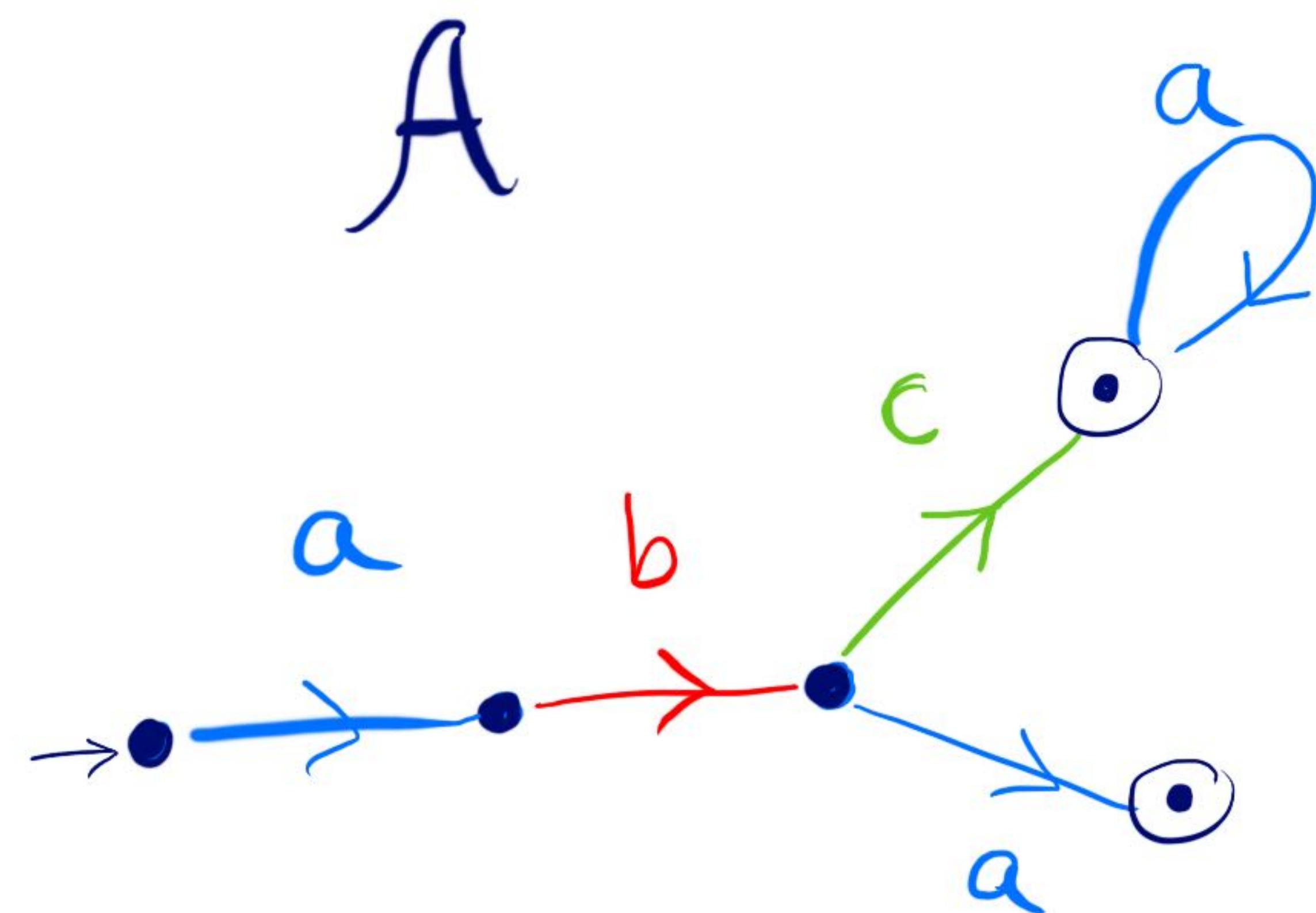
Language accepted: $L(A) = \{ \text{words in } S^* \text{ read as paths } \rightarrow \bullet \rightsquigarrow \odot \text{ in } A \}$

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Directed graph:

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Language accepted: $L(A) = \{ \text{words in } S^* \text{ read as paths } \rightarrow \bullet \rightsquigarrow \odot \text{ in } A \}$

A language is regular if $L = L(A)$ for some A .

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0. Defs / notation

0.3. Automatic groups

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0. Defs / notation

0.3. Automatic groups

- Finitely generated $G = \langle S \rangle$

$$S^* \xrightarrow{\pi} G$$

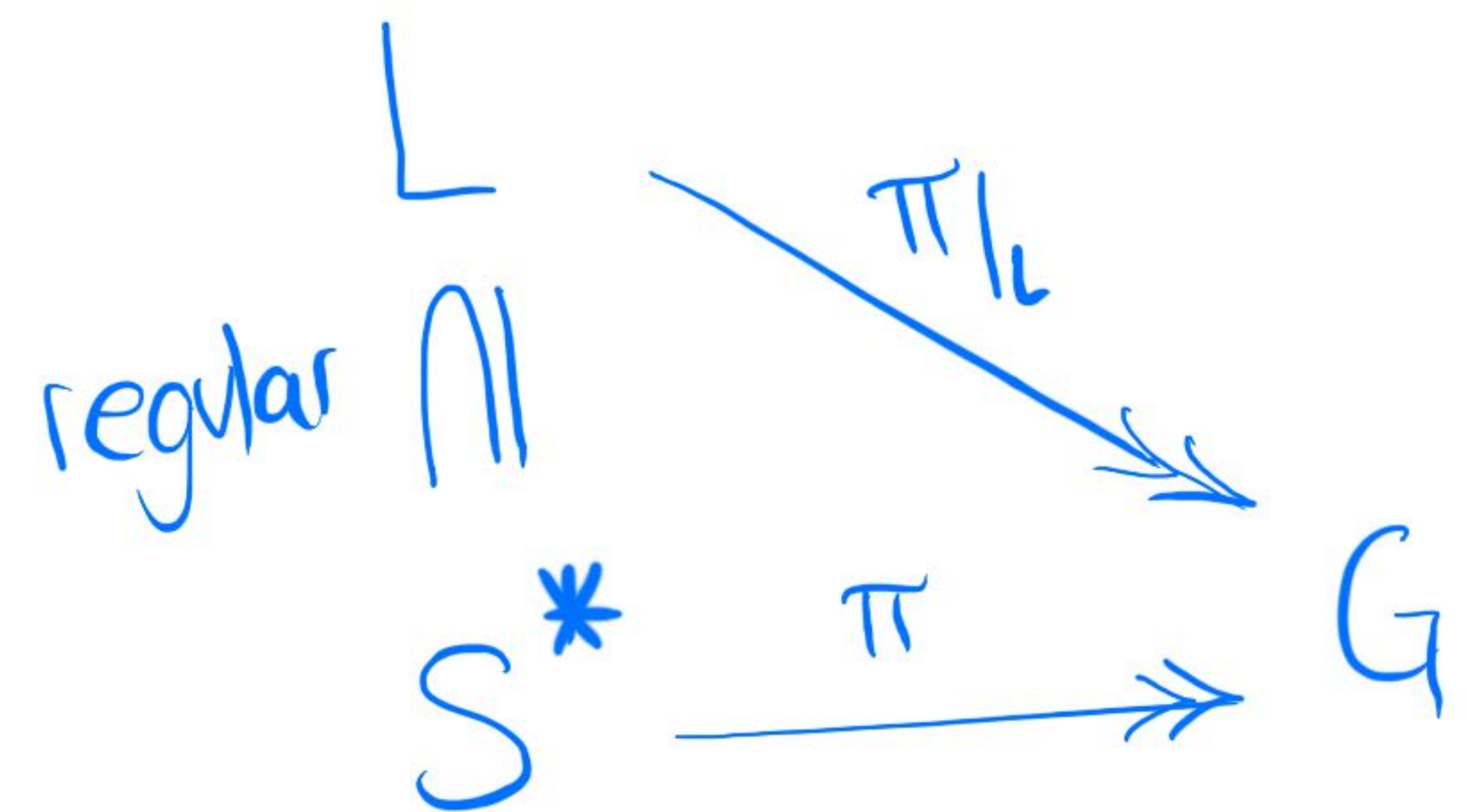
Folding (nice) rational
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0. Defs / notation

0.3. Automatic groups

- Finitely generated $G = \langle S \rangle$

+ regular language $L \subseteq S^*$, $L \twoheadrightarrow G$



Folding (nice) rational
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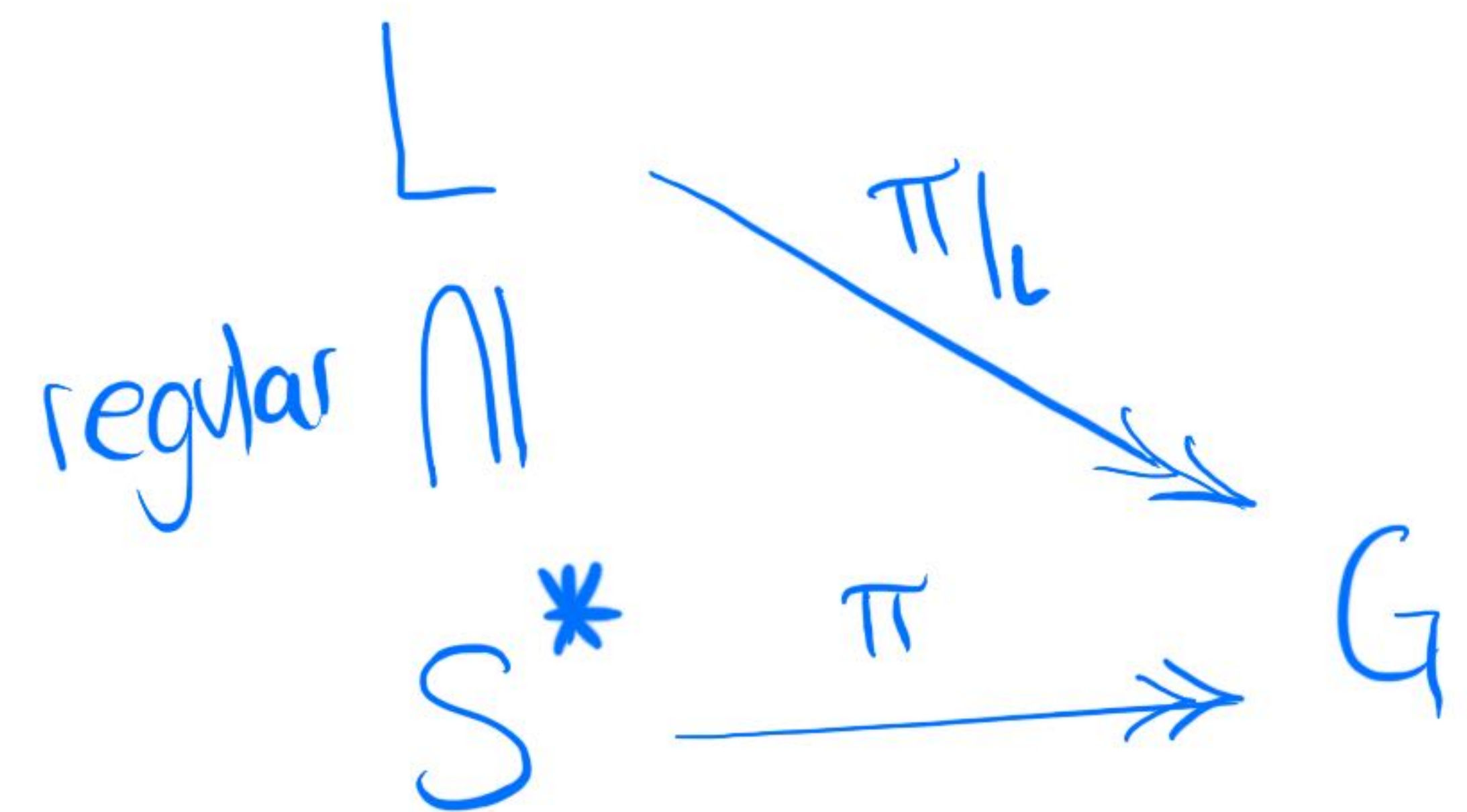
0. Defs / notation

0.3. Automatic groups

• Finitely generated $G = \langle S \rangle$

+ regular language $L \subseteq S^*$, $L \twoheadrightarrow G$

+ set of FSA $\{M_s\}_{s \in S}$ (multipliers)



Folding (nice) rational
subsets of automatic groups.

1. Context

1.1. Stallings automata

$$H = \langle h_1, \dots, h_k \rangle \leq F(S)$$

$$w \in F(S) \dots w \in H?$$

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1. Context

1.1. Stallings automata

- They are FSA that answer this

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1. Context

1.1. Stallings automata

- They are FSA that answer this
- Each $H \leq F(S)$ is uniquely associated

$$H = \langle h_1, \dots, h_k \rangle \leq F(S)$$

$$w \in F(S) \dots w \in H?$$



a FSA:

St_H

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subsets of automatic groups.

1. Context

1.1. Stallings automata

- They are FSA that answer this
- Each $H \leq F(S)$ is uniquely associated
- $L(St_H) = H$ (up to free reduction)

$$H = \langle h_1, \dots, h_k \rangle \leq F(S)$$

$$w \in F(S) \dots w \in H?$$



a FSA:

St_H

1. Context

1.1. Stallings automata

- They are FSA that answer this
- Each $H \leq F(S)$ is uniquely associated
- $L(St_H) = H$

$$H = \langle h_1, \dots, h_k \rangle \leq F(S)$$
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a FSA:

St_H

How is St_H built from H ?

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1. Context

1.1. Stallings automata

How is St_H built from H ?

Ingredients : $H = \langle h_1, h_2, \dots, h_k \rangle$

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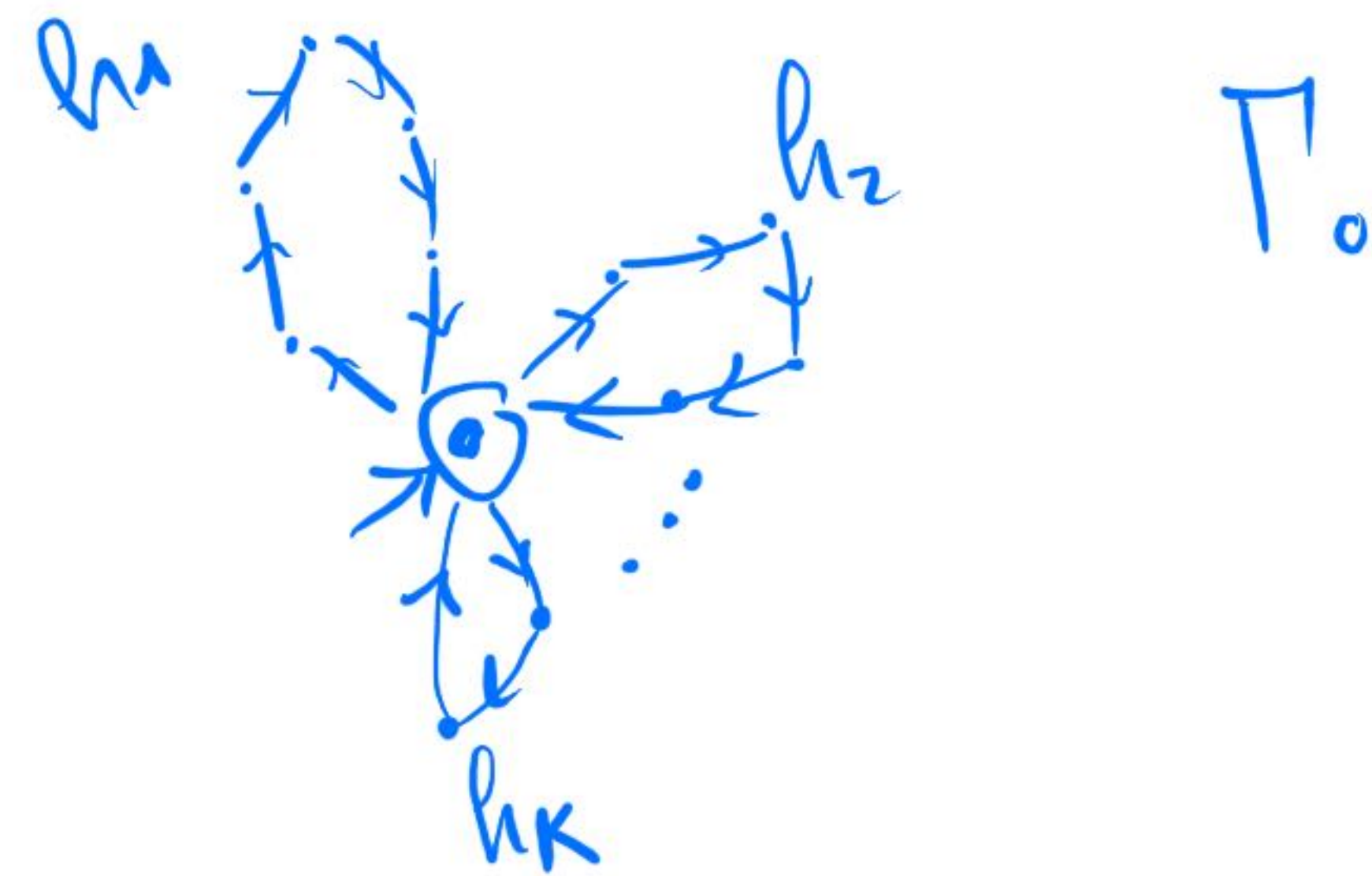
1. Context

1.1. Stallings automata

How is St_H built from H ?

Ingredients : $H = \langle h_1, h_2, \dots, h_k \rangle$

Process : ① Build flower FSA Γ_0



1. Context

1.1. Stallings automata

How is St_H built from H ?

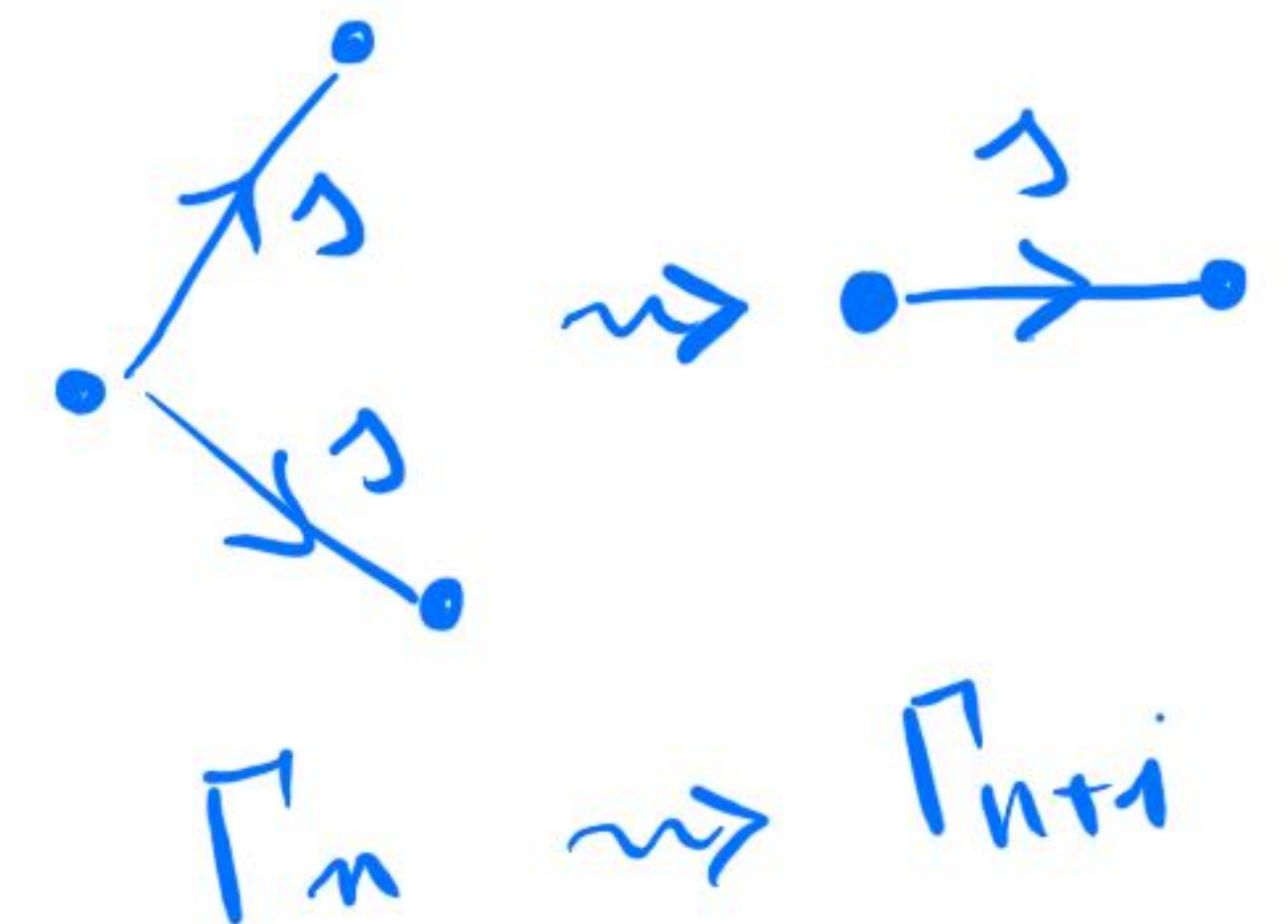
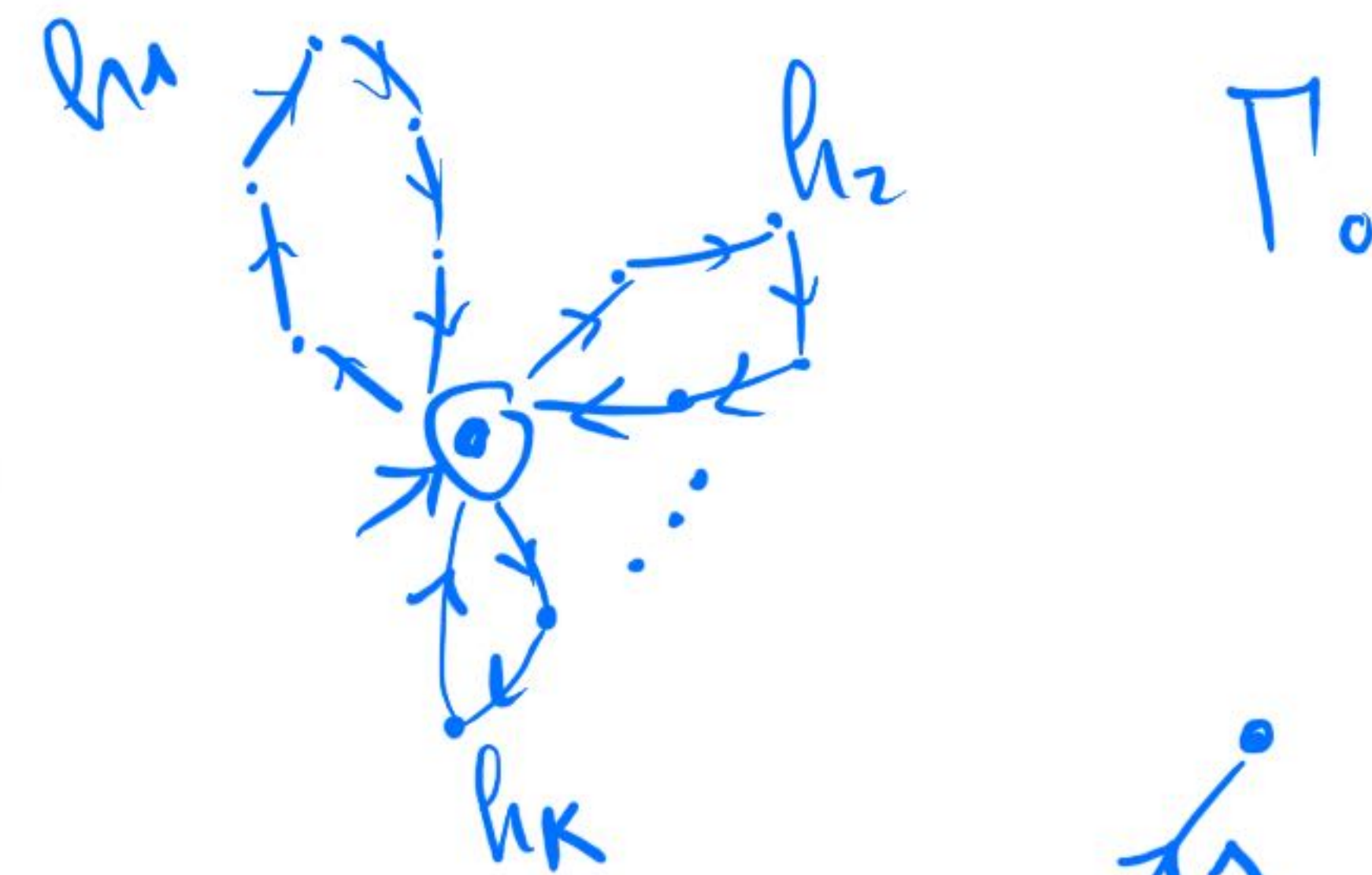
Ingredients : $H = \langle h_1, h_2, \dots, h_k \rangle$

Process : ① Build flower FSA

Γ_0

② Iteratively perform foldings $\Gamma_1, \dots, \Gamma_\ell$

edge identification



1. Context

1.1. Stallings automata

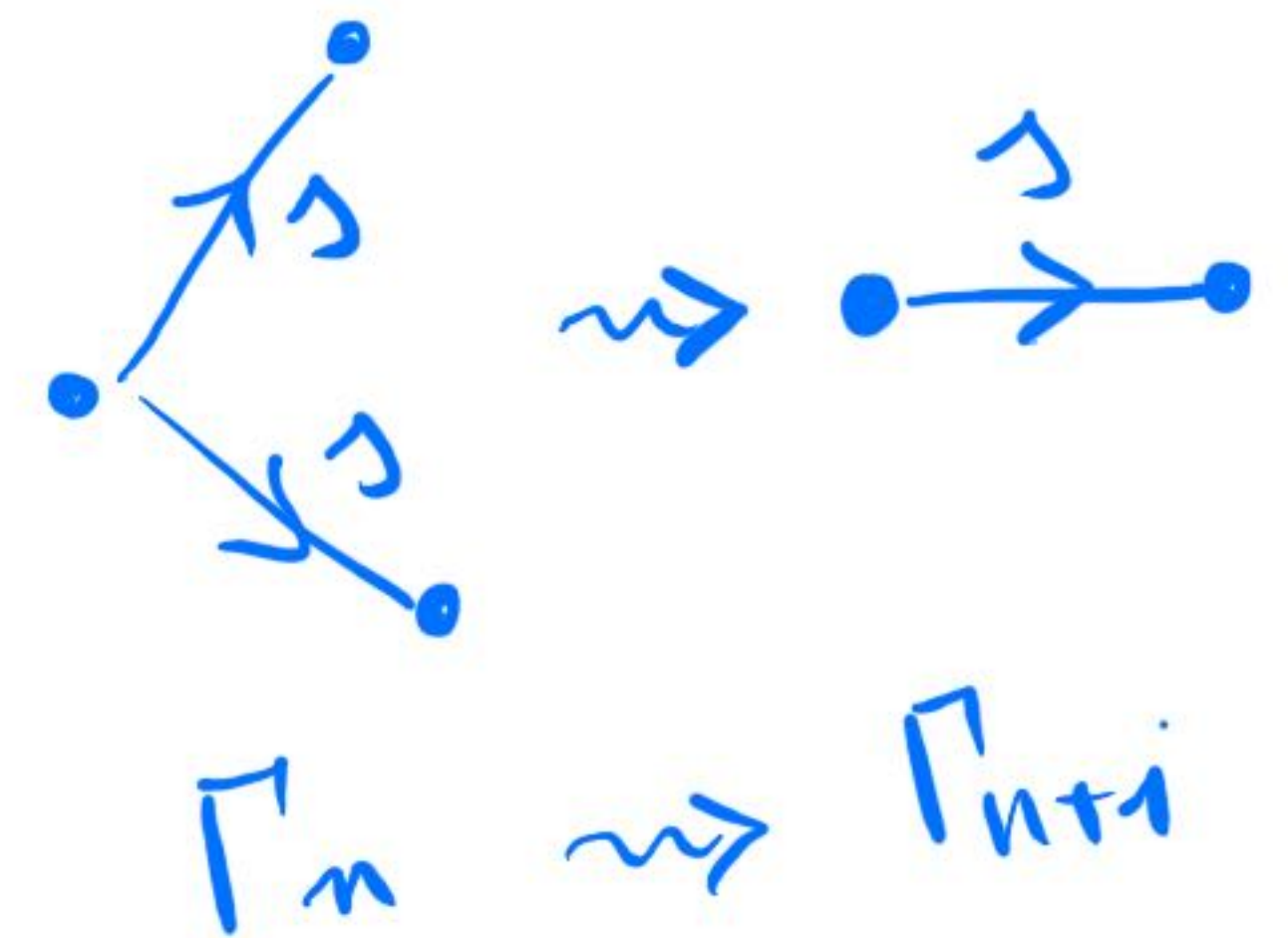
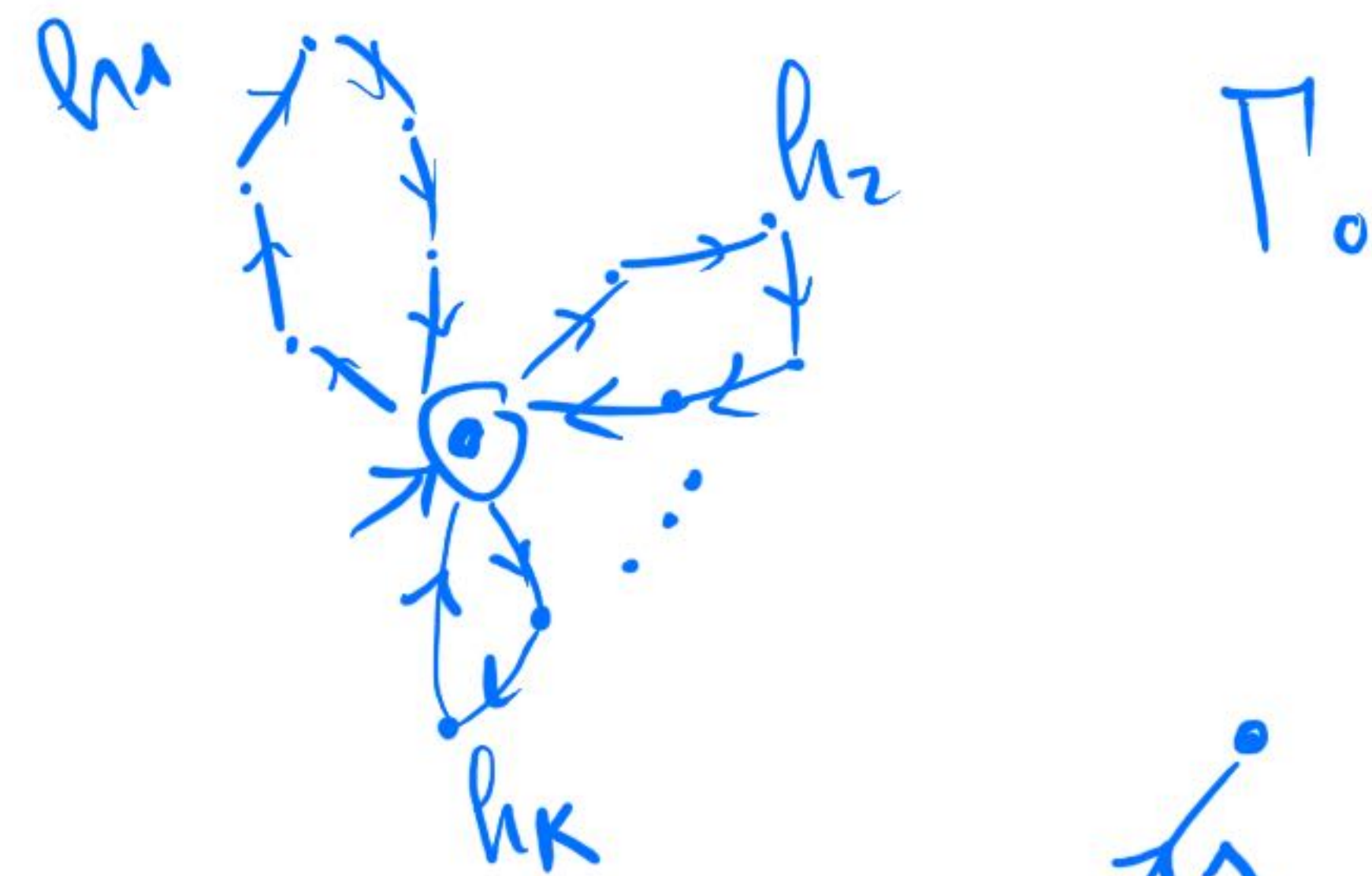
How is St_H built from H ?

Ingredients : $H = \langle h_1, h_2, \dots, h_k \rangle$

Process : ① Build flower FSA Γ_0

② Iteratively perform foldings $\Gamma_1, \dots, \Gamma_\ell$
edge identification

... until no more can be done : $\Gamma_\ell = St_H$



1. Context

1.1. Stallings automata How is St_H built from H ?

Ingredients : $H = \langle h_1, h_2, \dots, h_k \rangle$

Process : ① Build flower FSA Γ_0
② Iteratively perform foldings $\Gamma_1, \dots, \Gamma_e$
... until no more can be done : $\Gamma_e = St_H$

Simple idea, works great! (membership, index, intersections...)

1. Context

1.1. Stallings automata How is St_H built from H ?

Ingredients : $H = \langle h_1, h_2, \dots, h_k \rangle$

Process : ① Build initial object Γ_0

② Iteratively perform foldings $\Gamma_1, \dots, \Gamma_e$

... stop after finitely many : $\Gamma_e = St_H$

What about other groups ? many adaptations, same flavour.

Let's see two

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1. Context

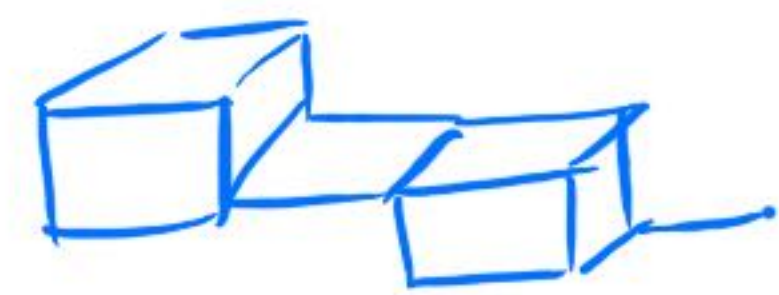
1.2. Ben-Zvi - Kropholler - Lyman '22

Folding (nice) rational
subsets of automatic groups.

1. Context

1.2. Ben-Zvi - Kropholler - Lyman '22

$$H \leq G = \pi_1(\underbrace{\text{npc cube complex}}_{\text{f.g.}})$$



n-cubes
nicely glued

Goal: Build St_H to solve membership problem of H .

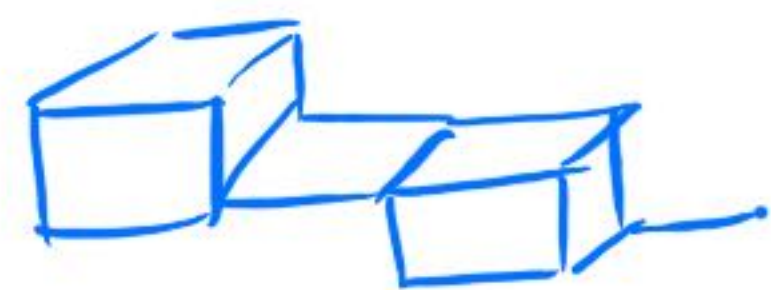
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1.2. Ben-Zvi - Kropholler - Lyman '22

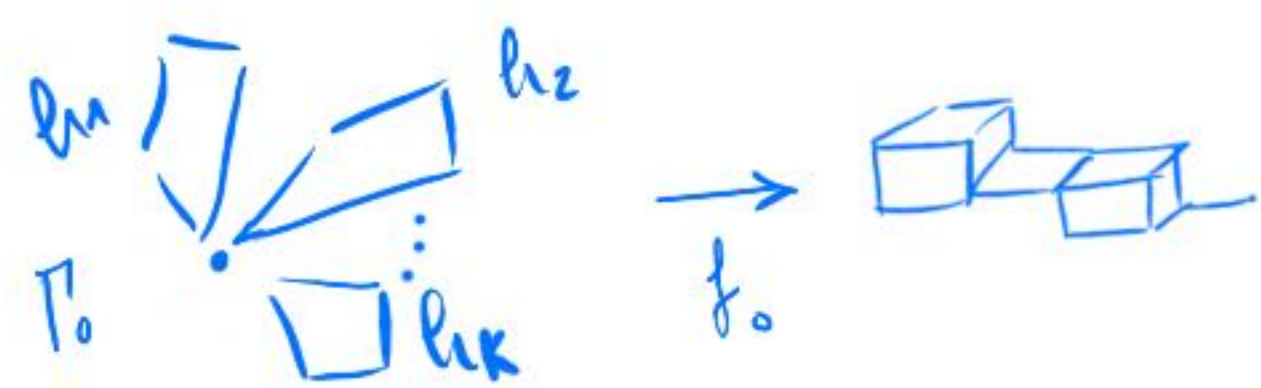
$H \leq G = \pi_1(\text{npc cube complex})$
f.g.



Goal: Build St_H to solve membership problem of H .

Ingredients: $H = \langle h_1, \dots, h_k \rangle$

Process:



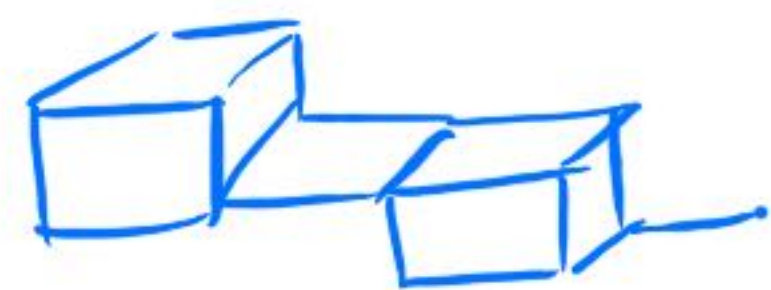
① Build flower cube complex Γ_0

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1. Context

1.2. Ben-Zvi - Kroppoller - Lyman '22

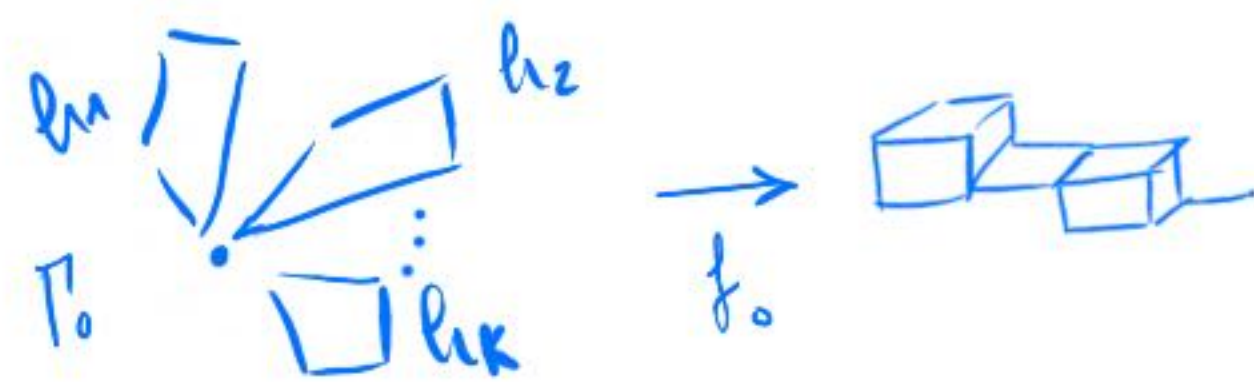
$H \leq G = \pi_1(\text{npc cube complex})$
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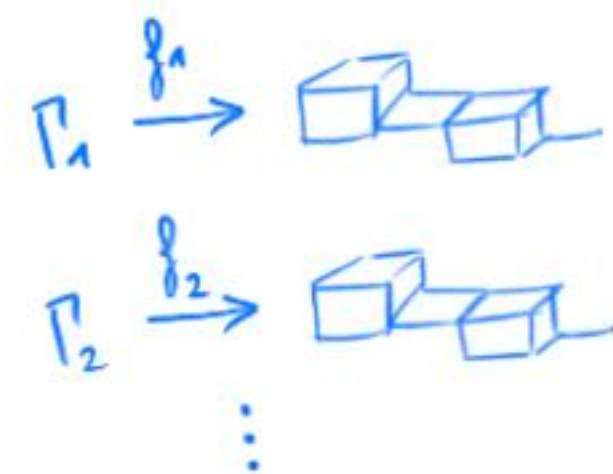
Goal: Build St_H to solve membership problem of H .

Ingredients: $H = \langle h_1, \dots, h_k \rangle$

Process:



① Build flower cube complex Γ_0



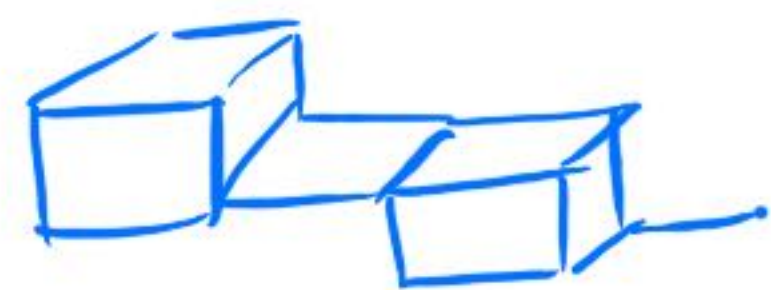
② Iteratively perform foldings
(n-cube identification + n-cube filling)

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1. Context

1.2. Ben-Zvi - Kropholler - Lyman '22

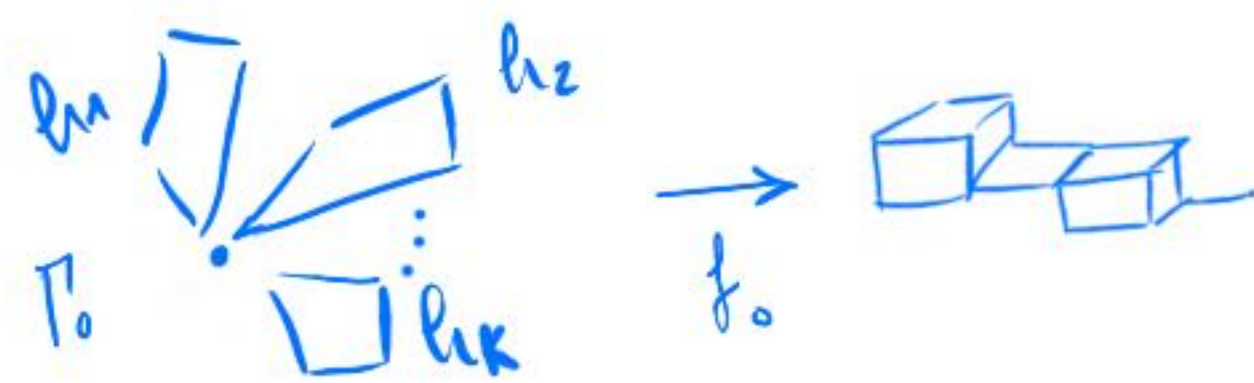
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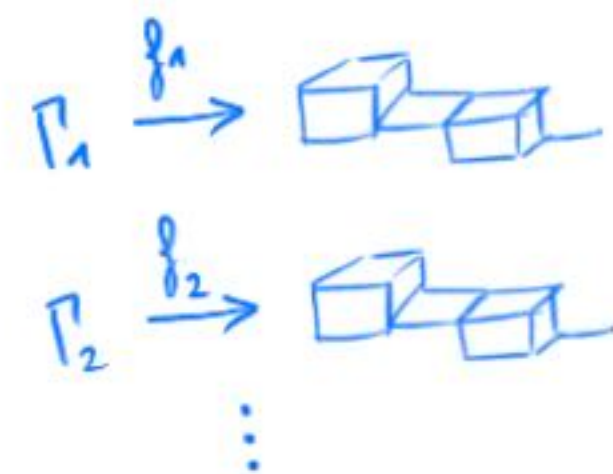
Ingredients: $H = \langle h_1, \dots, h_k \rangle$

Process:



① Build flower cube complex Γ_0

② Iteratively perform foldings
(n-cube identification + n-cube filling)



③ Until no more can be done. $\Gamma_e = St_H$

$St_H \rightarrow \text{cube complex}$ induces $H \leq \pi_1(\text{cube complex})$

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1. Context

1.2. Ben-Zvi - Kropholler - Lyman '22

$H \leq G = \pi_1(\text{npc cube complex})$
f.g.

Goal: Build St_H to solve membership problem of H .

Ingredients: $H = \langle h_1, \dots, h_k \rangle$

Process:

① Build flower cube complex Γ_0

② Iteratively perform foldings
(n-cube identification + n-cube filling)

③ Until no more can be done. $\Gamma_e = St_H$

$St_H \rightarrow$  induces $H \leq \pi_1(\text{cube complex})$

1.3. Kharlampovich - Miasnikov - Weil '17

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subsets of automatic groups.

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Ingredients: $H = \langle h_1, \dots, h_k \rangle$

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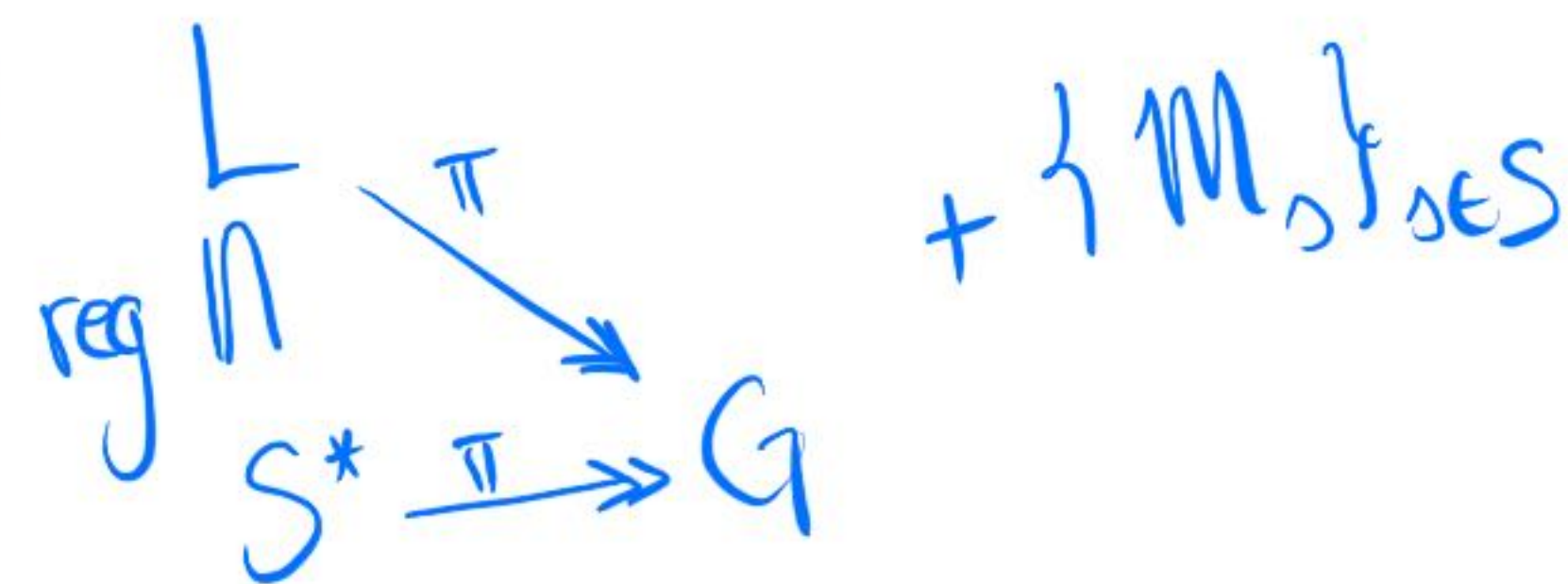
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$St_H \rightarrow$  induces $H \leq \pi_1(\text{cube complex})$

1.3. Kharlampovich - Miasnikov - Weil '17

$H \leq G$ automatic, finite presentation $\langle S | R \rangle$
f.g.



1.2. Ben-Zvi - Kropholler - Lyman '22

$$H \underset{\text{f.g.}}{\leq} G = \pi_1(\text{npc cube complex})$$

Goal: Build St_H to solve membership problem of H .

Ingredients : $H = \langle h_1, \dots, h_k \rangle$

Process :

- ① Build flower cube complex Γ_0

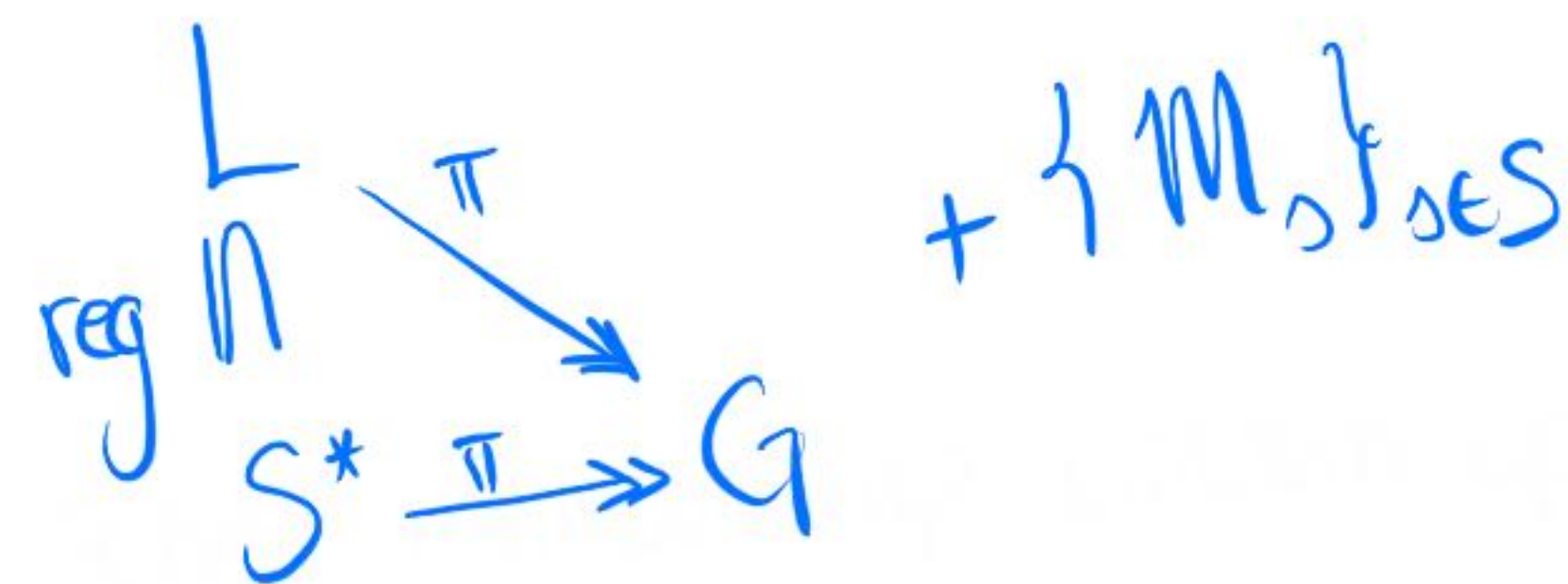
- ② Iteratively perform foldings
(n-cube identification + n-cube filling)

- ③ Until no more can be done. $\Gamma_e = St_H$

$$St_H \rightarrow \begin{array}{c} \square \\ \square \end{array} \quad \text{induces} \quad H \leq \pi_1(\begin{array}{c} \square \\ \square \end{array})$$

1.3. Kharlampovich-Miasnikov - Weil '17

$H \leq G$ automatic, finite presentation $\langle S \mid R \rangle$
f.g. $\downarrow$ $L = \pi$ $\rightarrow$ 3 m.f.s



Goal: same

Ingredients : $H = \langle h_1, \dots, h_k \rangle$, $G \cong \langle S \mid R \rangle$

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1. Context

1.2. Ben-Zvi - Kropholler - Lyman '22

$H \leq G = \pi_1(\text{npc cube complex})$
f.g.

Goal: Build St_H to solve membership problem of H .

Ingredients: $H = \langle h_1, \dots, h_k \rangle$

Process:

① Build flower cube complex Γ_0

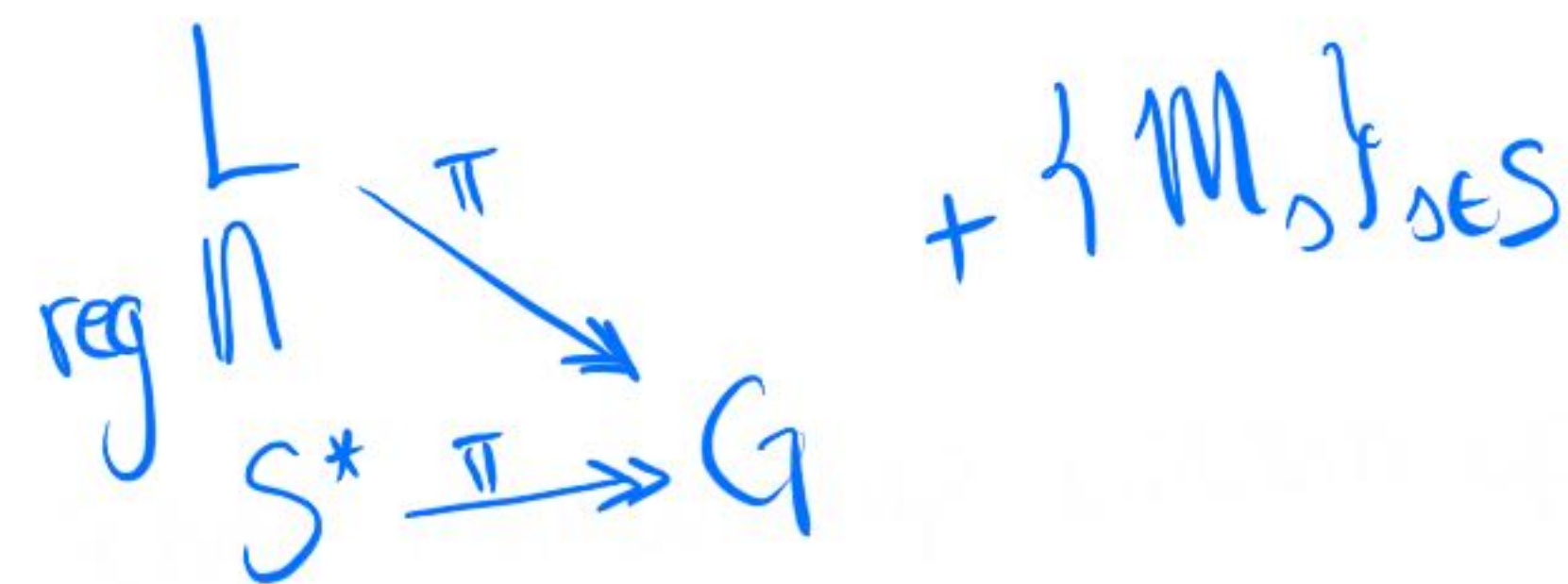
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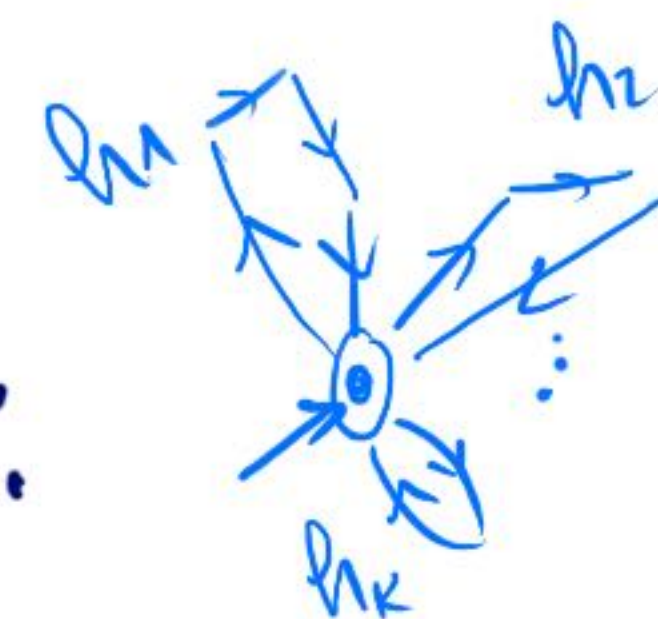


Goal: same

Ingredients: $H = \langle h_1, \dots, h_k \rangle$, $G \cong \langle S | R \rangle$

Process:

① Build flower FSA Γ_0 :



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Process:

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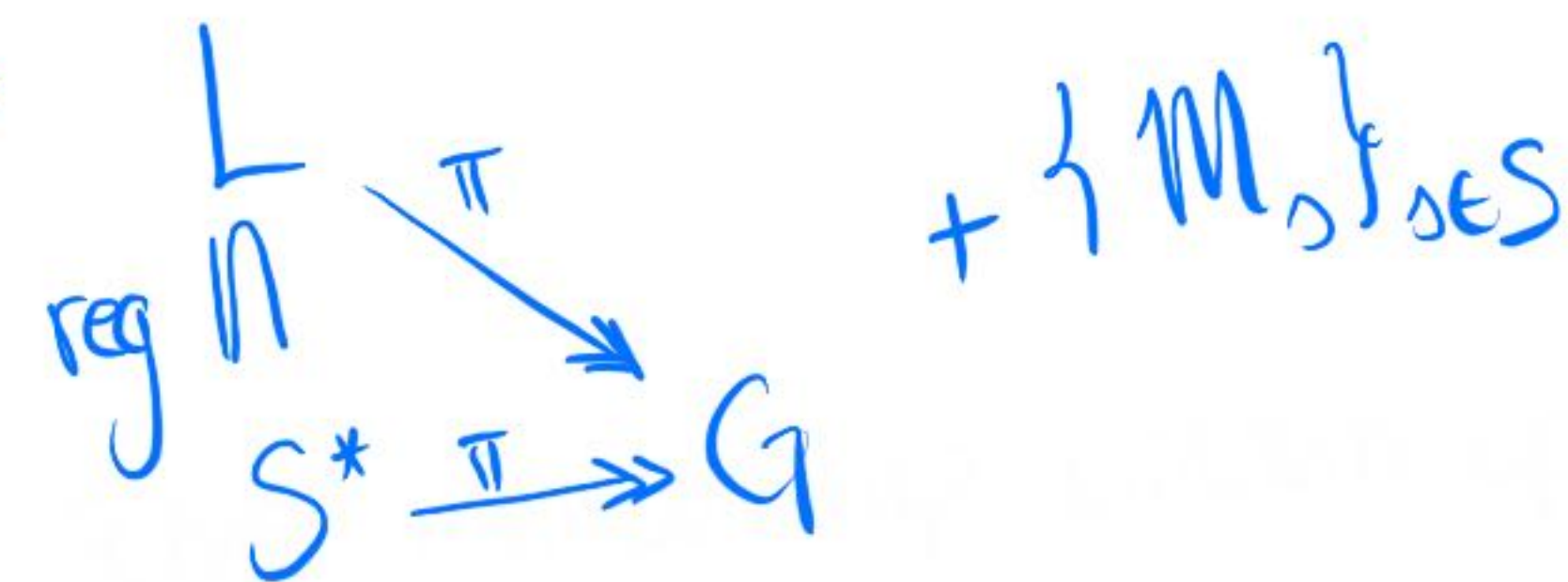
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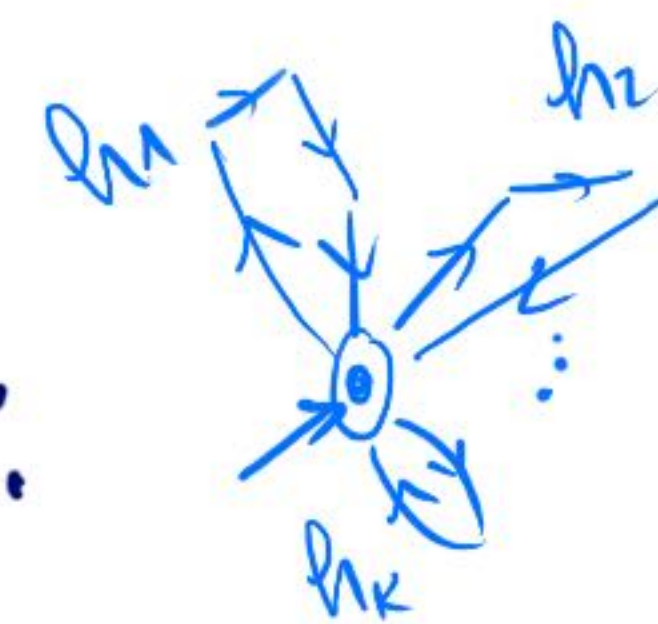


Goal: same

Ingredients: $H = \langle h_1, \dots, h_k \rangle$, $G \cong \langle S | R \rangle$

Process:

① Build flower FSA Γ_0 :



② Iteratively perform foldings:
(edge identification + gluing relations)



1. Context

1.2. Ben-Zvi - Kroppoller - Lyman '22

$$H \leq G = \pi_1(\text{npc cube complex})$$

f.g.

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Ingredients: $H = \langle h_1, \dots, h_k \rangle$

Process:

① Build flower cube complex Γ_0

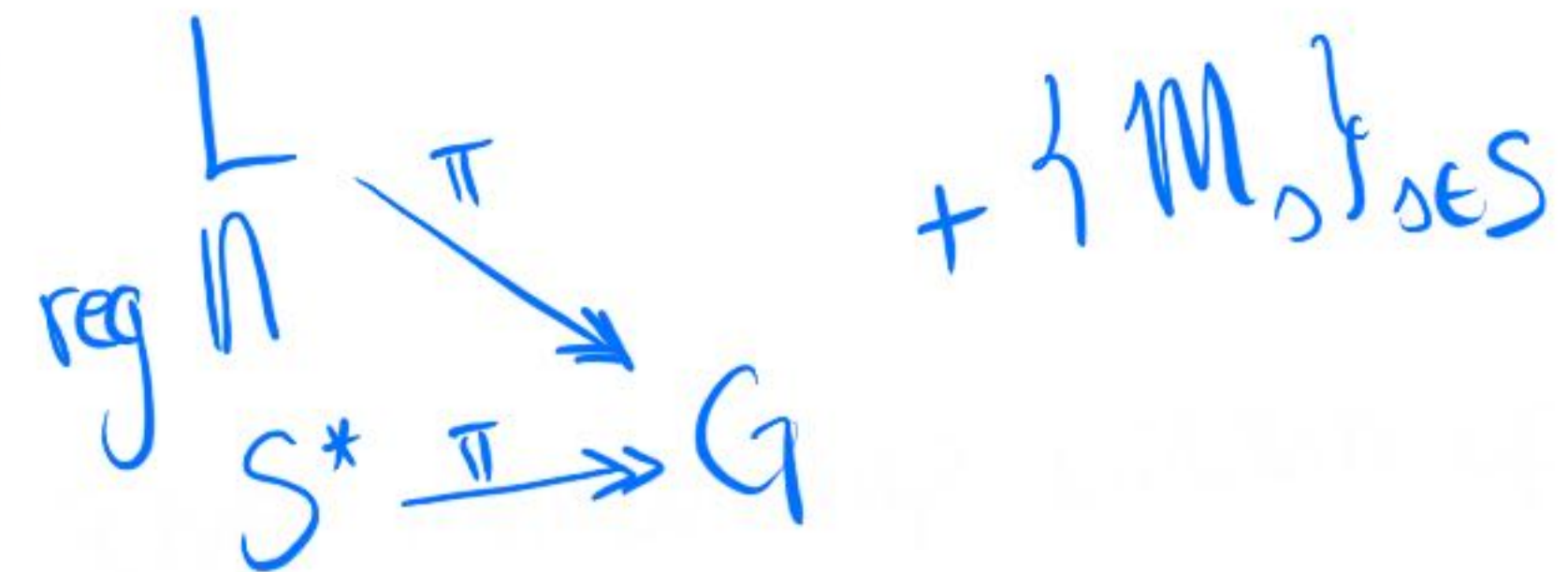
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Goal: same

Ingredients: $H = \langle h_1, \dots, h_k \rangle$, $G \cong \langle S | R \rangle$

Process:

① Build flower FSA Γ_0 :

② Iteratively perform foldings:
(edge identification + gluing relations)

③ Use multipliers $\{M_s\}_{s \in S}$ to know
when to stop $\Gamma_e = St_H$

1. Context

1.2. Ben-Zvi - Kropholler - Lyman '22

$$H \leq G = \pi_1(\text{npc cube complex})$$

f.g.

Goal: Build St_H to solve membership problem of H .

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Process:

① Build flower cube complex Γ_0

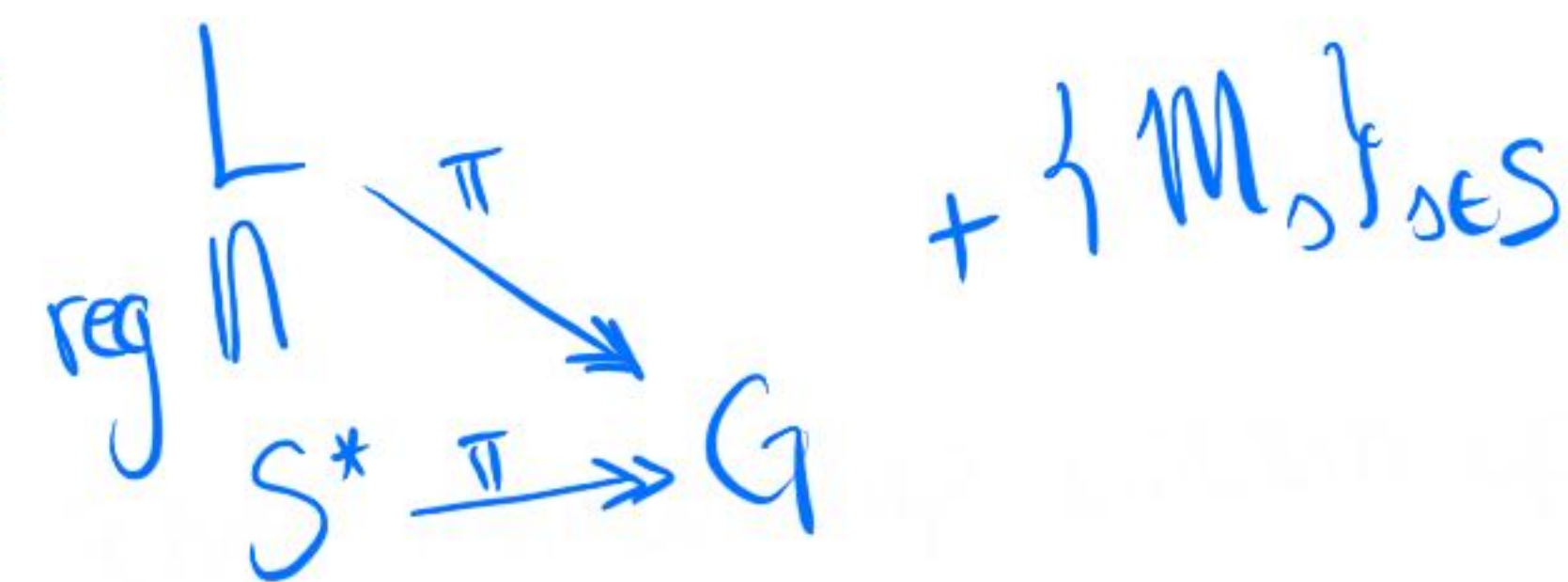
② Iteratively perform foldings
(n-cube identification + n-cube filling)

③ Until no more can be done. $\Gamma_e = St_H$

$St_H \rightarrow$  induces $H \leq \pi_1(\text{cube complex})$

1.3. Kharlampovich - Miasnikov - Weil '17

$H \leq G$ automatic, finite presentation $\langle S | R \rangle$
f.g.



Goal: same

Ingredients: $H = \langle h_1, \dots, h_k \rangle$, $G \cong \langle S | R \rangle$

Process:

① Build flower FSA Γ_0 :

② Iteratively perform foldings:
(edge identification + gluing relations)

③ Use multipliers $\{M_s\}_{s \in S}$ to know
when to stop $\Gamma_e = St_H$

$$L(St_H) = \{ w \in L : \pi(w) = H \}$$

Folding (nice) rational
subsets of automatic groups.

Gazteak RSME '25

1. Context

1.2. Ben-Zvi - Kropholler - Lyman '22

$H \leq G = \pi_1(\text{npc cube complex})$
f.g.

Goal: Build St_H to solve membership problem of H .

Ingredients: $H = \langle h_1, \dots, h_k \rangle$

Process:

① Build flower cube complex Γ_0

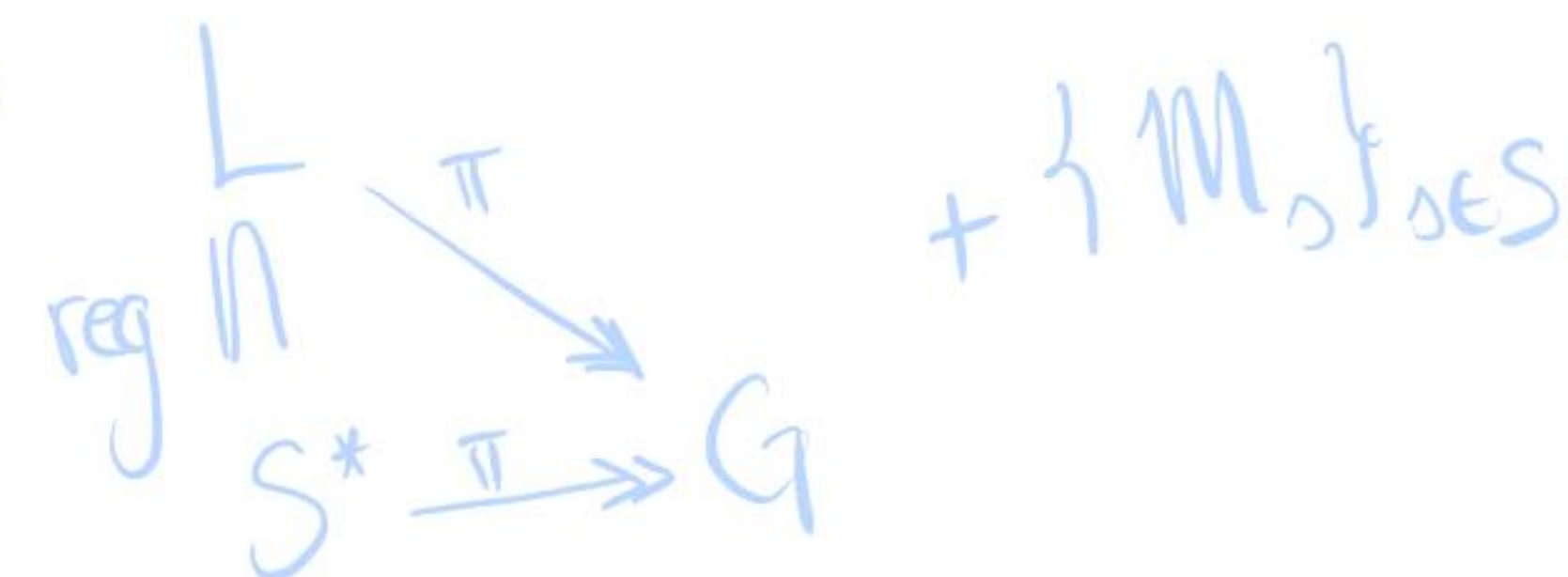
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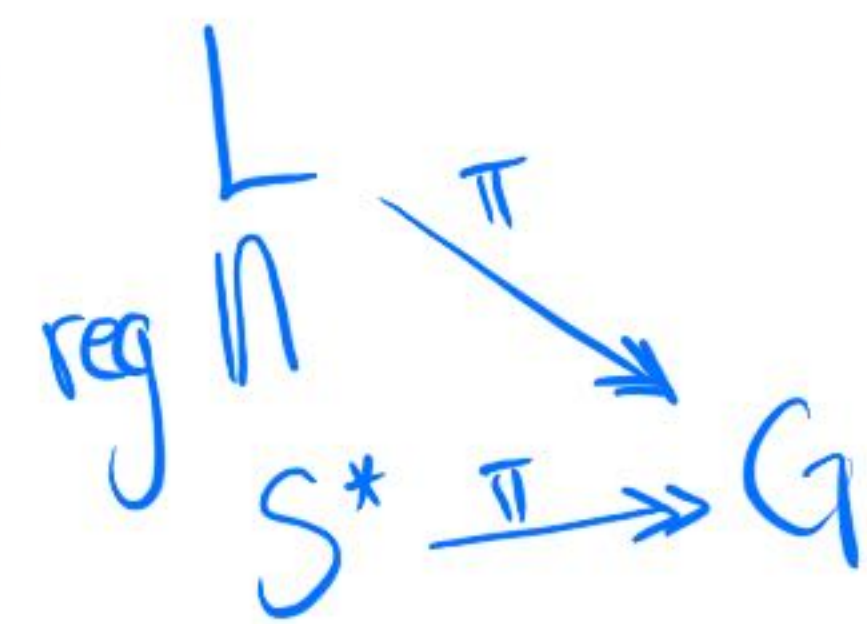
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I lied! Never works as well as with $F(S)$

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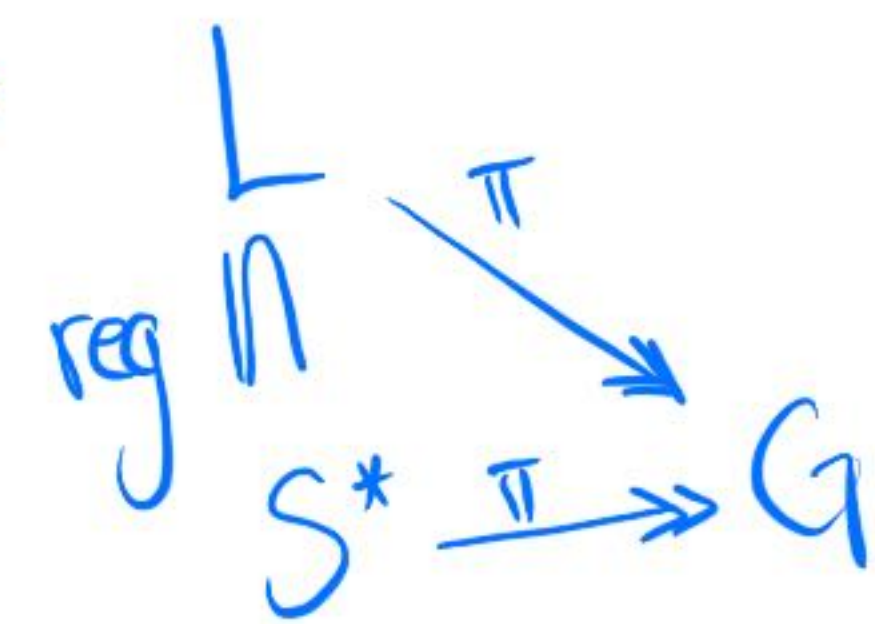
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Need to impose (convexity) constraints on H

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Gazteak RSME '25

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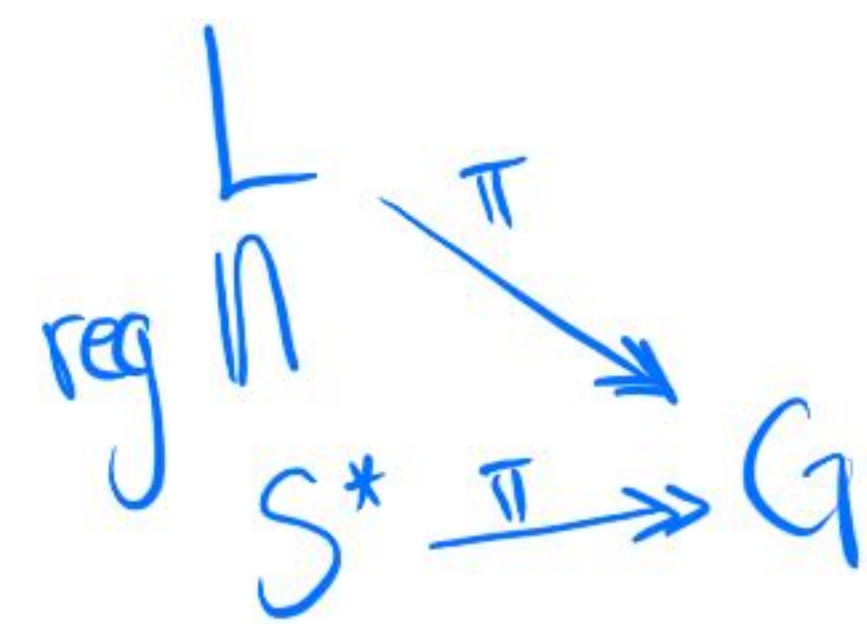
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$H \leq G = \pi_1(\text{npc cube complex})$
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Ingredients : $H = \langle h_1, \dots, h_k \rangle$
 H is quasiconvex

1.3. Kharlampovich - Miasnikov - Weil '17

$H \leq G$ automatic, finite presentation $\langle S | R \rangle$
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Folding (nice) rational
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Gazteak RSME '25

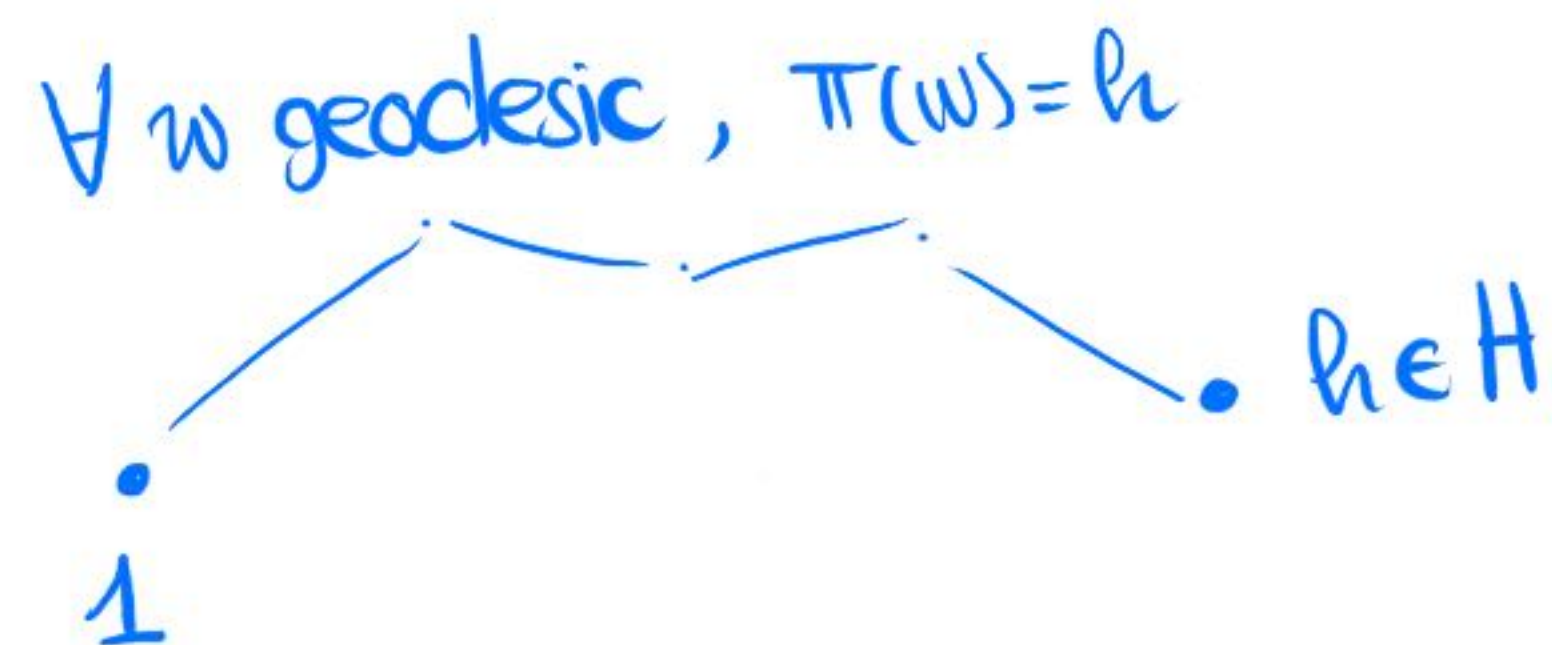
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f.g. $\langle S \rangle$

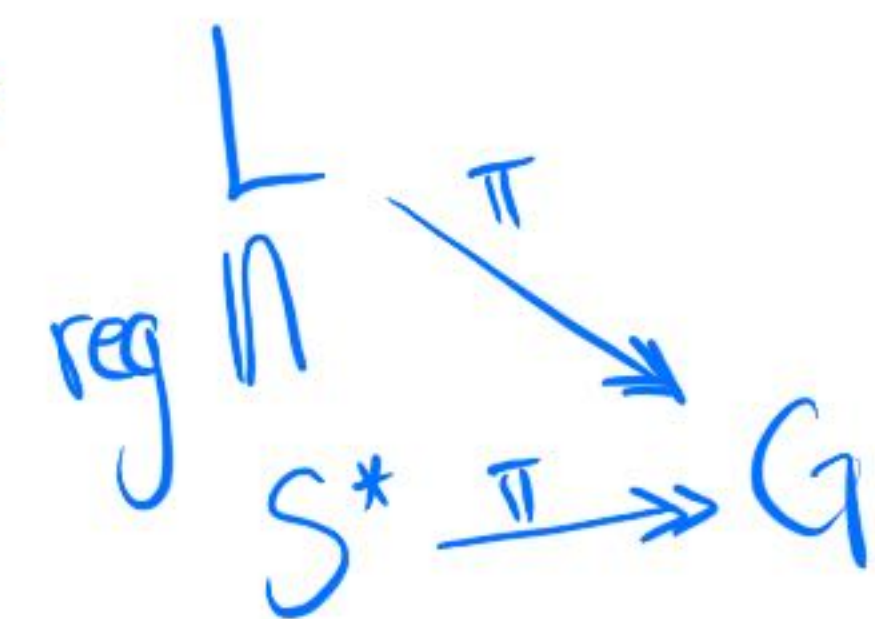
Ingredients : $H = \langle h_1, \dots, h_k \rangle$
 H is quasiconvex

$\text{Cay}(G, S)$
 $\exists K$



1.3. Kharlampovich - Miasnikov - Weil '17

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Folding (nice) rational
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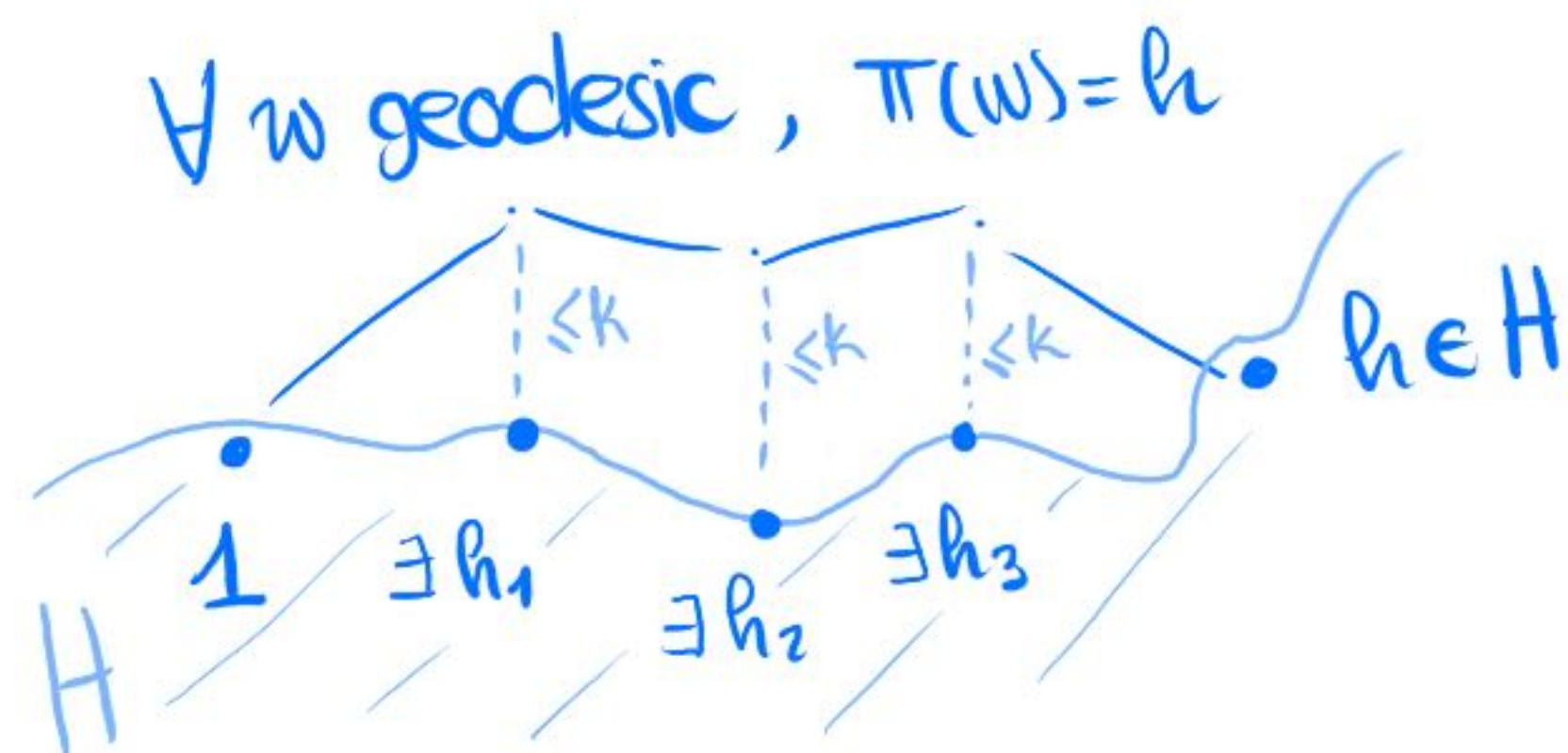
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f.g. $\langle S \rangle$

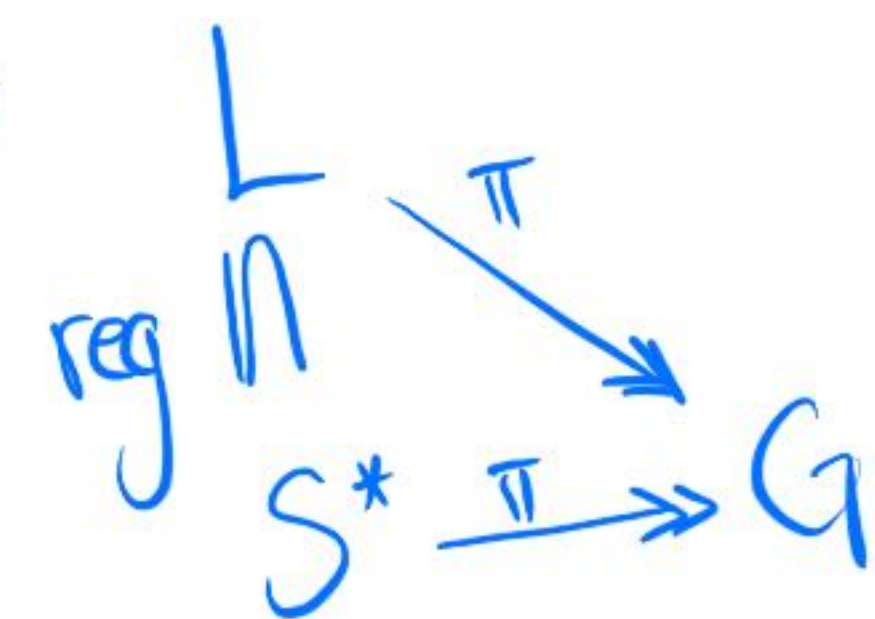
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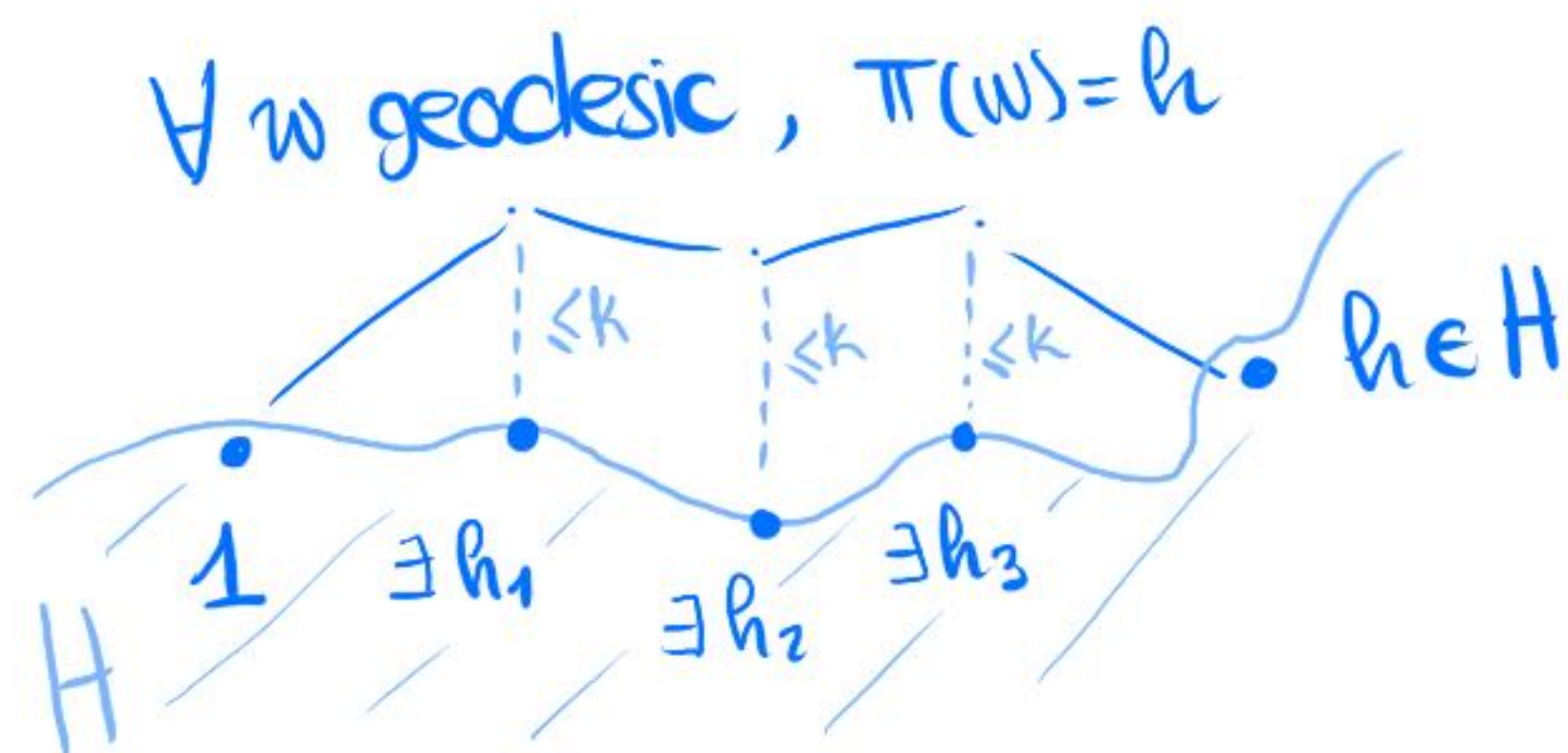
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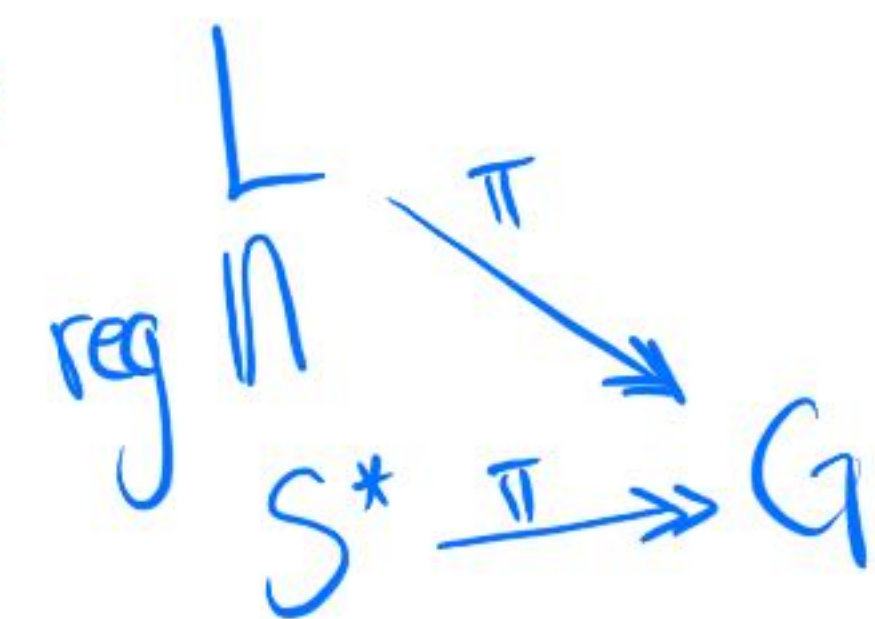
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f.g.



Ingredients : $H = \langle h_1, \dots, h_k \rangle$ $G \cong \langle S | R \rangle$
 H is L-quasiconvex

Need to impose (convexity) constraints on H

Folding (nice) rational
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Gazteak RSME '25

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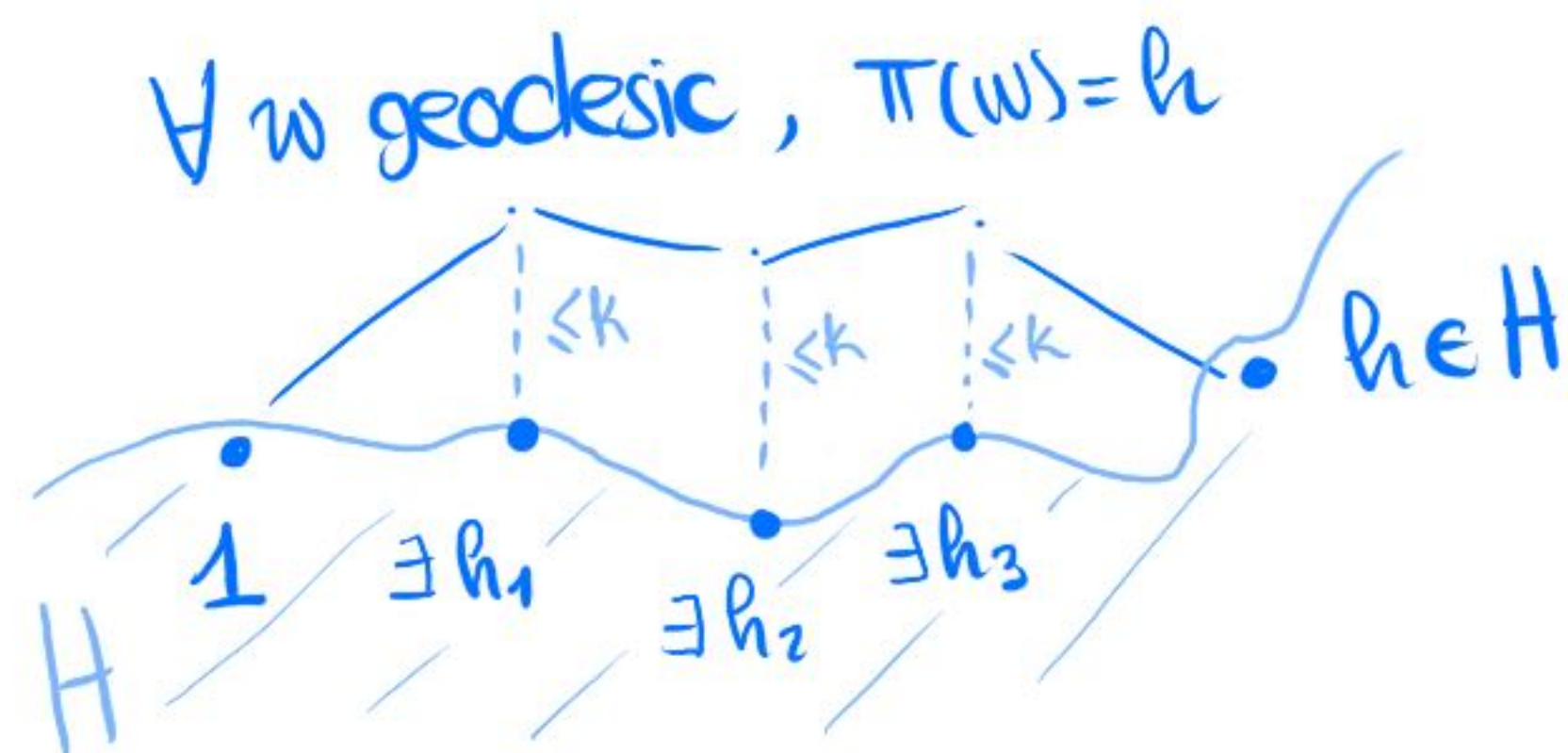
1.2. Ben-Zvi - Kropholler - Lyman '22

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f.g. $\langle S \rangle$

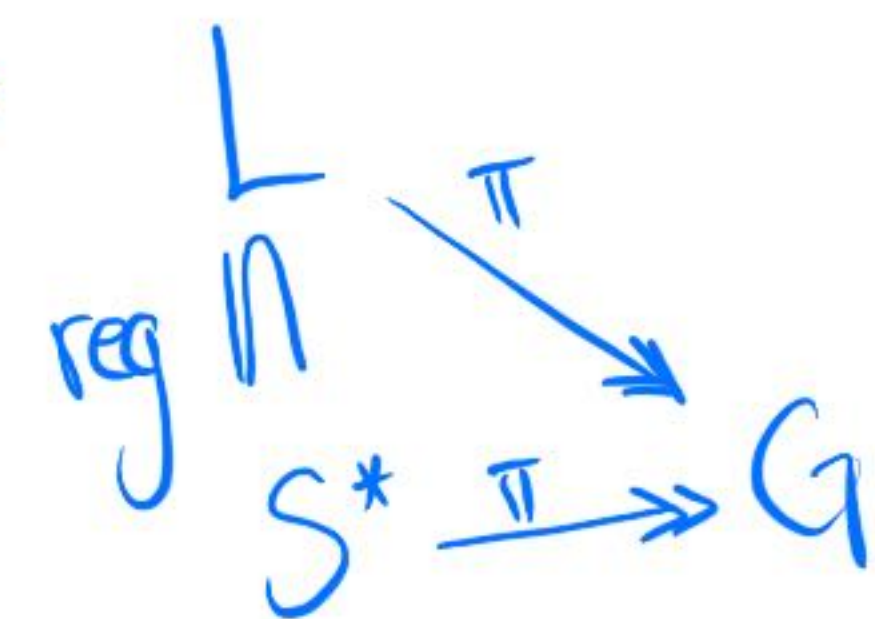
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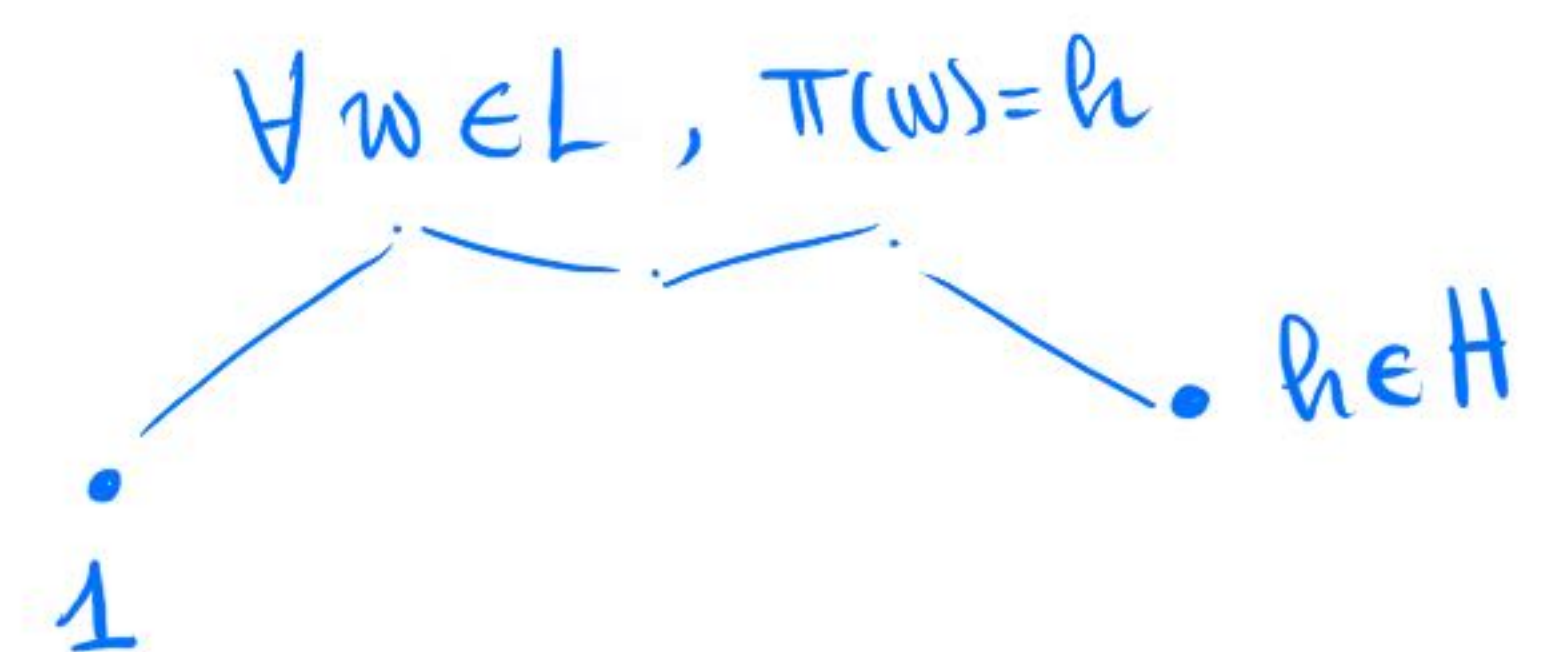
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Folding (nice) rational
subsets of automatic groups.

Gazteak RSME '25

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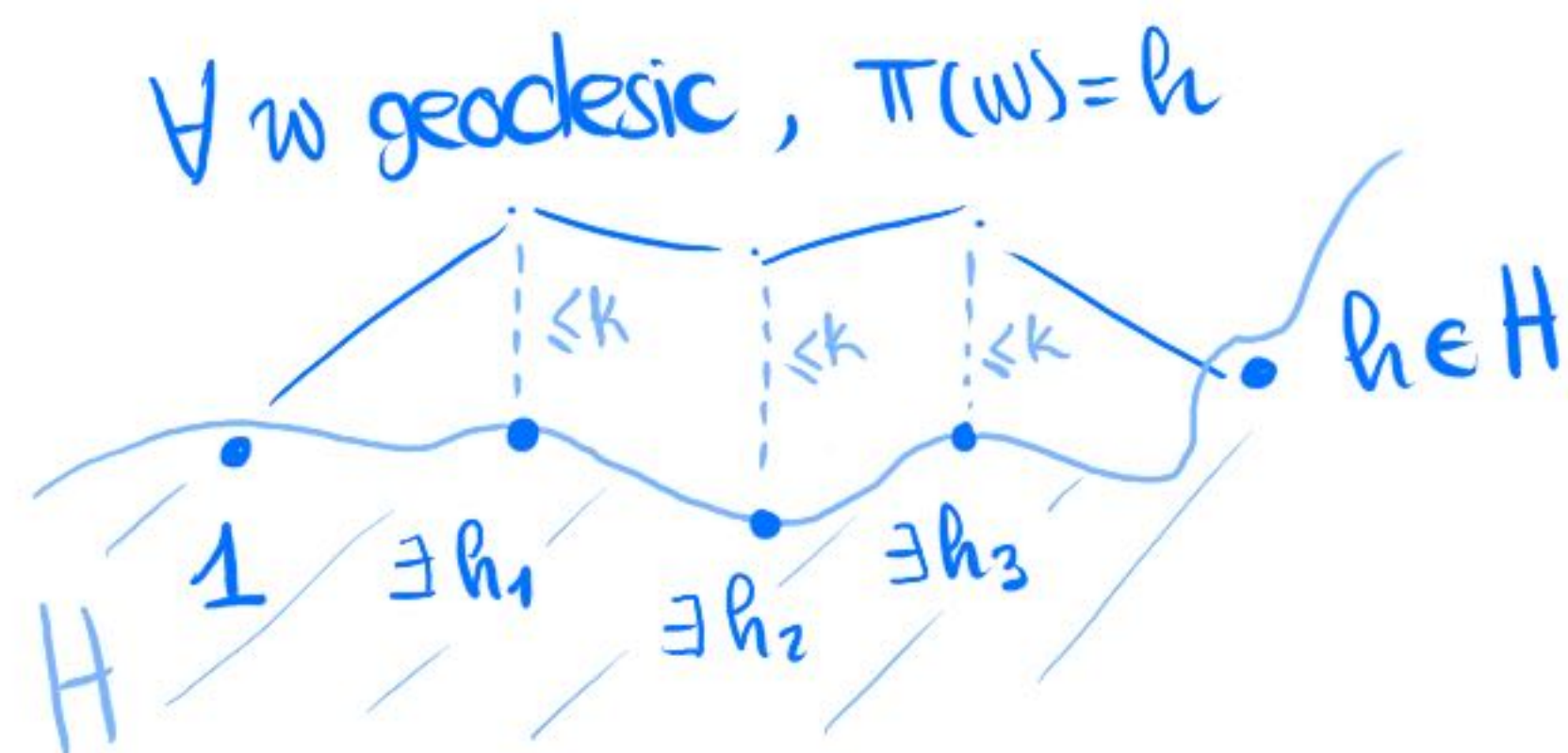
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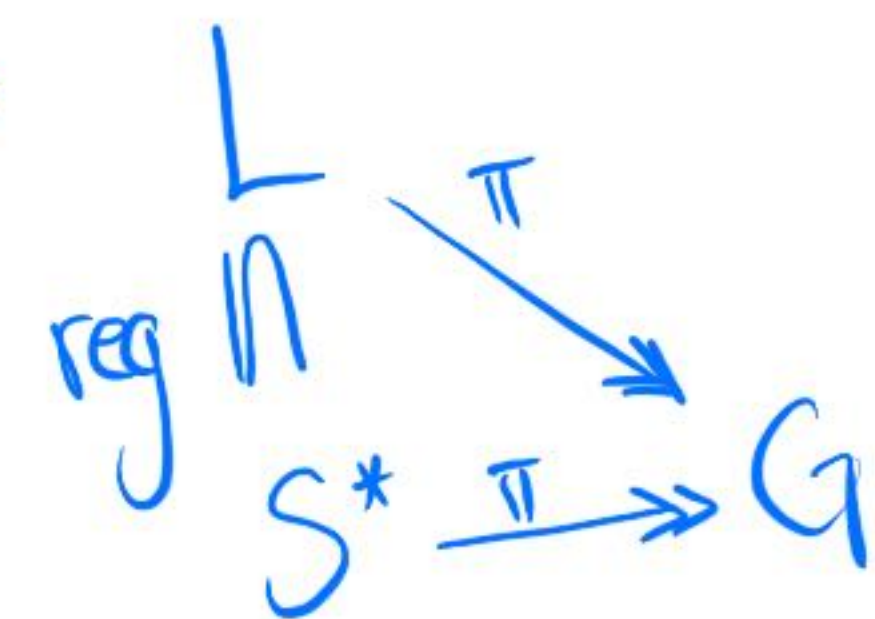
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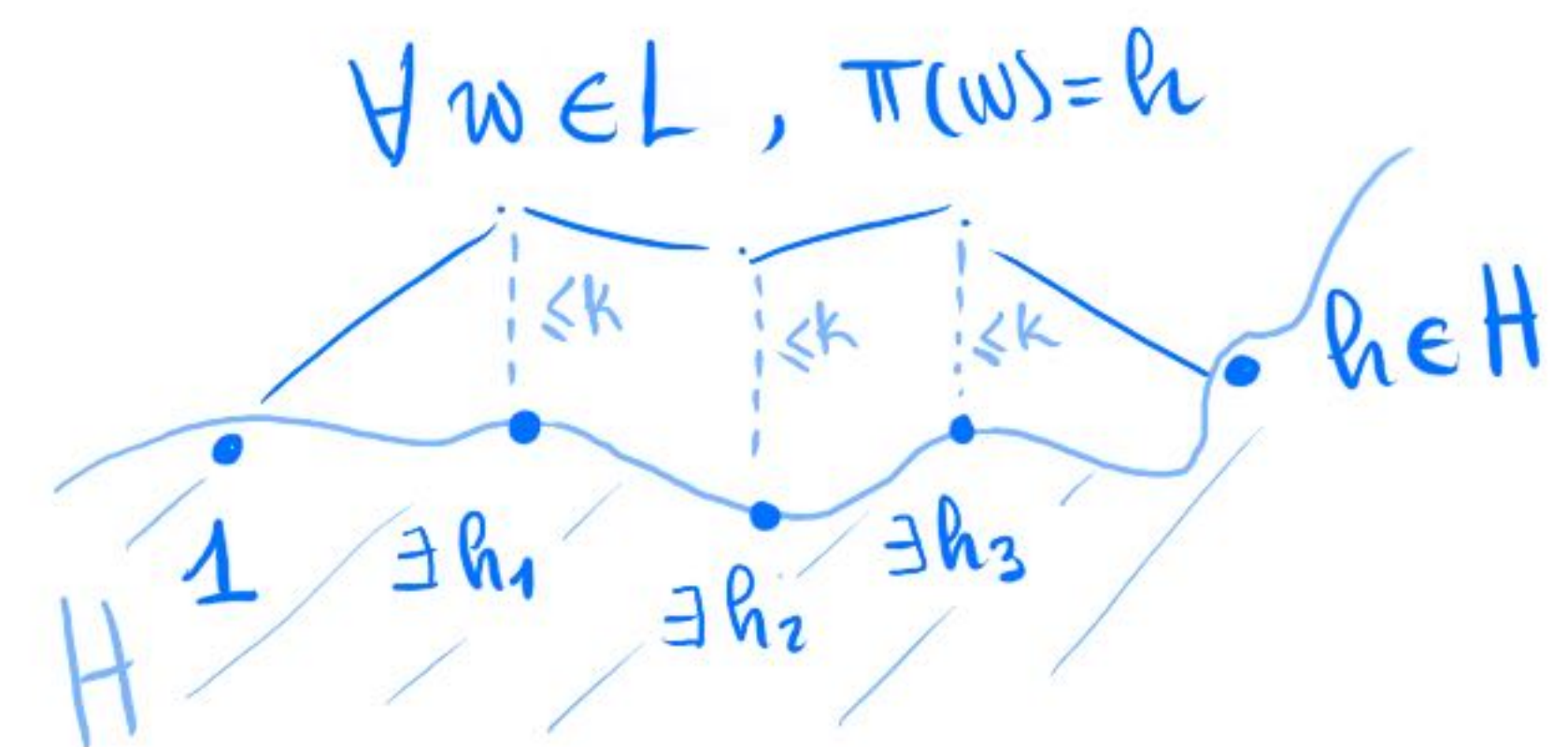
$$H \leq G \text{ automatic, finite presentation } \langle S | R \rangle$$

f.g.



Ingredients : $H = \langle h_1, \dots, h_k \rangle$ $G \cong \langle S | R \rangle$
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Folding (nice) rational
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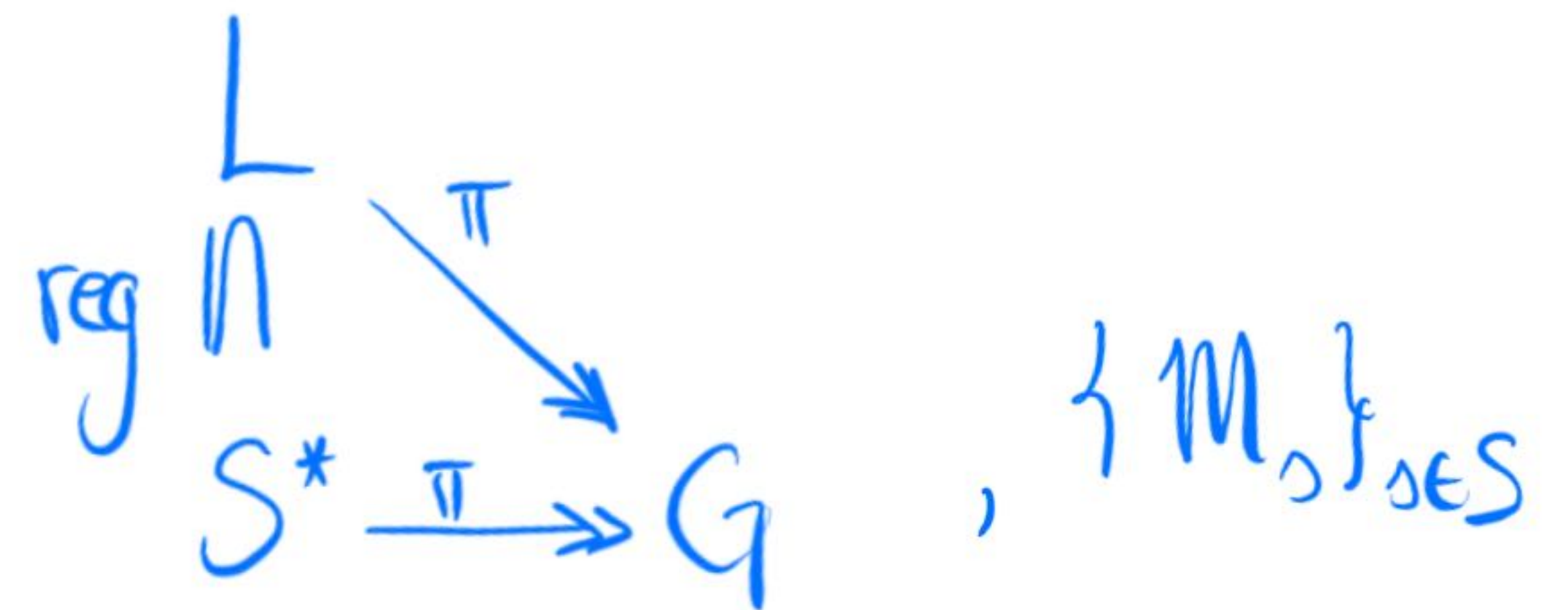
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Gazteak RSME '25

Folding (nice) rational
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G automatic,
fin. pres. $\langle S \mid R \rangle$

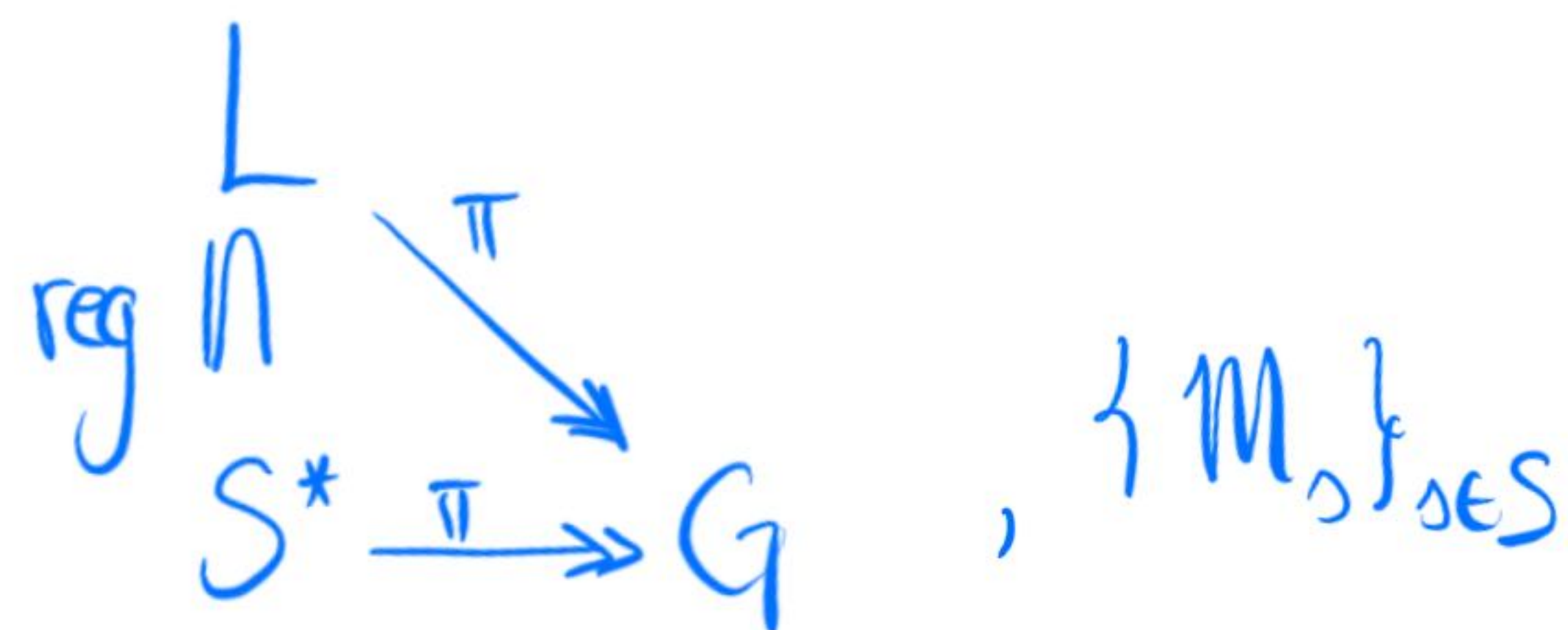


Folding (nice) rational
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$$2. \underset{\text{rational}}{K} \subseteq \underset{\text{automatic}}{G}$$

K rational subset $\subseteq G$ automatic,
 fin. pres. $\langle S \mid R \rangle$

Gazteak RSME '25

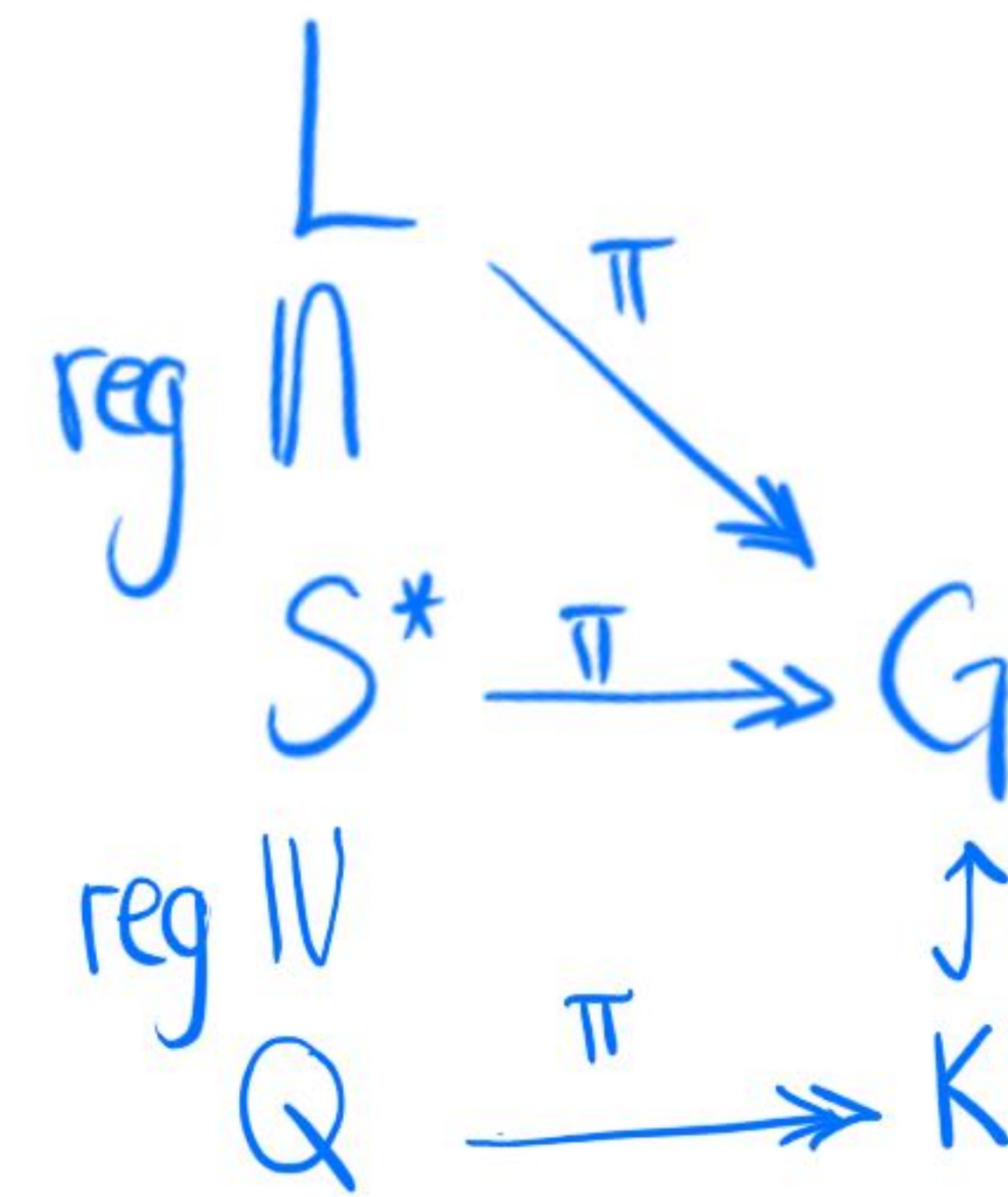


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"image of regular language"



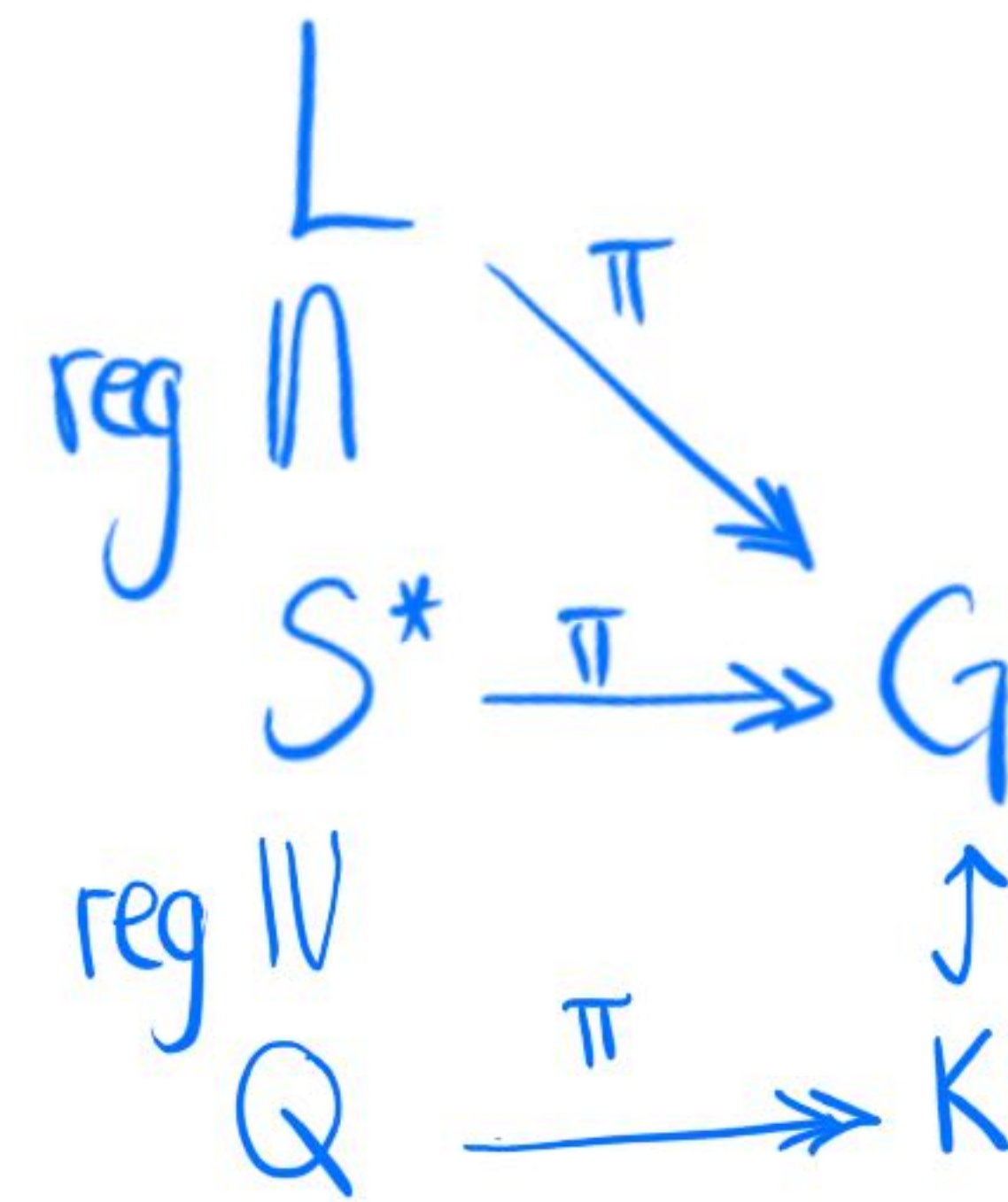
, $\{m_s\}_{s \in S}$

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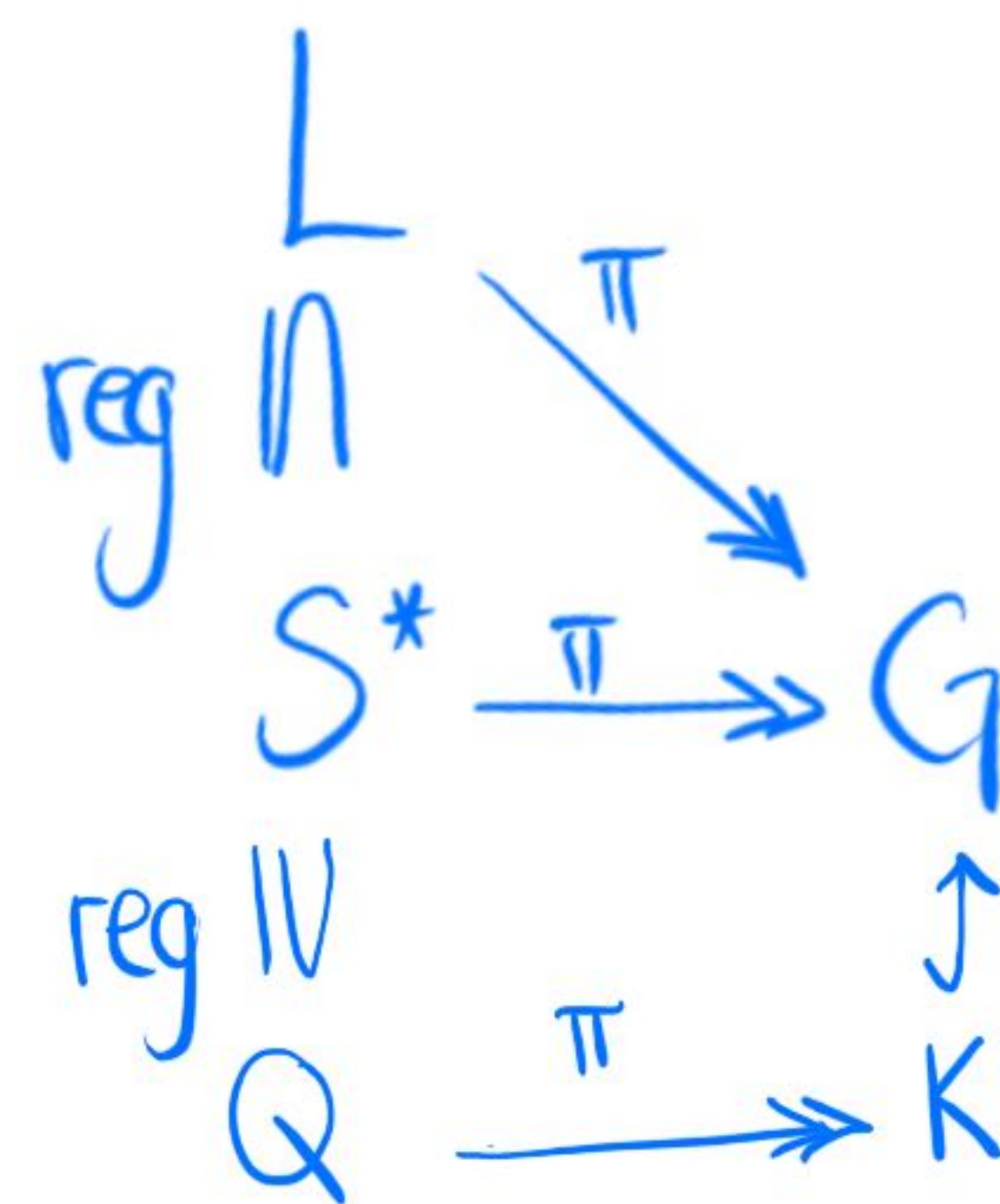
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(e.g., K submonoid)

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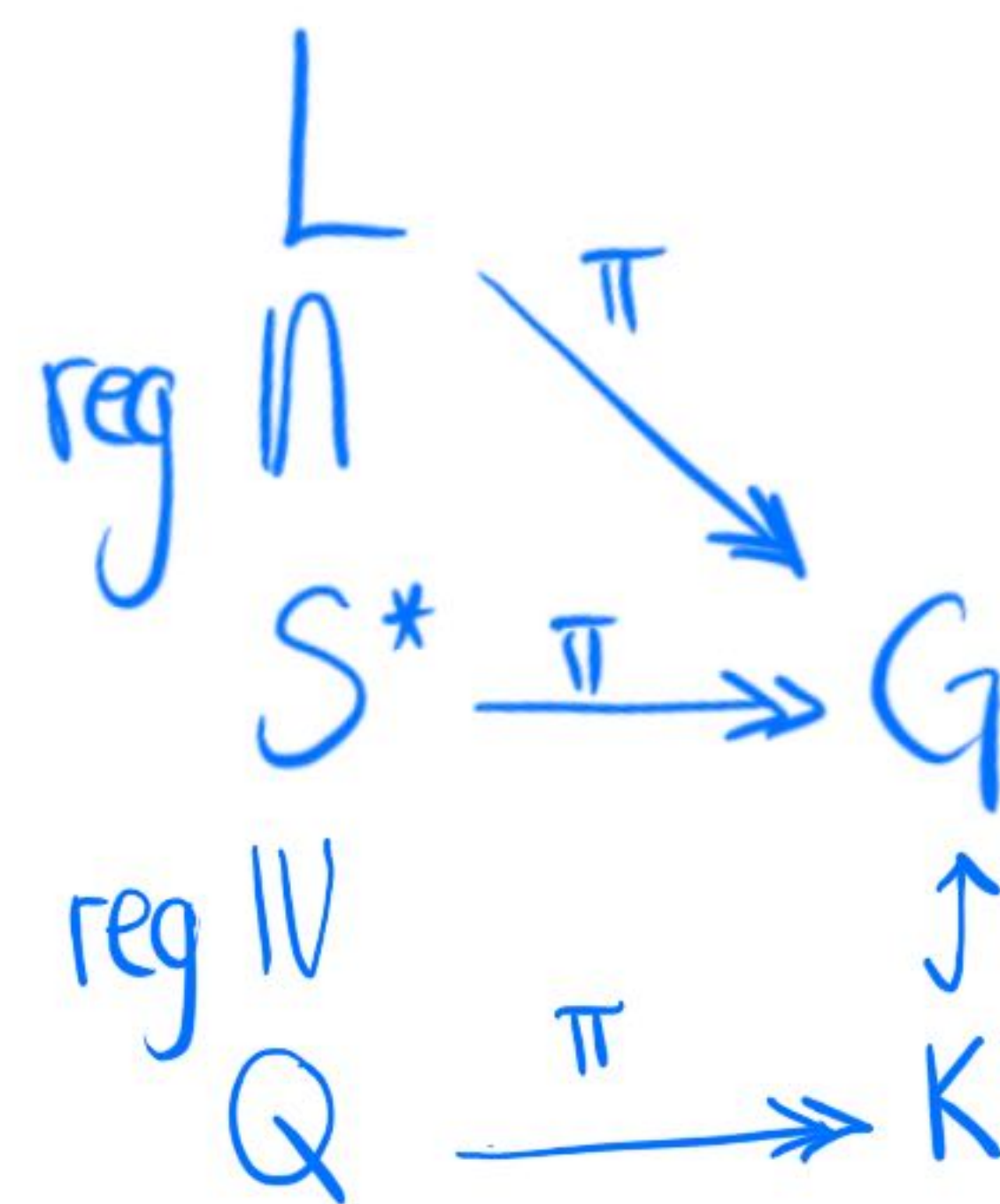
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K L -quasiconvexly-undistorted (nice)



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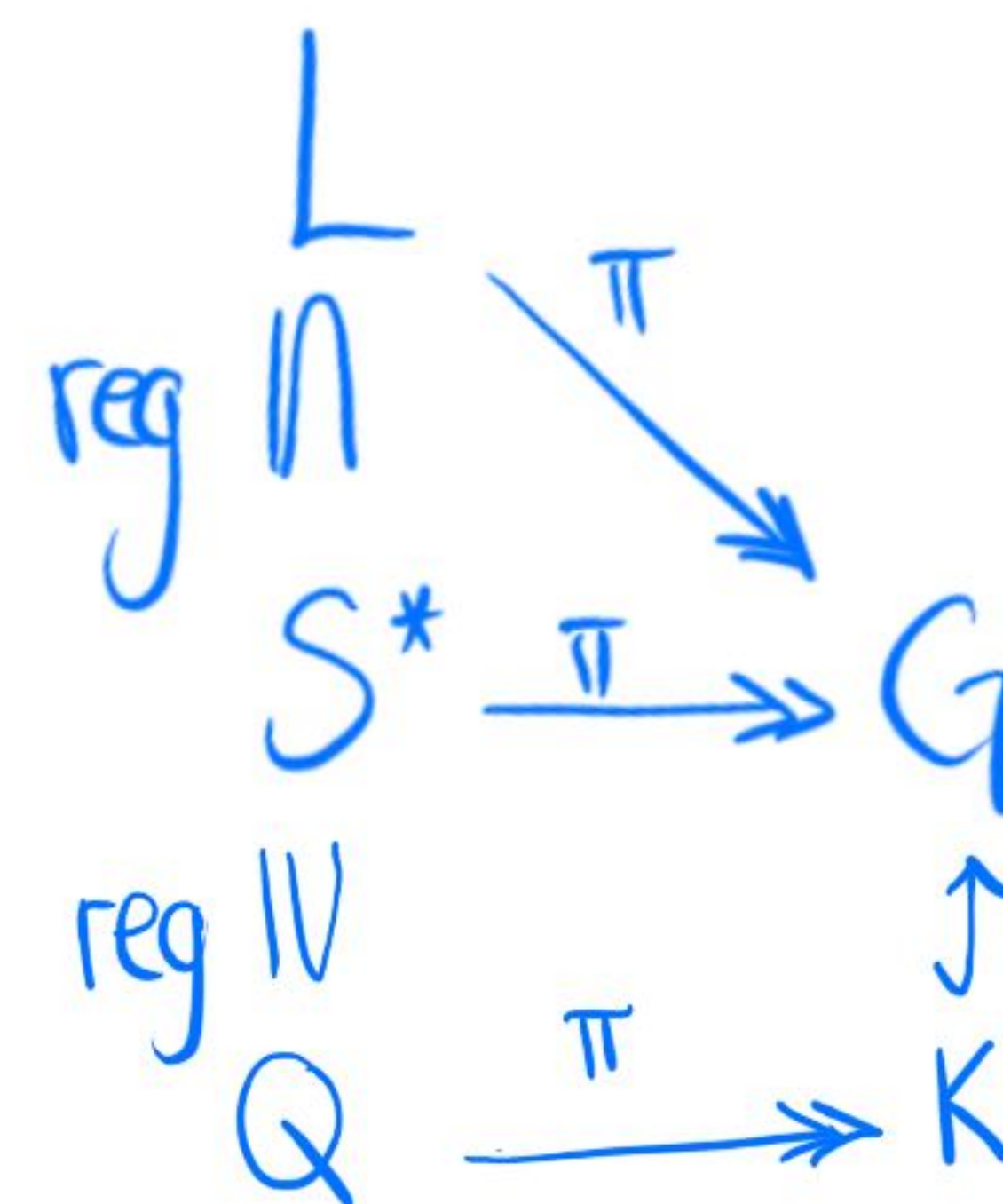
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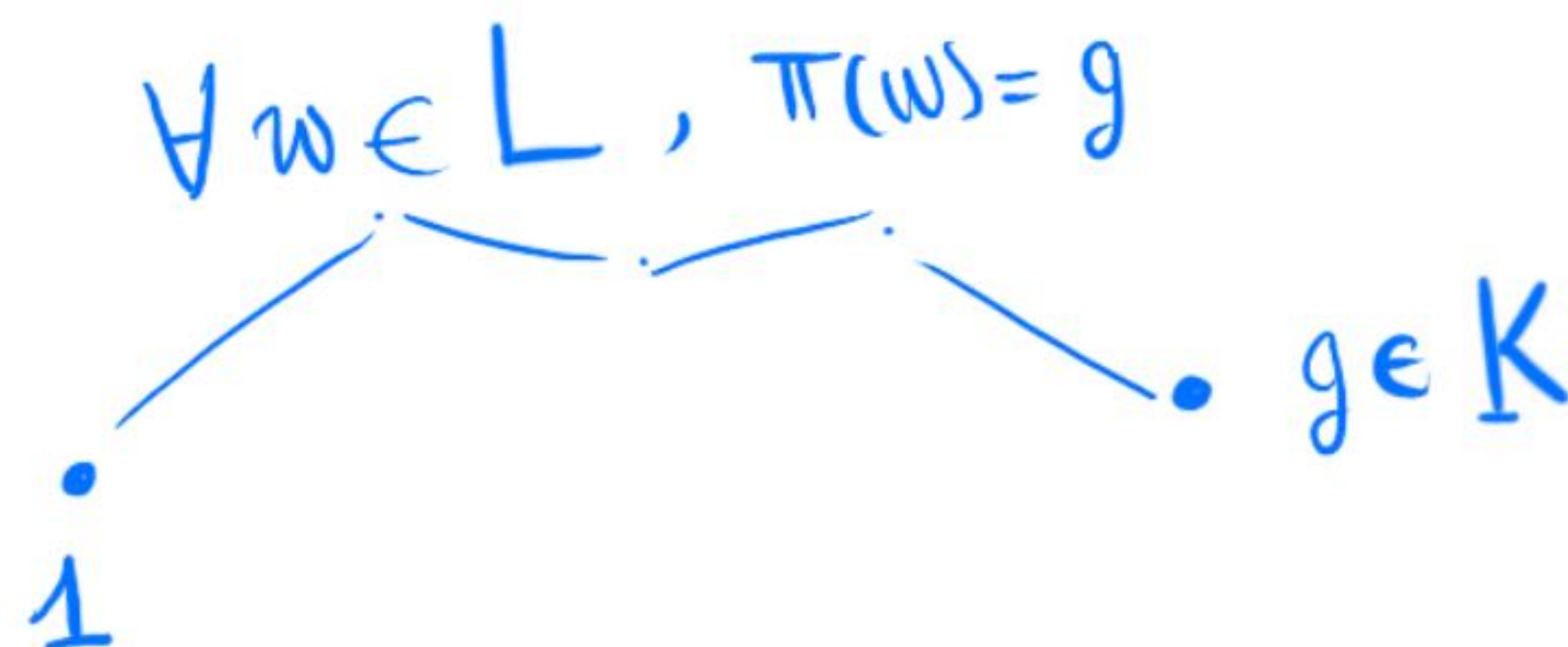


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$\text{Cay}(G, S)$

$\exists k, c :$



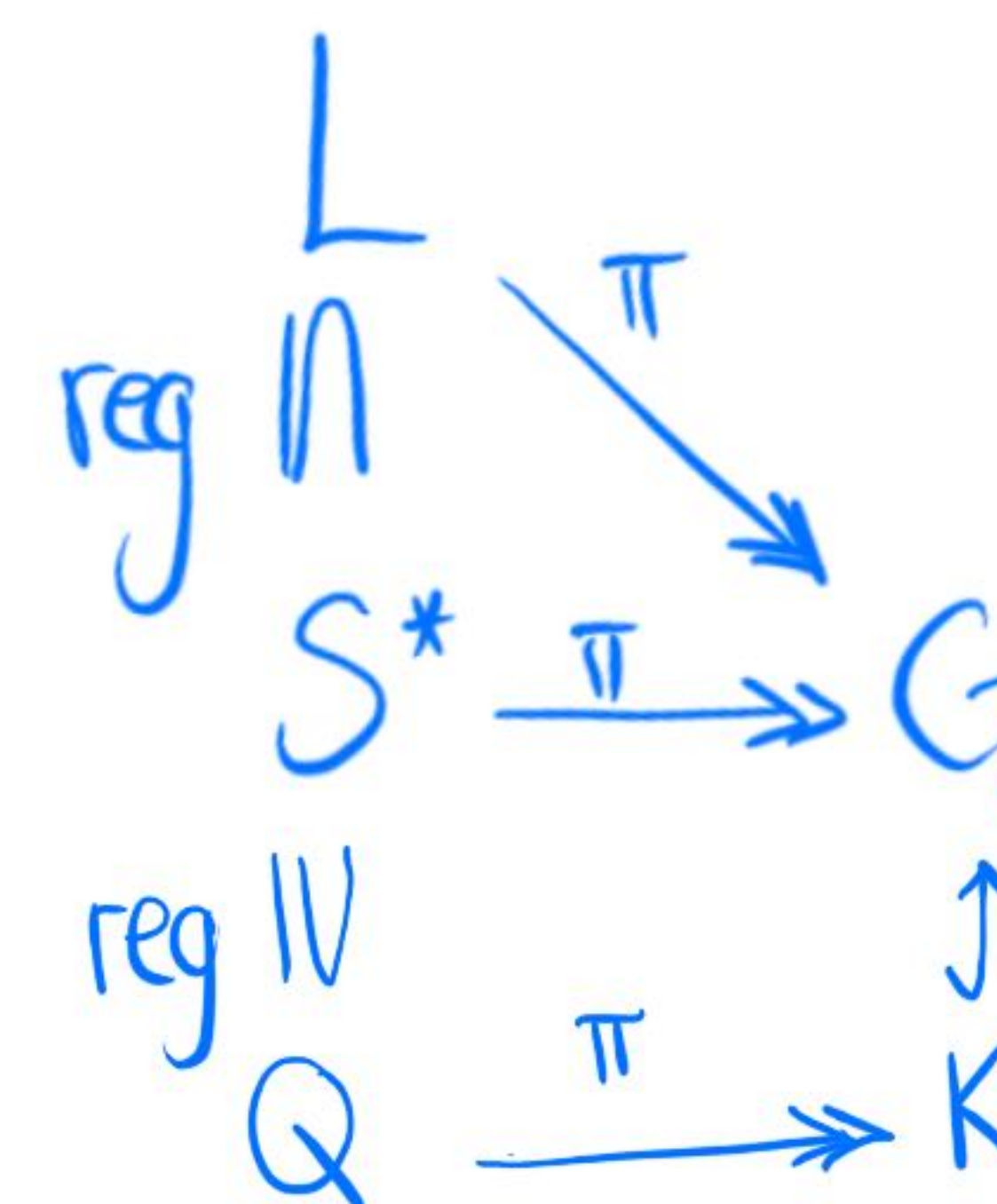
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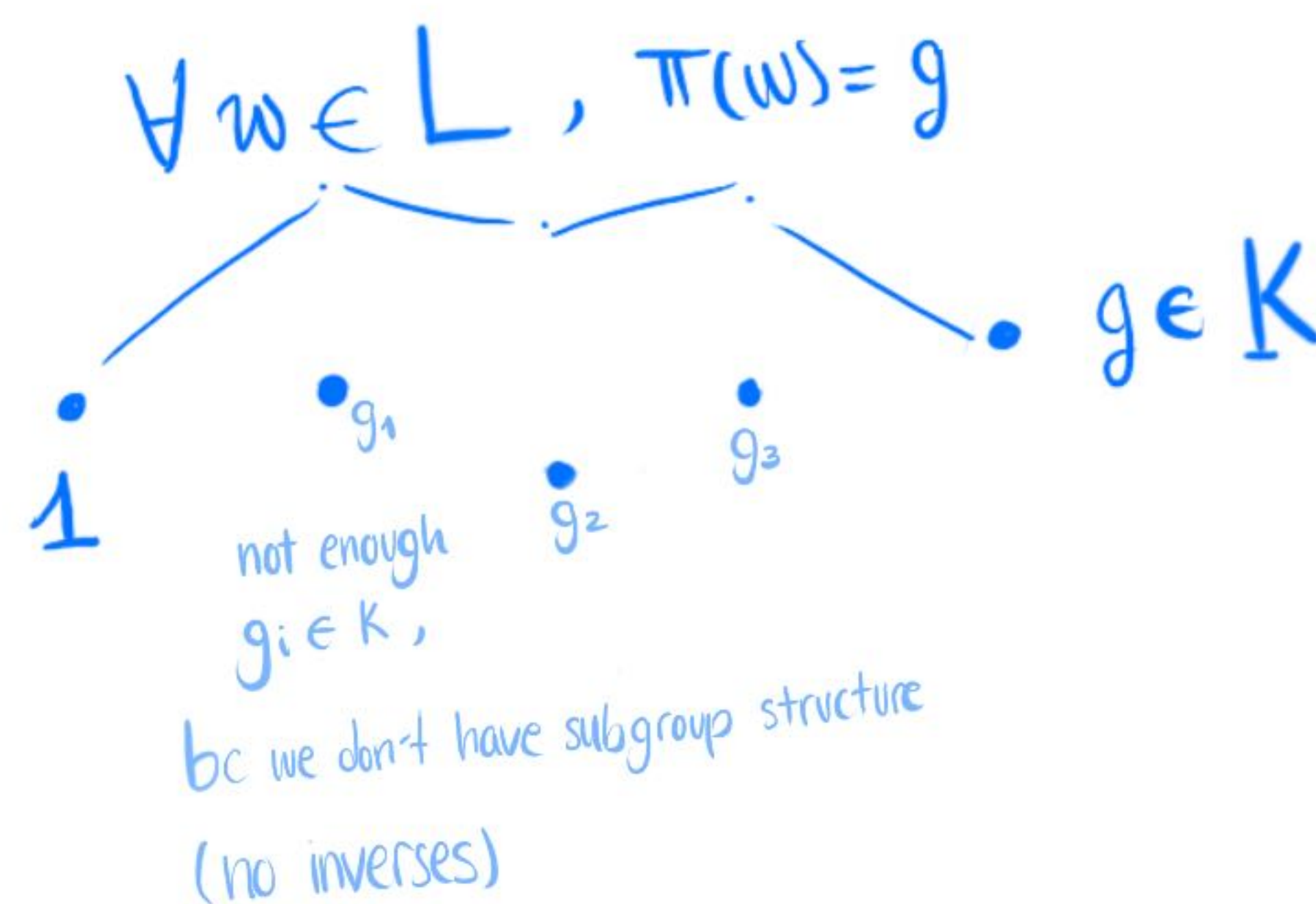


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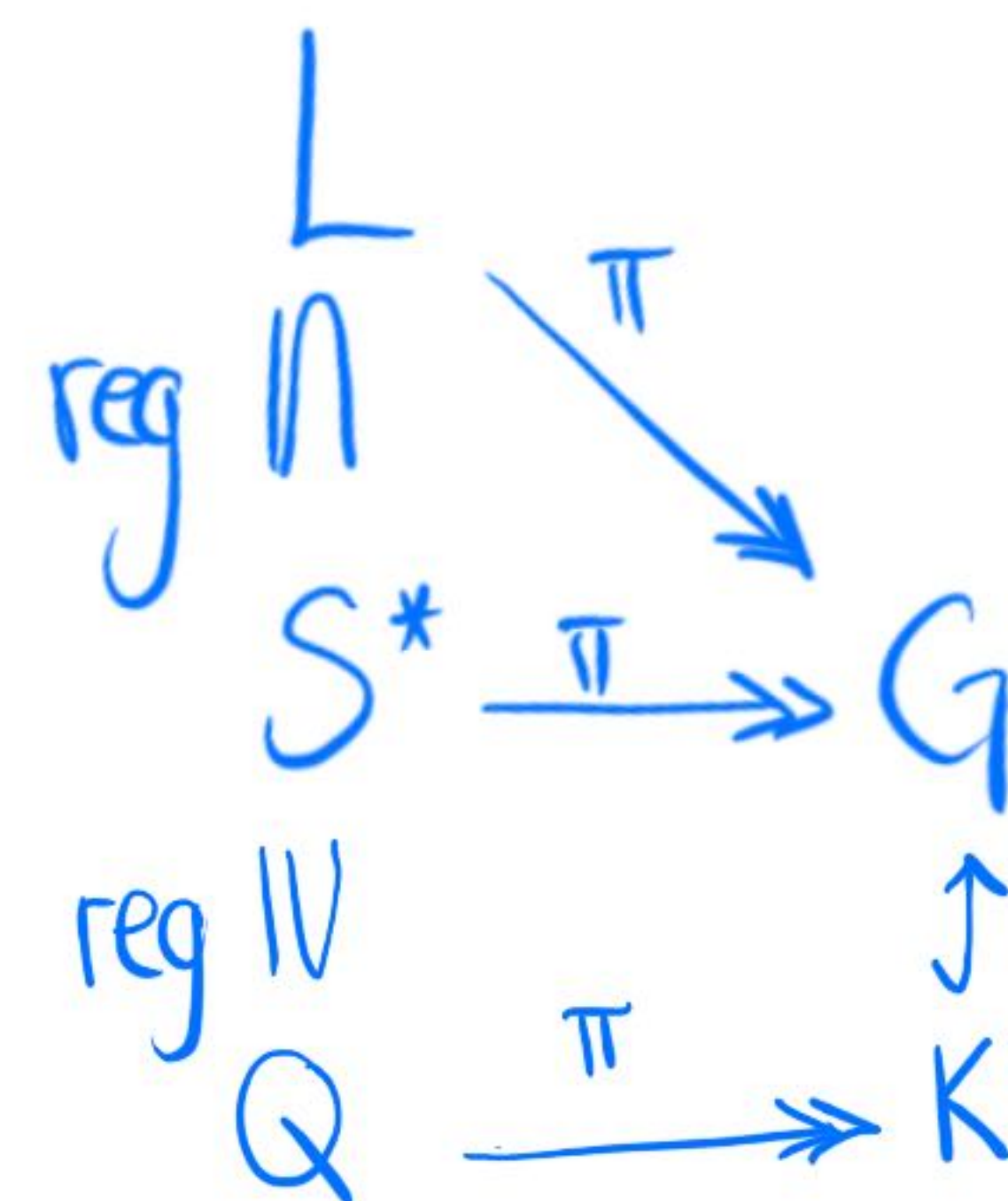
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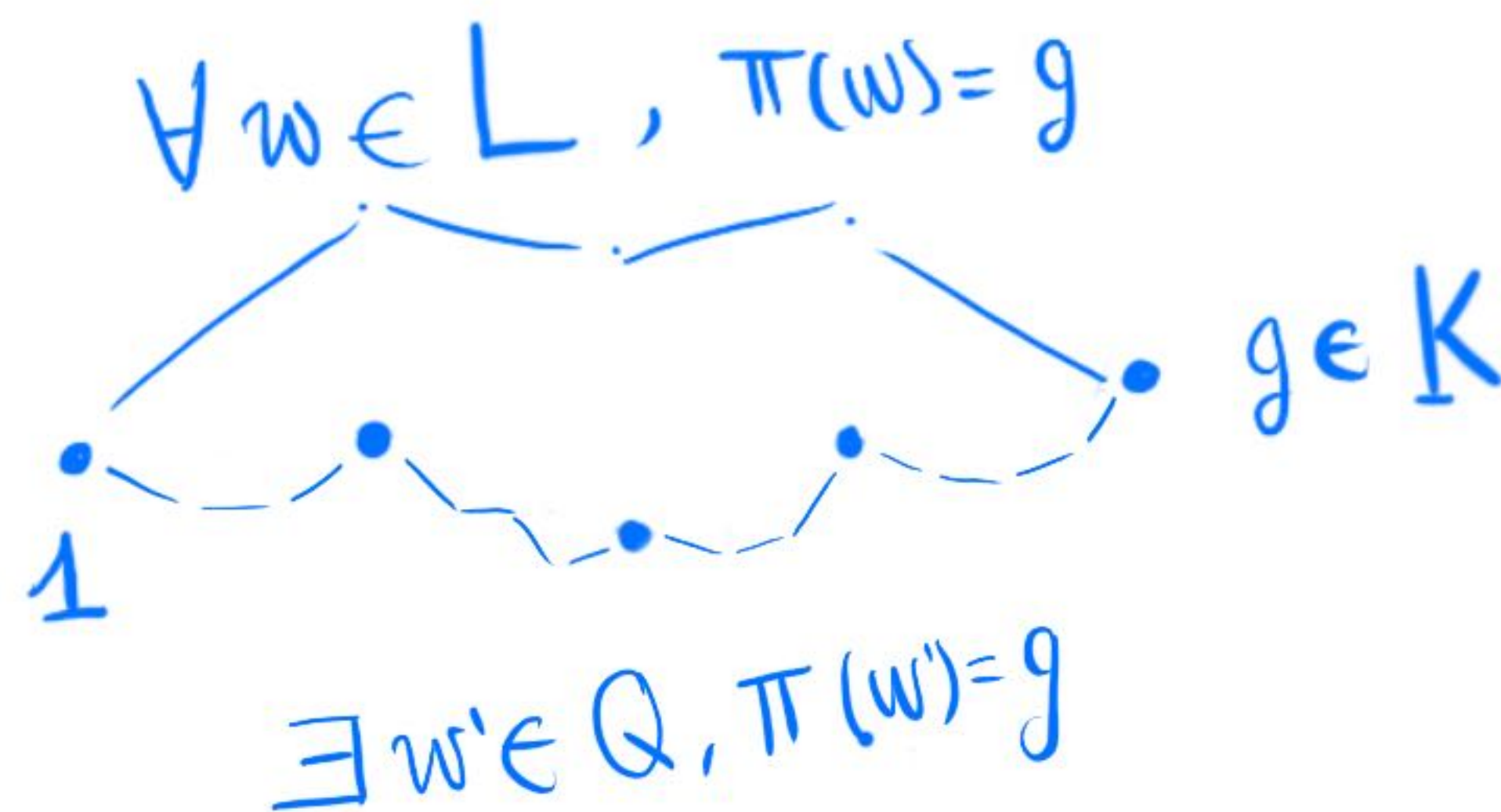


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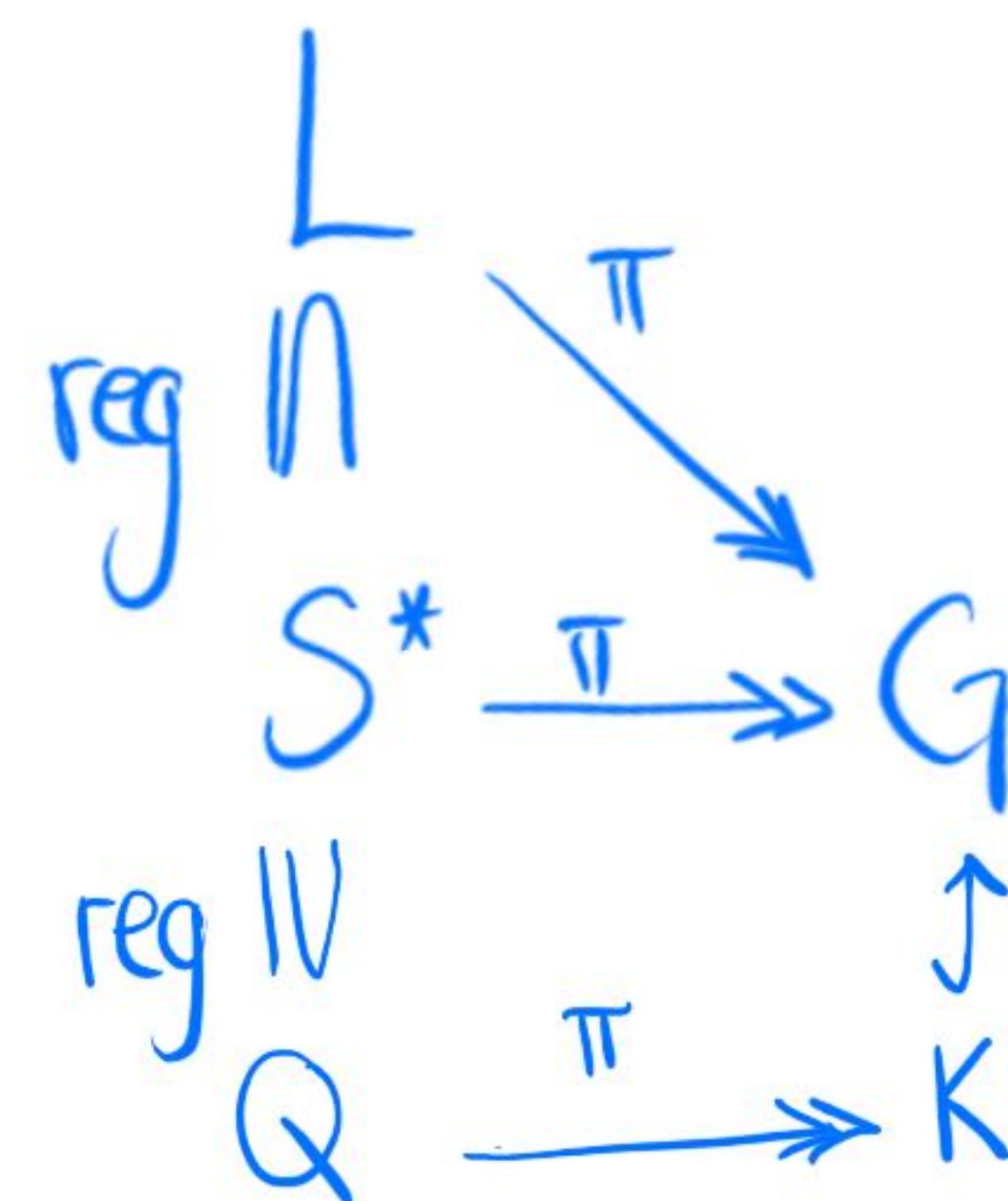
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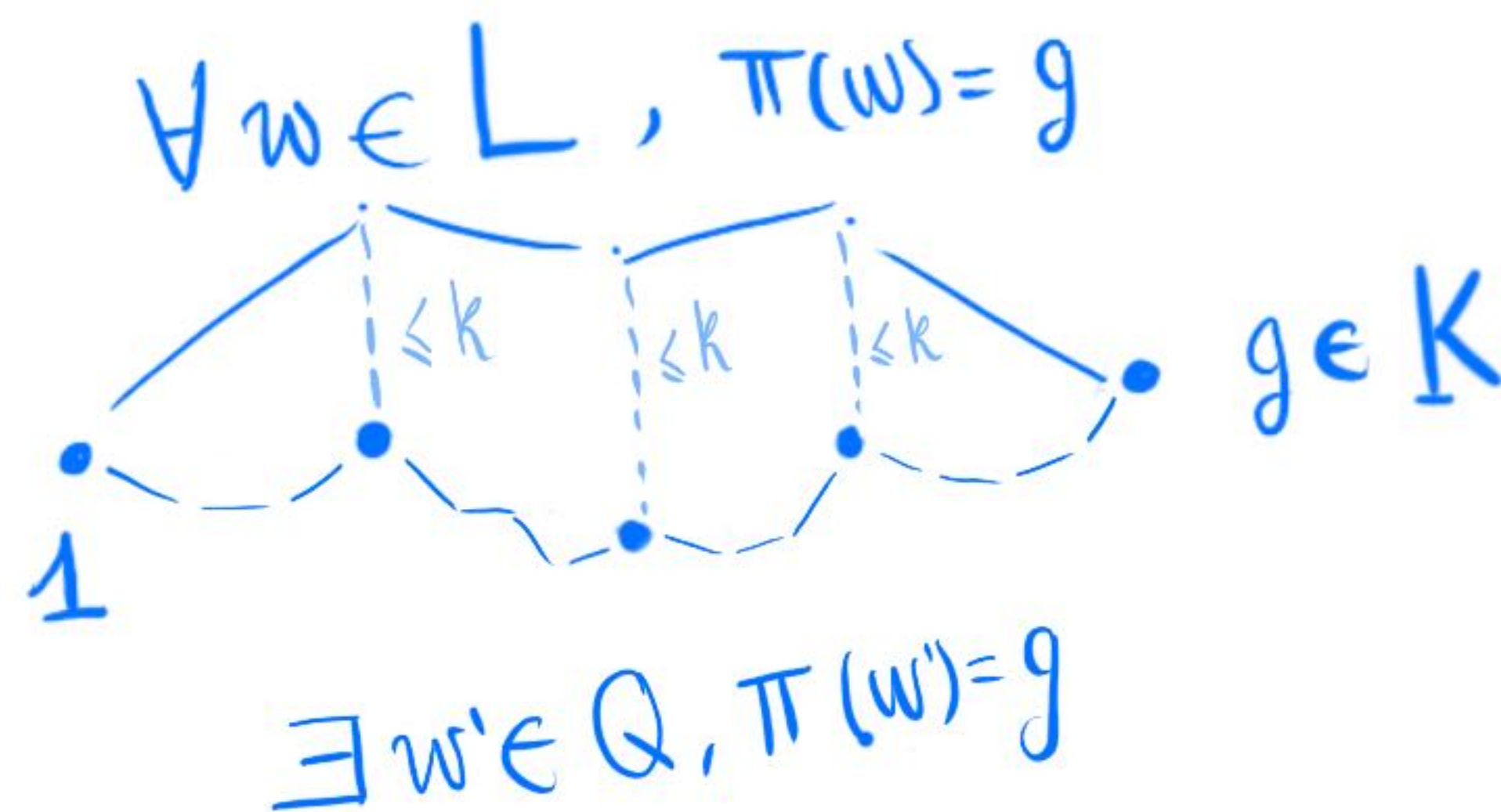


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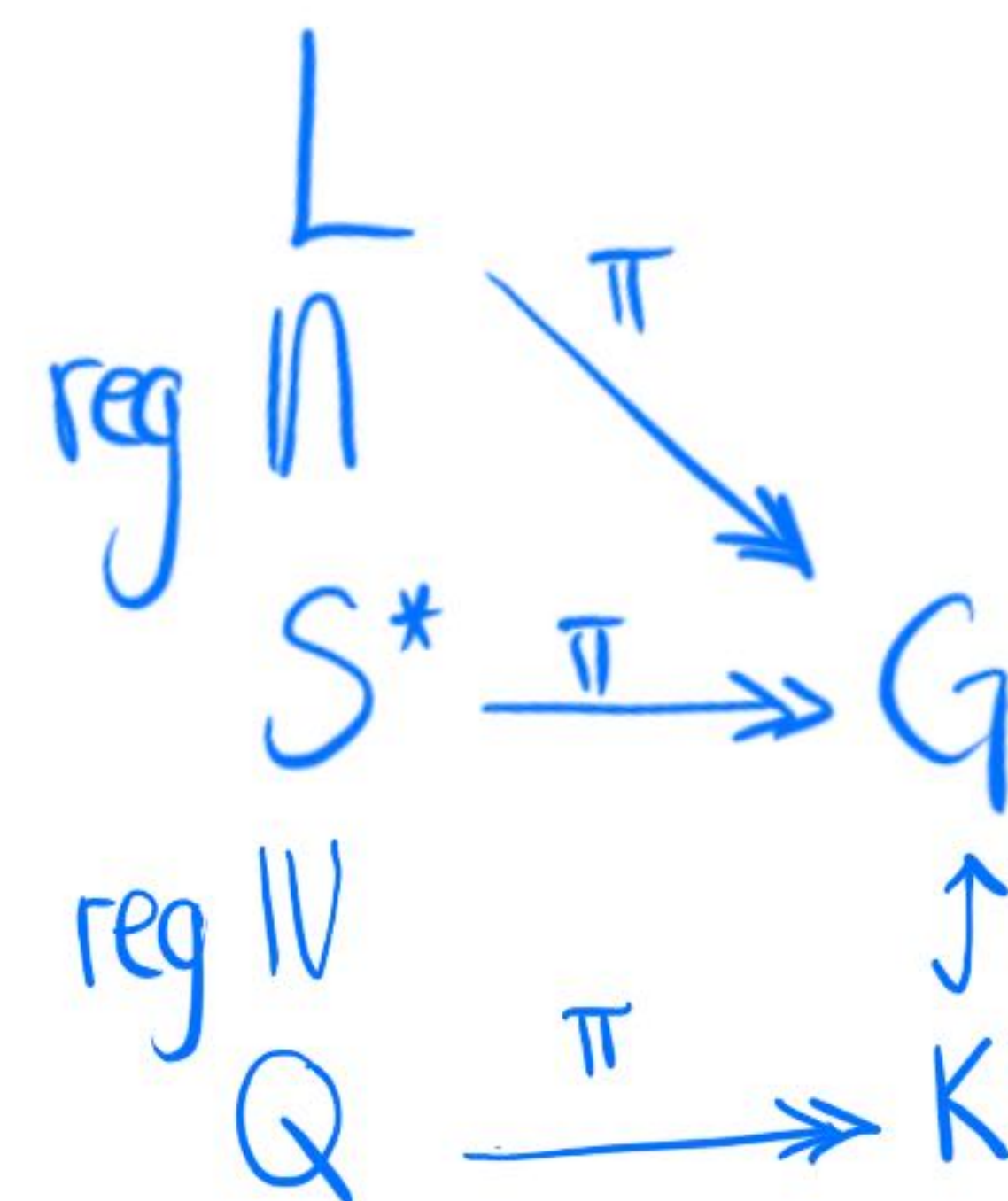
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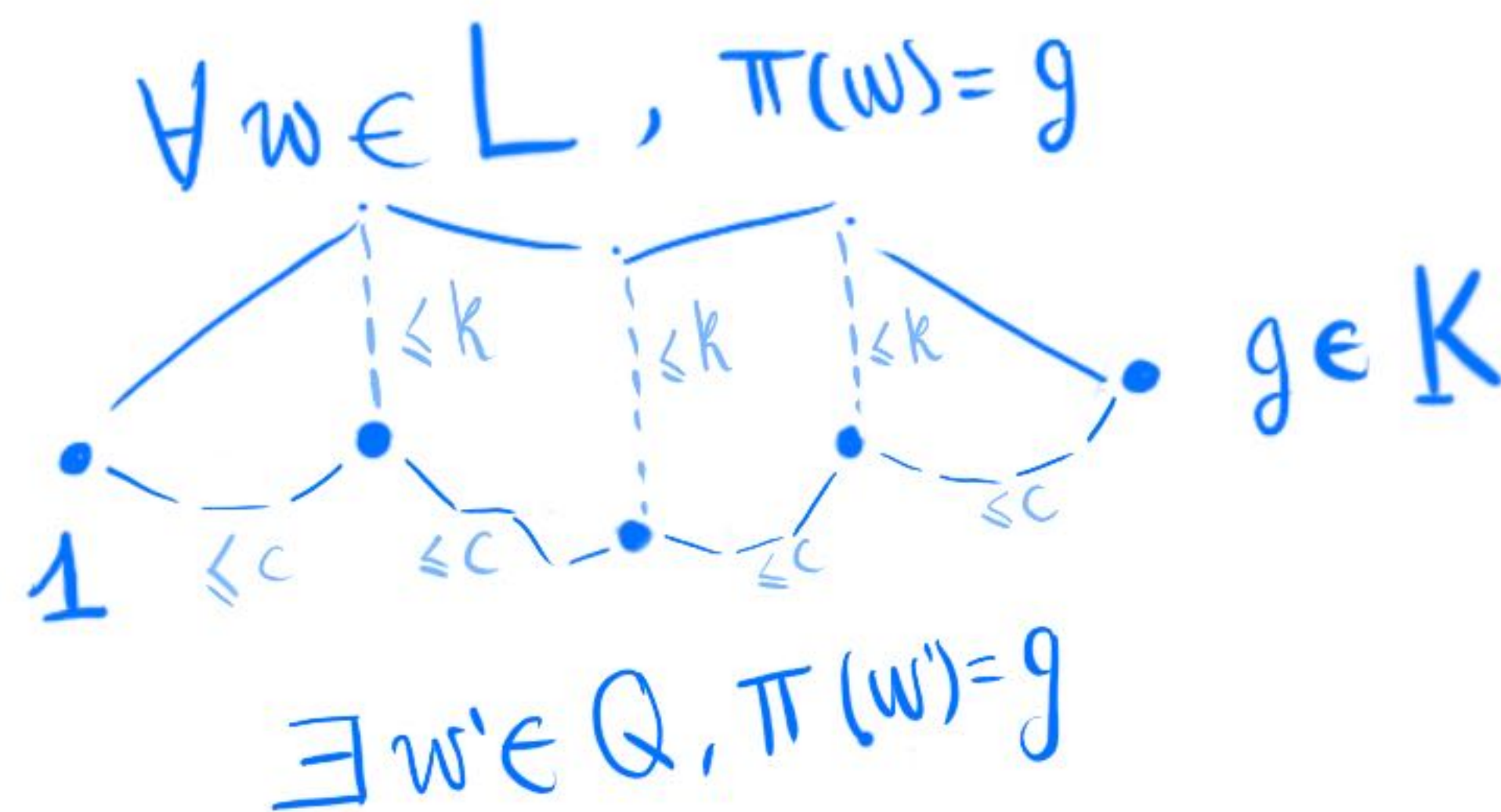


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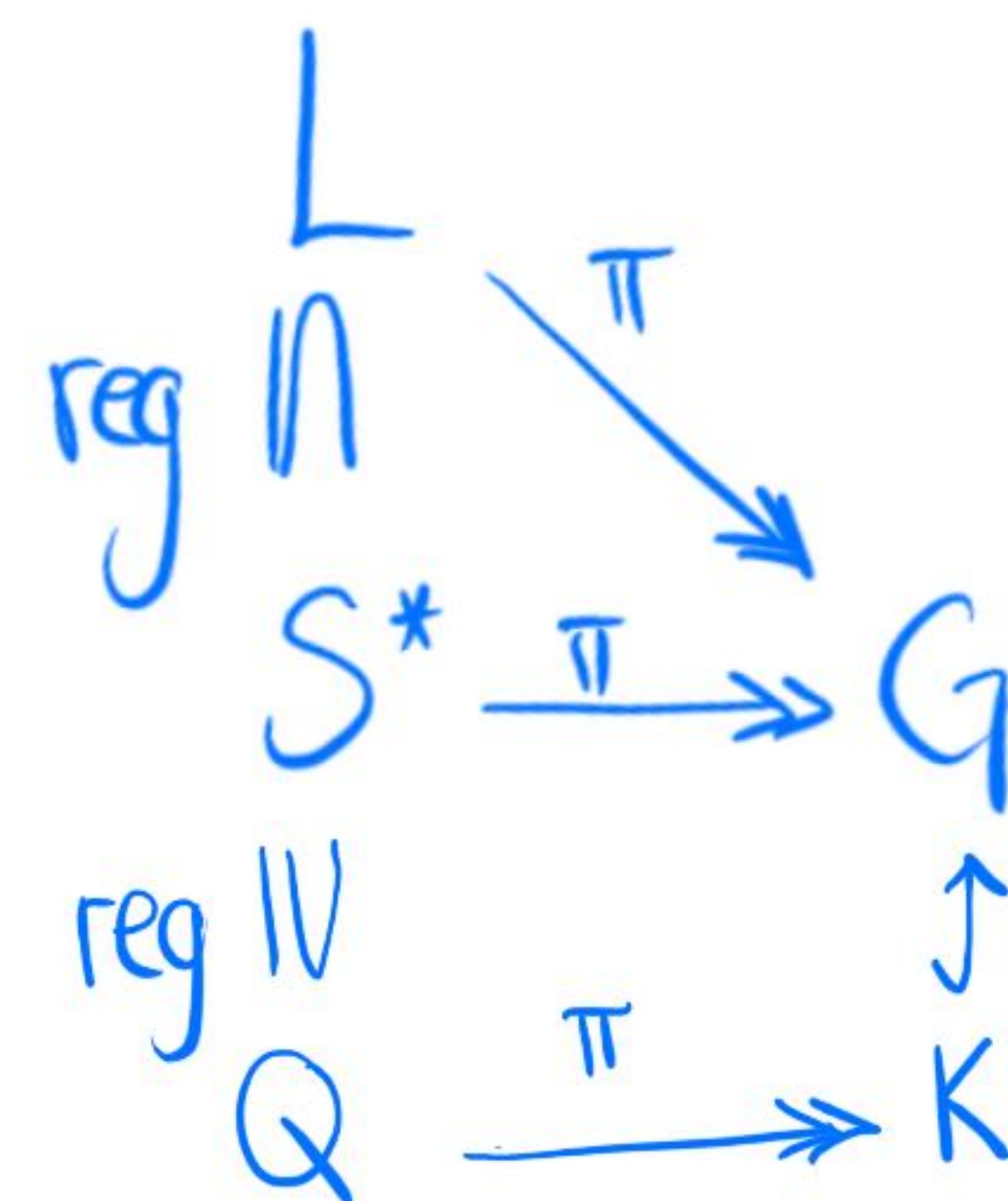
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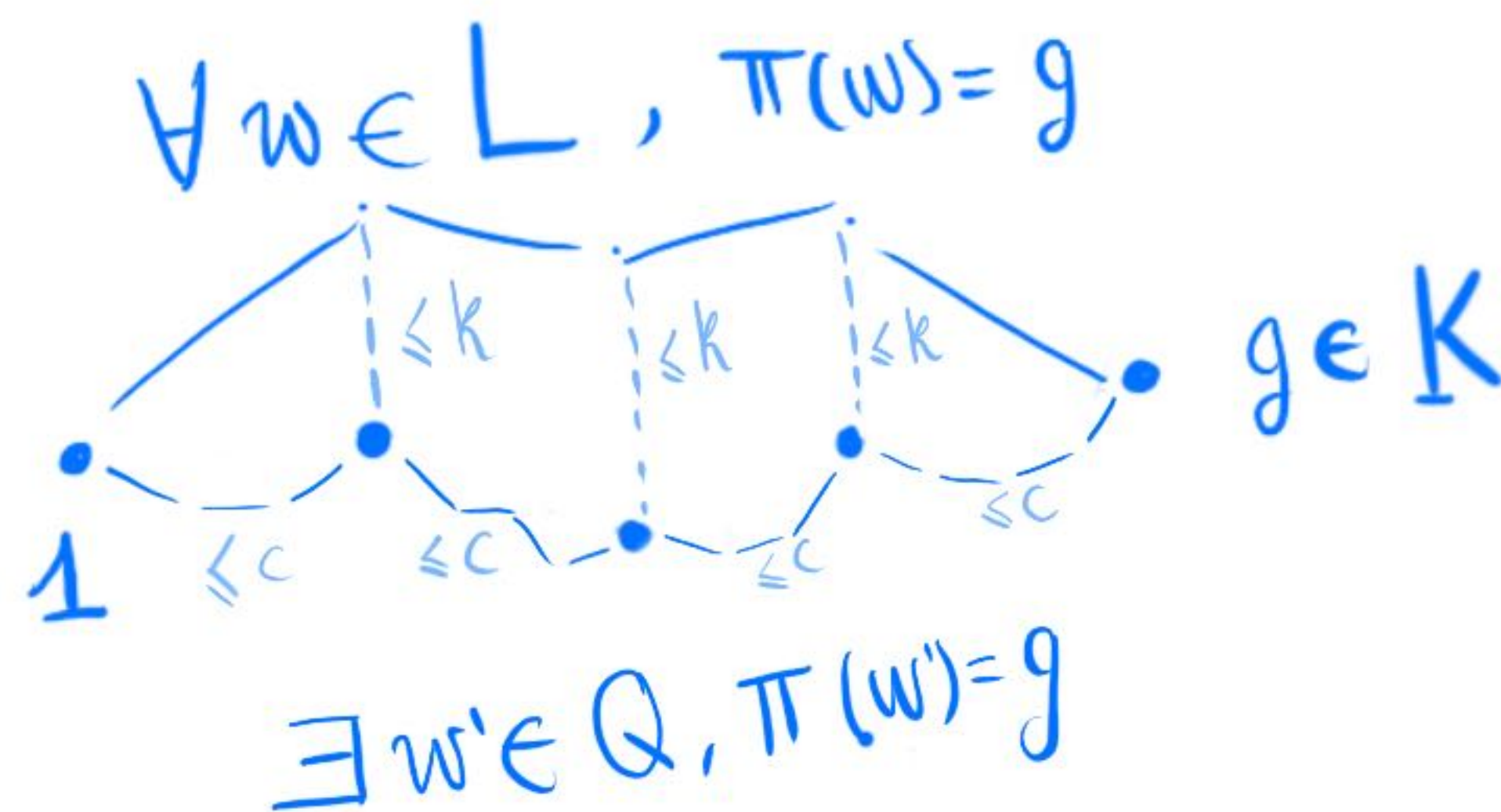


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K L -quasiconvexly - undistorted (nice)

$\text{Cay}(G, S)$

$\exists k, c:$



- w' fellow travels (asynchronously) with w

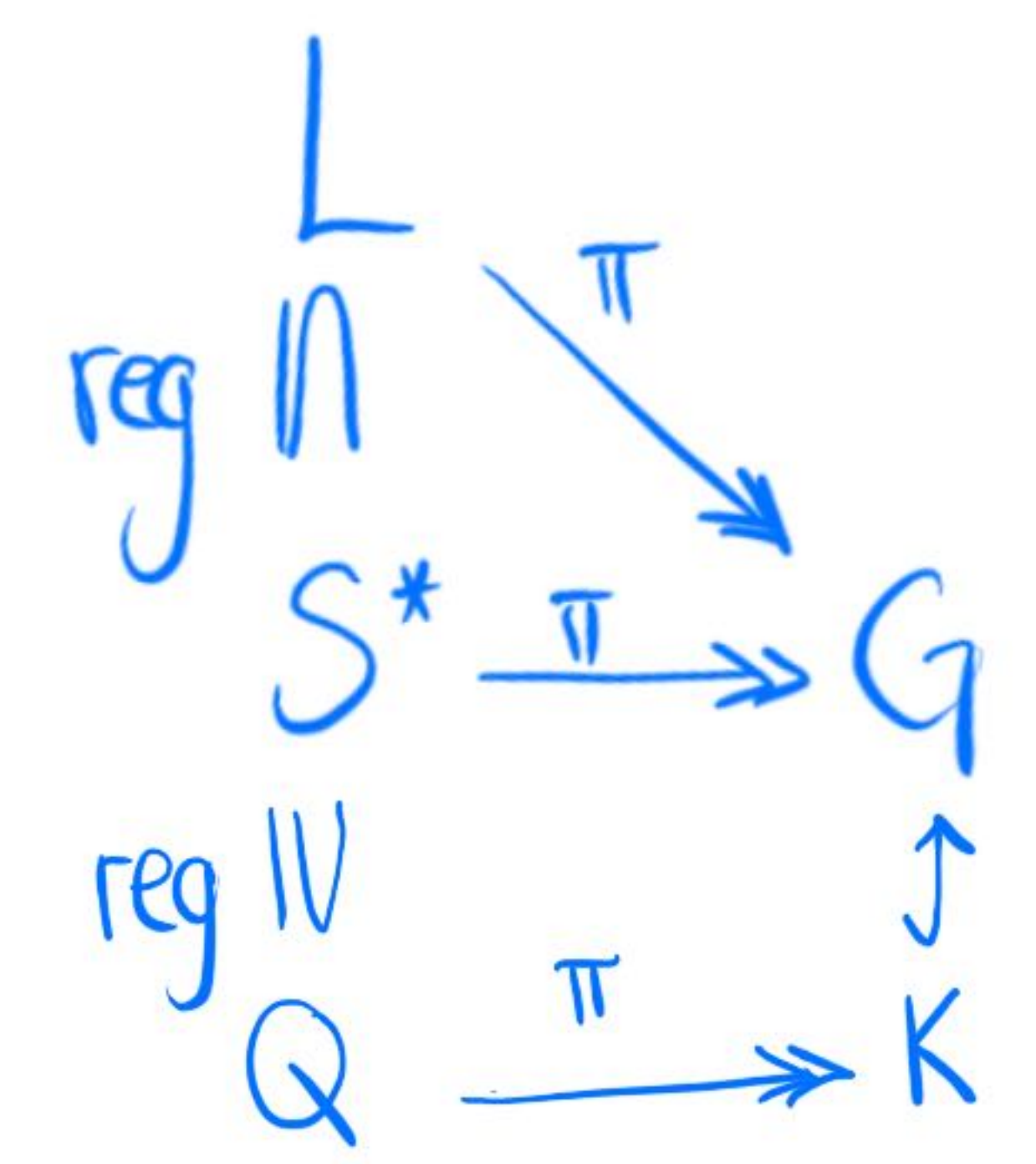
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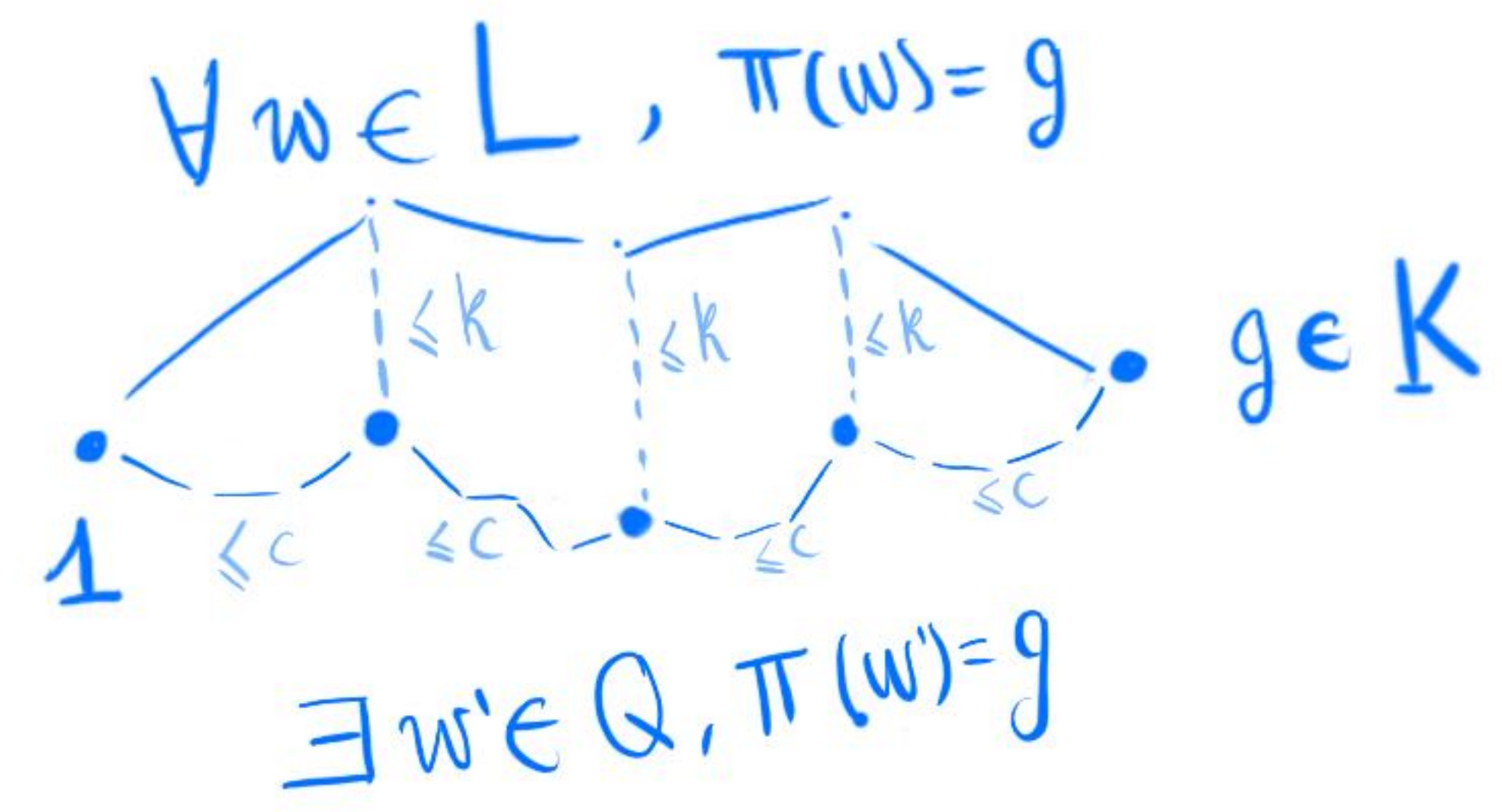


$\{m_s\}_{s \in S}$

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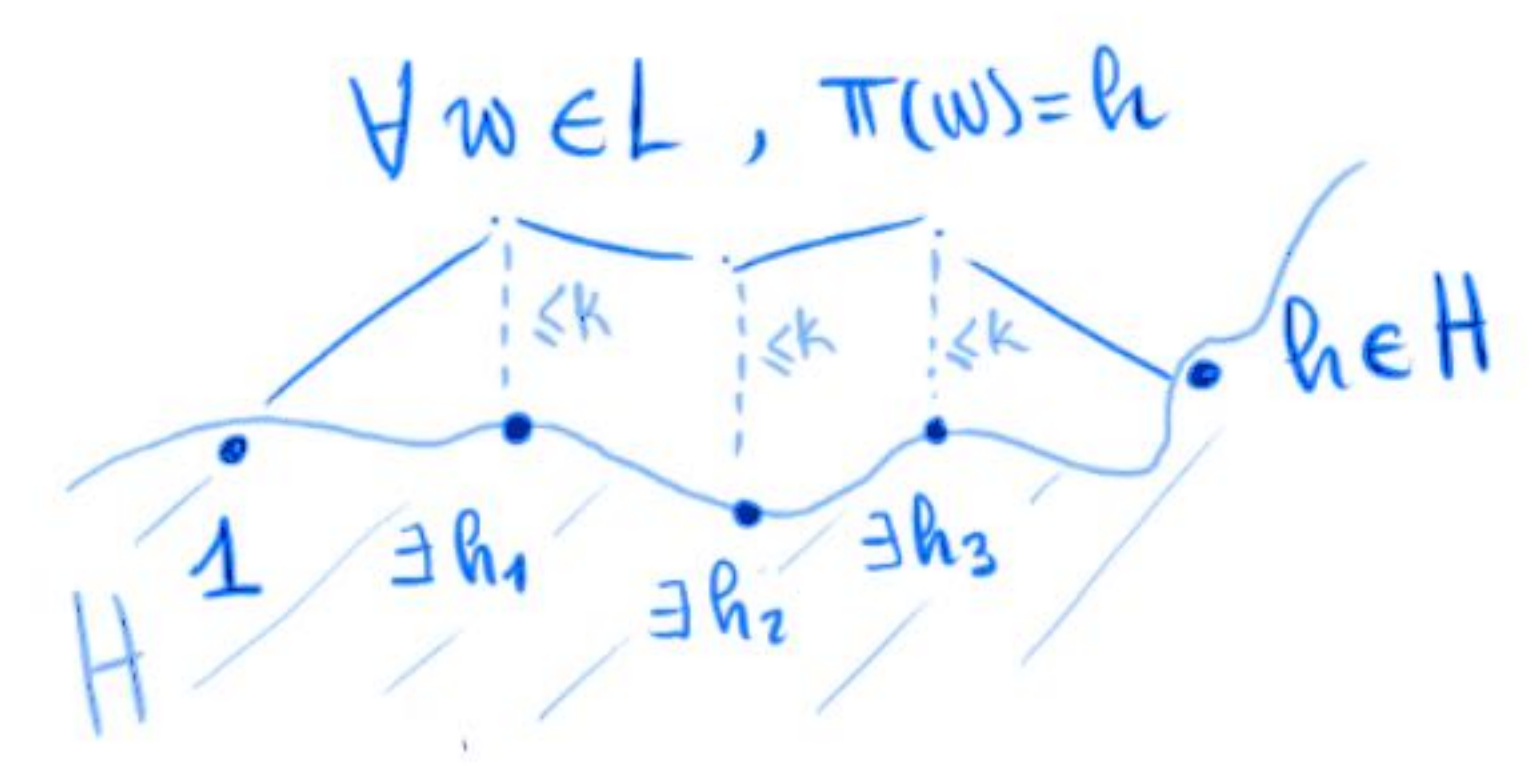
$\text{Cay}(G, S)$

$\exists k, c:$



• w' fellow travels (asynchronously) with w

• If $H \leq G$, we can build w' from $\{h_i\}_{i < |w|}$
 L -quasiconvex

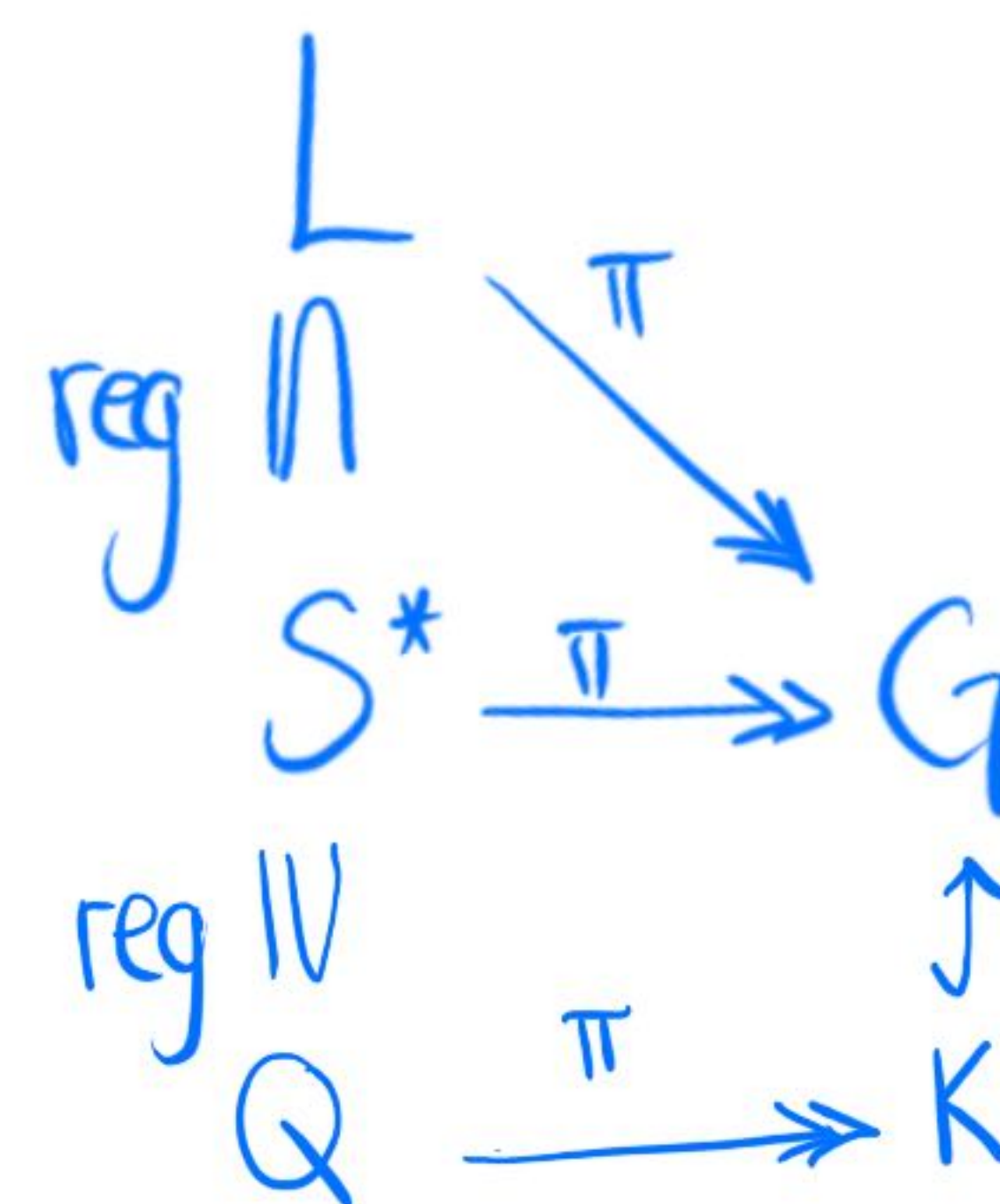


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, $\{m_s\}_{s \in S}$

K L -quasiconvexly - undistorted

Goal: Build St_K to solve membership problem of K

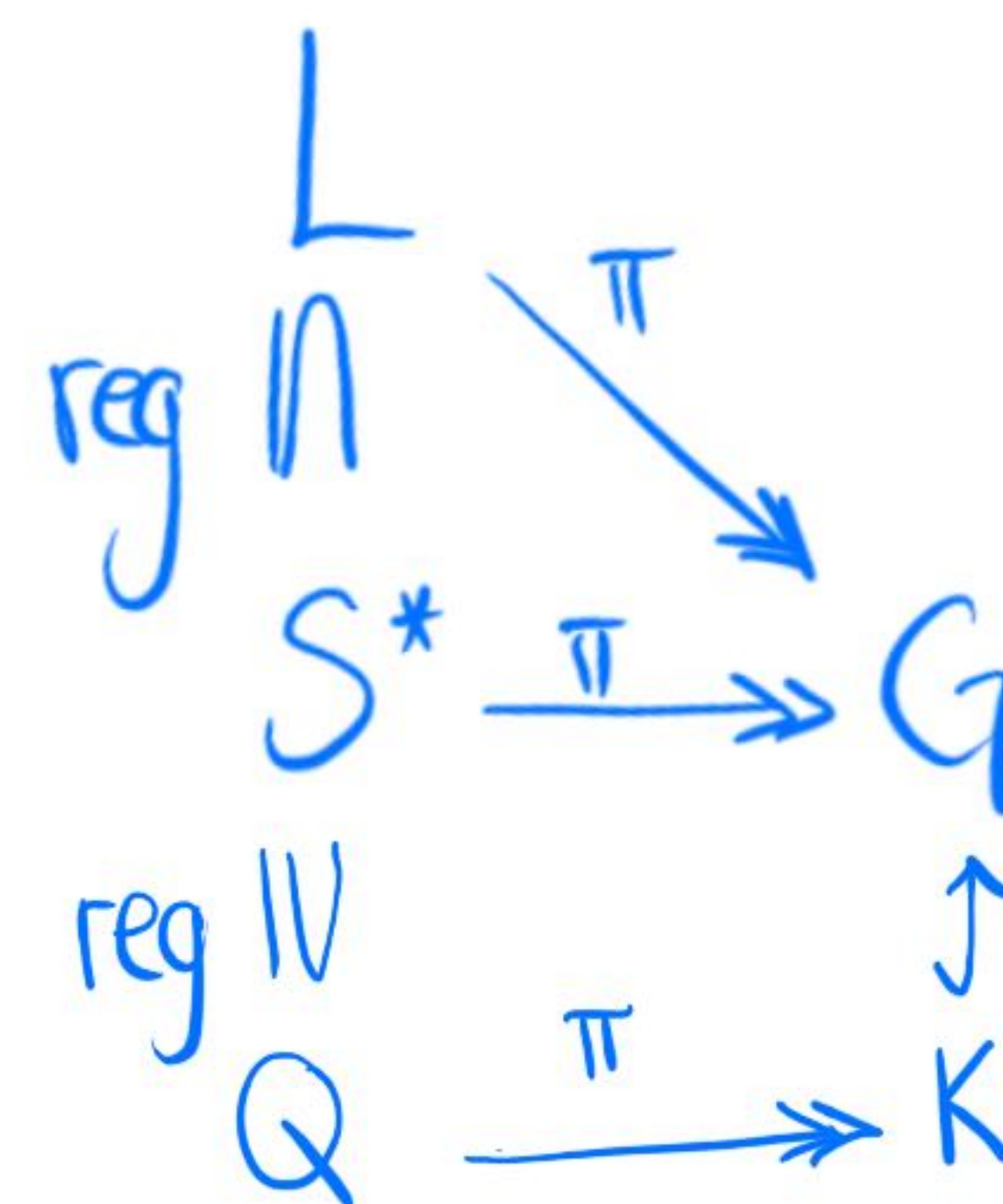
Ingredients: automaton A , presentation $G \cong \langle S | R \rangle$

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, $\{m_s\}_{s \in S}$

K L -quasiconvexly - undistorted

Goal: Build St_K to solve membership problem of K

Ingredients: automaton A , presentation $G \cong \langle S | R \rangle$

Process: ① Initial automaton $\Gamma_0 = A$

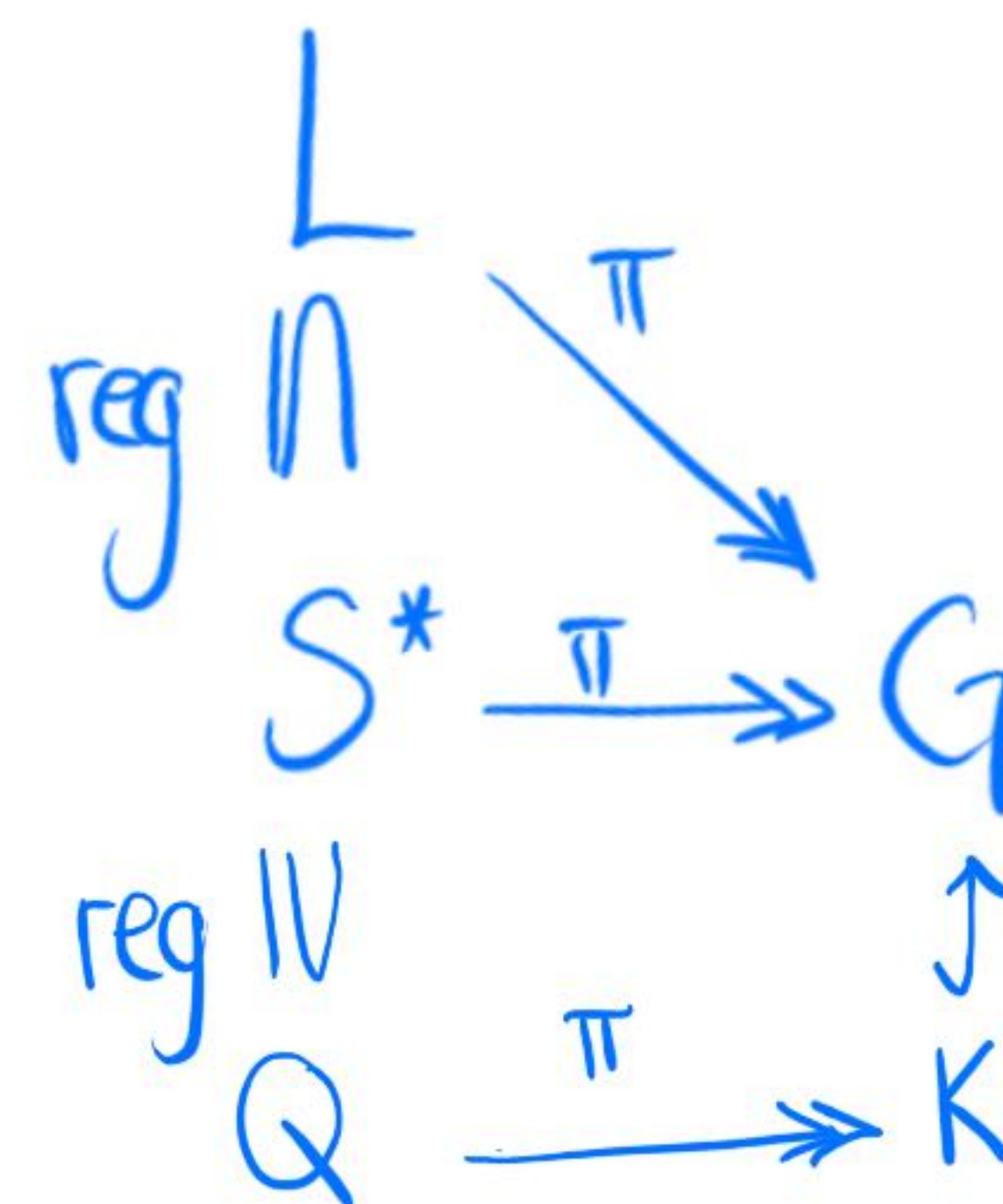
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Goal: Build St_K to solve membership problem of K

Ingredients: automaton A , presentation $G \cong \langle S | R \rangle$

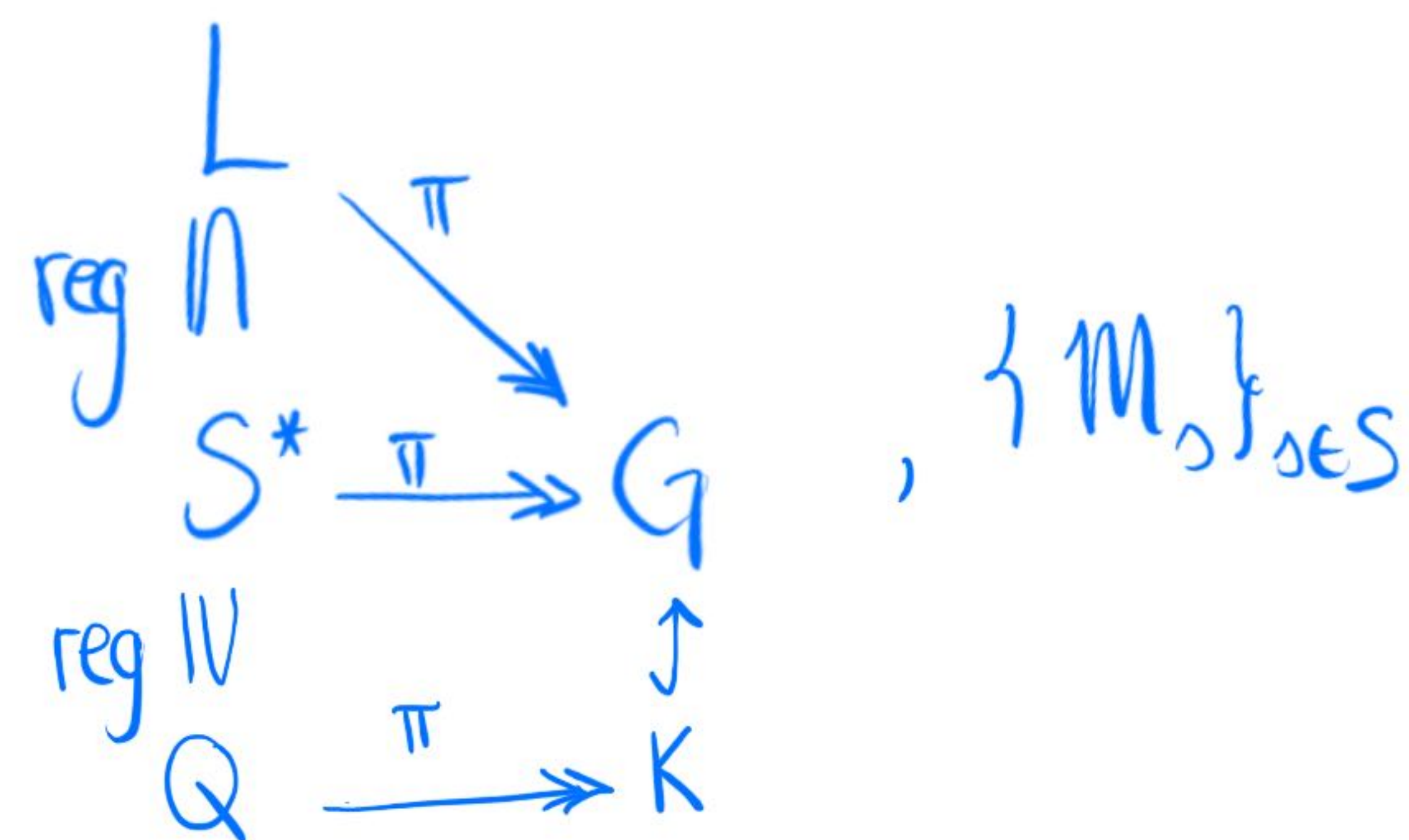
Process: ① Initial automaton $\Gamma_0 = A$

② Iteratively perform foldings $\Gamma_1, \Gamma_2, \dots, \Gamma_\ell$.

Folding (nice) rational
subsets of automatic groups.

$$2. \underset{\text{rational}}{K} \subseteq \underset{\text{automatic}}{G}$$

K rational subset $\subseteq G$ automatic,
fin. pres. $\langle S | R \rangle$
 $\exists A$ FSA : $Q = L(A), K = \pi(Q)$



K L -quasiconvexly - undistorted

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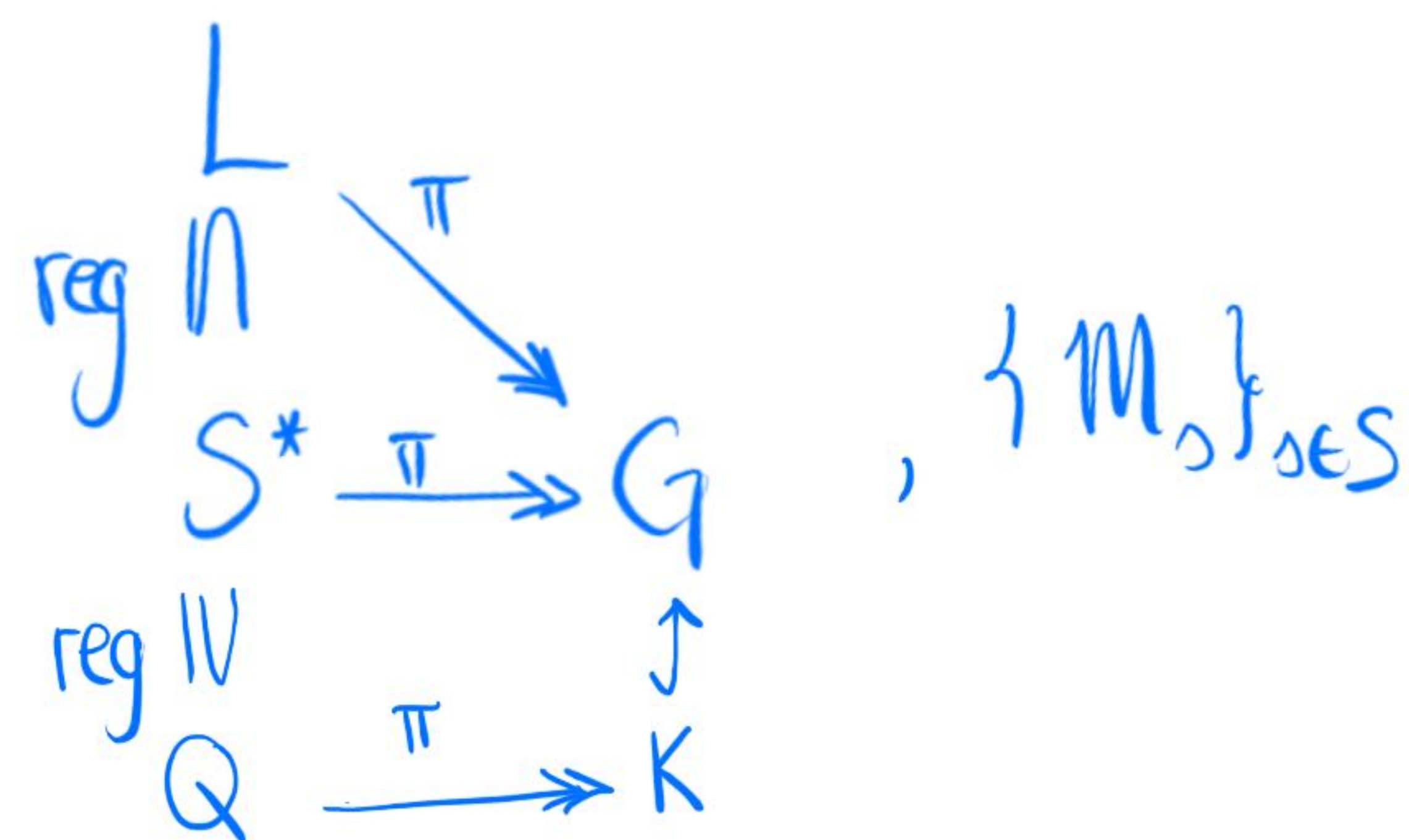
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similar to KMW
 Glue relations + determinise
 edge identification
 + adapt to work away from subgroups

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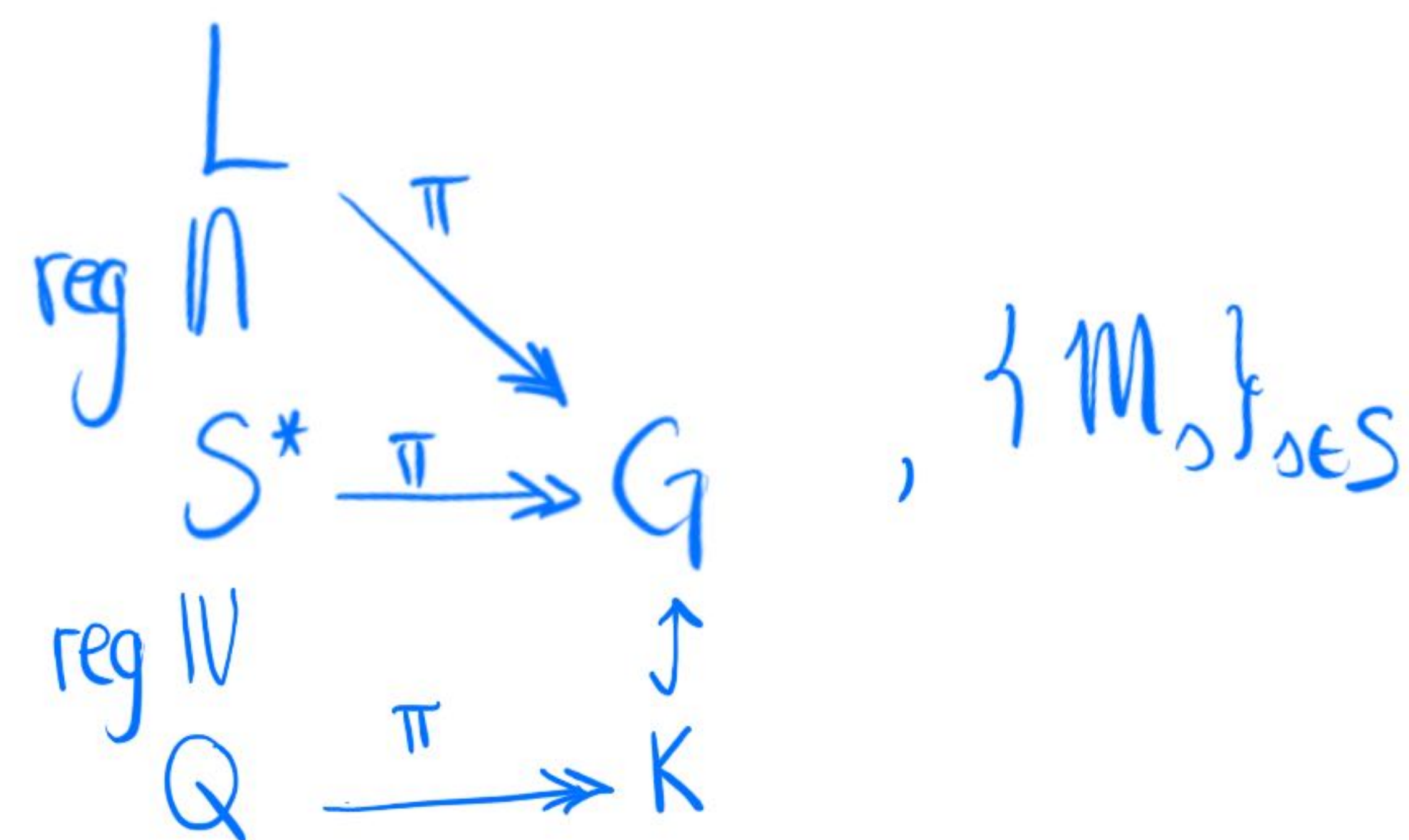
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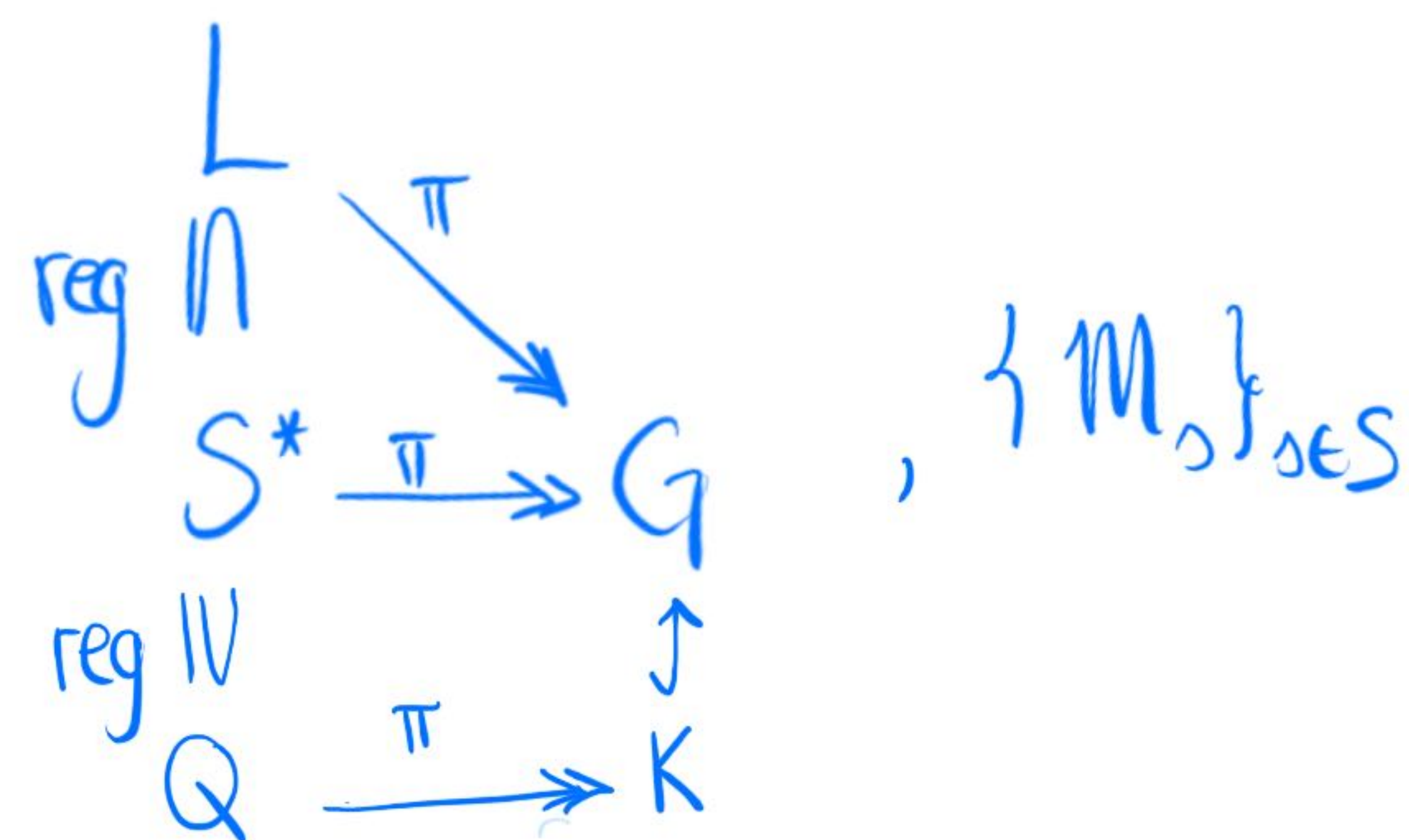
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$$L(St_K) = \{w \in L : \pi(w) \in K\}$$

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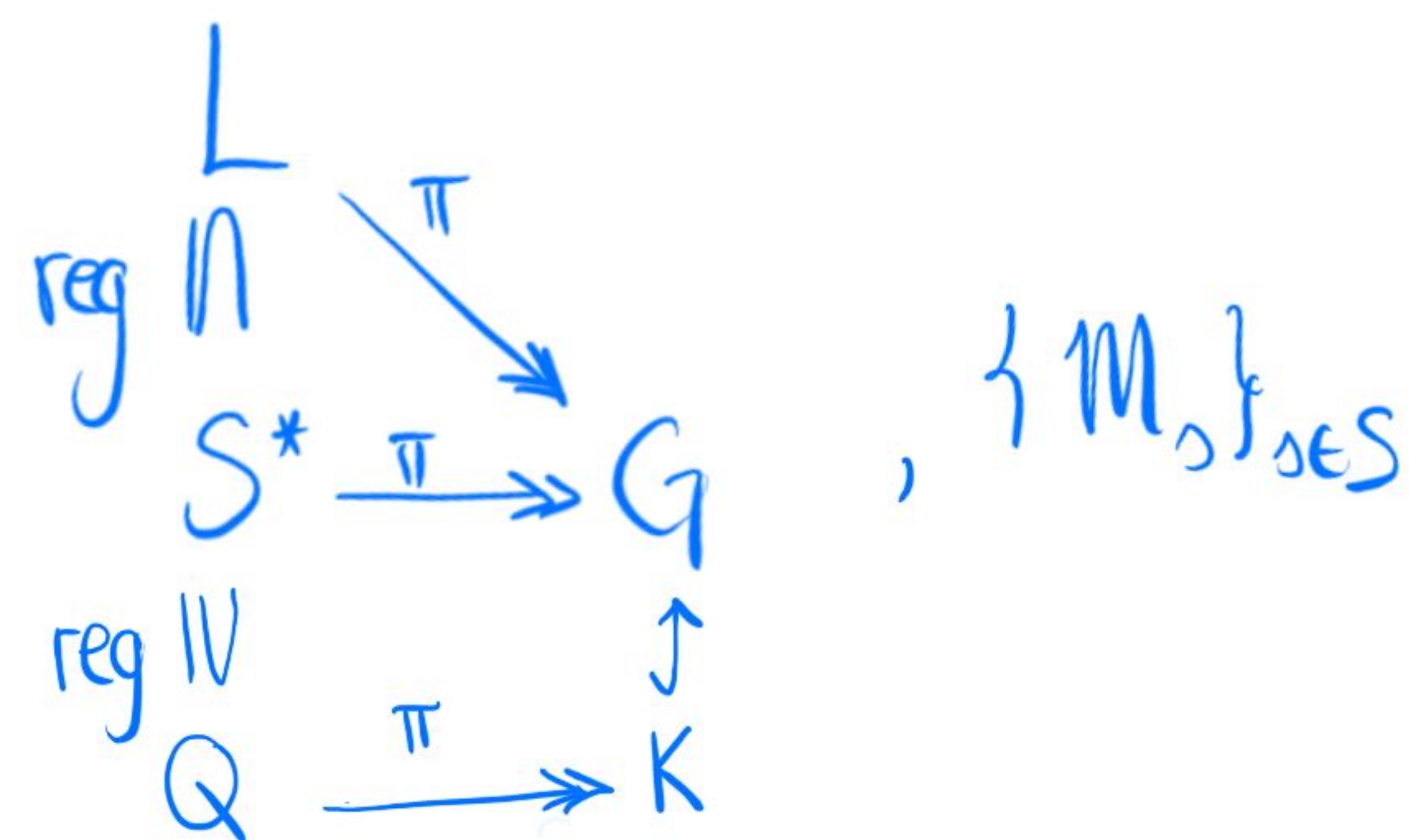
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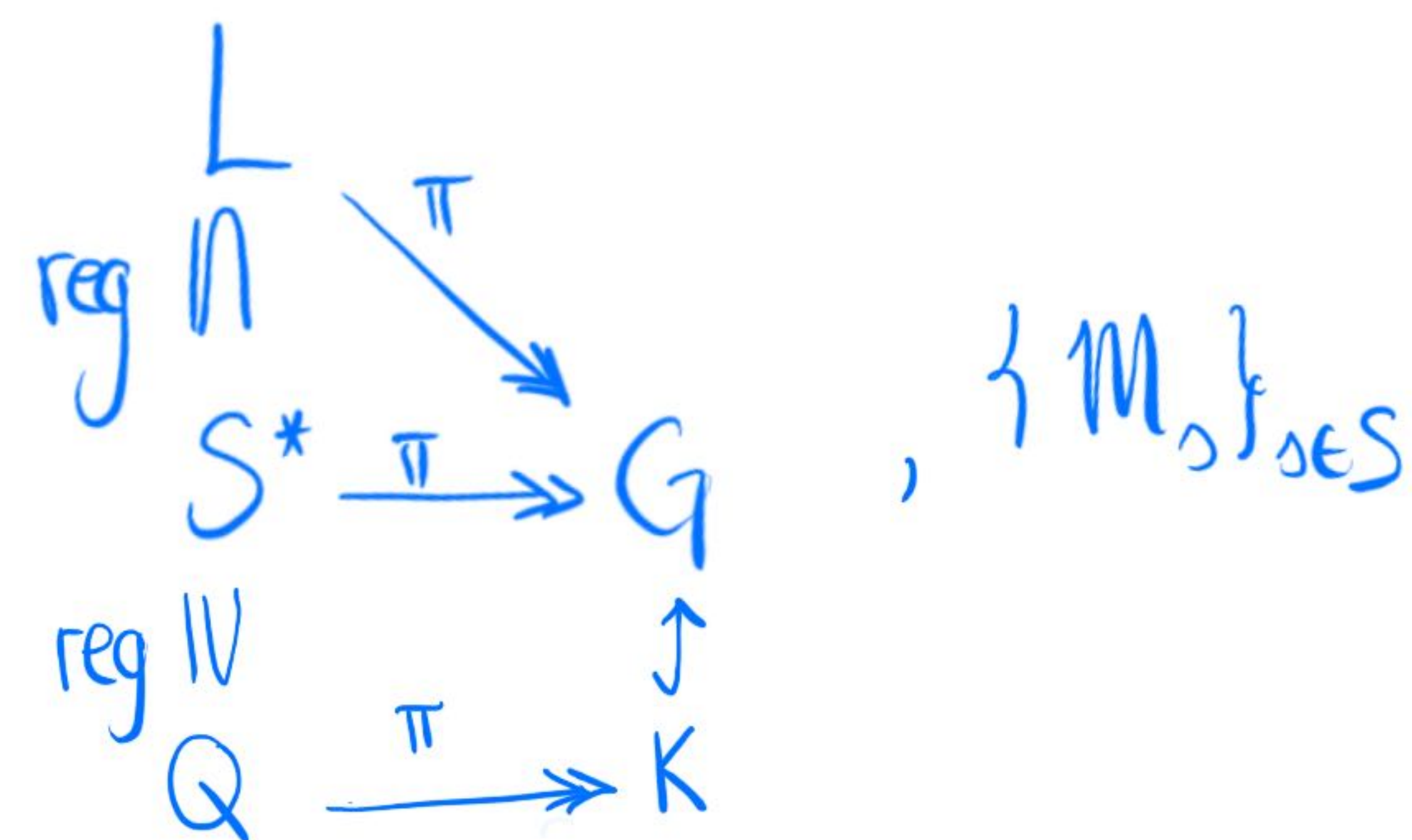
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- Can we do w/o c, k ?
(partial algorithm)
- Complexity?

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Now:

- Can we do w/o c, k ?
(partial algorithm)
- Complexity?
- (Automatic) applications

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Gazteak RSME '25

¡Gracias!